On Arithmetic Subgroups of Split Semisimple Groups

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INTRODUCTION

Let k be an algebraic number field and $\mathfrak{o}$ the ring of integers of k. Let G be a connected linear algebraic group defined over k and G_k the group of rational points of G. If we denote by $G_{\mathfrak{o}}$ the group of integer points of G relative to a given embedding of G into a linear group GL_n, a subgroup Γ' of G_k is called an arithmetic subgroup if $\Gamma' \cap G_{\mathfrak{o}}$ is of finite index in Γ' and in $G_{\mathfrak{o}}$. For every non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$, the subgroup $\Gamma_{\mathfrak{q}}$ of $G_{\mathfrak{o}}$ formed of elements g such that $g \equiv 1 \pmod{\mathfrak{q}}$ is such a subgroup of G_k. An arithmetic subgroup Γ' of G_k is called a congruence subgroup if Γ' contains the subgroup $\Gamma_{\mathfrak{q}}$ of $G_{\mathfrak{o}}$ for some non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$. These definitions of arithmetic and congruence subgroups do not depend on the choice of the embedding of G into a linear group.

The problem of congruence subgroups for G_k consists in knowing if every arithmetic subgroup of G_k is a congruence subgroup, in other words if every subgroup of finite index in $G_{\mathfrak{o}}$ contains $\Gamma_{\mathfrak{q}}$ for some non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$. The main goal of this Memoir is to solve this problem for simple simply connected groups G split over k and of rank ≥ 2. We also treat an analogous problem for groups $G_{\mathfrak{o}}$ where $\mathfrak{o}$ is a Dedekind ring said to be of arithmetic type.

In Chapter I, we study the properties of an exact sequence of locally compact groups linked to the problem of congruence groups. Let G be a simple simply connected group split over k. Let us put a topology T_a [resp. T_c] on G_k by taking as a basis of neighborhoods of the identity element the arithmetic [resp. congruence] subgroups of G_k,

and let us denote by $\hat{G}_k$ [resp. $\overline{G}_k$] the completion of G_k for T_a [resp. T_c]; these completions are locally compact totally disconnected groups. Since T_a is finer than T_c, we have a canonical morphism π from $\hat{G}_k$ into $\overline{G}_k$; the morphism π is surjective and proper. The congruence group problem reduces to knowing whether the kernel $C(G_k)$ of π reduces to the identity element. It is known, by the strong approximation theorem, that $\overline{G}_k$ is identified with the adelified group $G_{A_k^f}$, where A_k^f denotes the ring of finite adeles of k, that is to say the restricted product of the completions k_v of k relative to the rings $\mathfrak{o}_v$ of integers in k_v, with v ranging over the set of finite places of k. On the other hand, the extension $\hat{G}_k$ of $G_{A_k^f}$ is trivial above the group G_k, regarded as a discrete group, contained in $G_{A_k^f}$.

We prove for the groups G_σ a reduction theorem on the rank of G, analogous to a stability theorem of H. Bass [2] for the groups $SL_n(\mathfrak{o})$, and we deduce from this that, if G is of rank ≥ 2, the kernel $C(G_k)$ of π is central in $\hat{G}_k$. Furthermore, the extension $\hat{G}_k$ of $G_{A_k^f}$ possesses a universal property that characterizes it among all the central extensions of $G_{A_k^f}$.

The group schemes over $\mathbf{Z}$ associated by C. Chevalley [15] with semisimple groups will play a fundamental role in the first chapter as in the second, and moreover they will allow us to subject groups of various types to a unified treatment, if only to avoid matrix calculations.

Chapter II is devoted to the study of central extensions of G_{k_t} where G is a split simply connected simple group over an infinite topological field k_t; the group G_{k_t} is endowed with the topology defined from that of k_t. If A is a topological abelian group, the classes of central extensions of G_{k_t} by A form, with Baer multiplication, the cohomology group $H^2(G_{k_t}, A)$. We reduce the determination of the central

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extensions of G_{k_t} by A to that of certain cocycles on $k_t^\times$ with values in A, following the method of R. Steinberg [30] and C. Moore [26]; it is therefore the Bruhat decomposition of G_{k_t} that plays a primary role there.

We shall call a Steinberg cocycle any map c from $k_t^\times \times k_t^\times$ into A satisfying the following equations and conditions:

(S1) $c(x,y)c(xy,z) = c(x,yz)c(y,z)$;

(S2) $c(1,1) = 1$; $\quad c(x,y) = c(x^{-1},y^{-1})$;

(S3) $c(x,y) = c(x,(1-x)y) \quad$ if $x \neq 1$;

(S0) *The function $c(x,y)$ is continuous in $k_t^\times \times k_t^\times$, and, when we agree that $c(0,1) = 1$, the function $c(x,1-xy)$ of (x,y), defined on a neighborhood of $(0,0)$ in $k_t \times k_t$, is continuous at $(0,0)$.*

Let then $S(k_v^\times, A)$ be the group of Steinberg cocycles on $k_v^\times$ with values in A, and $S^0(k_v^\times, A)$ its subgroup formed by the elements c which are bilinear as maps from $k_v^\times \times k_v^\times$ into A. The main theorem of Chapter II states that the group $H^2(G_{k_v}, A)$ is isomorphic to $S(k_v^\times, A)$ or to $S^0(k_v^\times, A)$ according to whether G is isomorphic or not to a symplectic group. For $G = \mathrm{SL}_2$, this result was essentially obtained by C. Moore [26].

Let k_v be a locally compact non-discrete field, and μ_v the group of roots of unity contained in k_v. If $k_v \neq \mathbf{C}$, the symbol of norm remainders $(x,y)_{m,k_v}$, m being the order of μ_v, is a bilinear Steinberg cocycle c^0 on $k_v^\times$ with values in μ_v. Moreover, as $\mathbf{R}$ is an ordered field, one defines as follows a Steinberg cocycle c_0, non-bilinear, on $\mathbf{R}^\times$ with values in $\mathbf{Z}$: $c_0(x,y) = 1$ if $x < 0, y < 0$; $c_0(x,y) = 0$ otherwise.

In Chapter III, we first recall some generalities on the central extensions of locally compact groups; notably the notions, due to C. Moore [26], of universal covering, fundamental group, and relative fundamental group. Then we combine the theorems of the preceding chapters with two arithmetic results of C. Moore [26] on the determination of Steinberg cocycles for a local field and on the uniqueness of reciprocity formulas in a global field.

Using the first of these results, we see that, if k_v is an ultrametric local field [resp. the field $\mathbf{R}$], the extension of G_{k_v} corresponding to the canonical Steinberg cocycle c^0 [resp. c_0 or c^0 according to whether G is isomorphic or not to a symplectic group] is a universal covering of G_{k_v}; for $k_v = \mathbf{C}$, it is well known that $G_{\mathbf{C}}$ is itself simply connected in the topological sense.

Let k be a global field and A_k the ring of adeles of k. C. Moore's theorem on reciprocity formulas implies that the relative fundamental group of the adelized group G_{A_k} with respect to its subgroup G_k is isomorphic to the group μ_k of roots of unity contained in k; which essentially means that there exists a central extension $\tilde{G}_{A_k}$ of G_{A_k} by μ_k which is trivial over G_k, and that every central extension of G_{A_k} by a locally compact abelian group B, trivial over G_k, is deduced from $\tilde{G}_{A_k}$ by a uniquely determined morphism from μ_k into B.

Such coverings of G_{k_v} and of G_{A_k} have been noticed and constructed for the groups SL_n and Sp_{2n} by A. Weil [32], T. Kubota [19], C. Moore [26] and H. Bass, J. Milnor and J.-P. Serre [4]. The case of the field $\mathbf{R}$ goes back, of course, to É. Cartan [11], who had moreover determined the fundamental group $\pi_1(G_{\mathbf{R}})$ for every simply connected simple group G over $\mathbf{R}$, split or not.

From these theorems, we derive a solution to the congruence subgroup problem for simple simply connected groups G of rank ≥ 2 and split over an algebraic number field k: the kernel $C(G_k)$ of the morphism π from $\hat{G}_k$ onto $\bar{G}_k$ is isomorphic to μ_k if k is totally imaginary; otherwise, $C(G_k)$ reduces to the neutral element. In particular, if k has at least one real conjugate, every arithmetic subgroup of G_k is a congruence subgroup. One also sees that, if G is of rank ≥ 2, every arithmetic subgroup of G_k contains a subgroup of finite index generated by unipotent elements.

This solution to the congruence subgroup problem had been obtained, for the groups SL_n $(n \geq 3)$ and Sp_{2n} $(n \geq 2)$ over $\mathbf{Q}$, independently by J. Mennicke ([22], [23]) and by H. Bass, M. Lazard, and J.-P. Serre [3], and then for these same groups over an arbitrary number field by Bass, J. Milnor, and Serre [4]. The case of SL_2 has recently been solved by Serre (see [29]).

This thesis was carried out under the direction of Mr. F. Bruhat, who suggested to me the idea of undertaking this study. I express all my gratitude to him. I would like to warmly thank Mr. J.-P. Serre for the advice he constantly gave me both during the preparation of my work and during its final writing. I also thank Mr. P. Samuel, who was kind enough to suggest a second thesis topic to me.

CHAPTER I
ARITHMETIC SUBGROUPS OF A SPLIT SIMPLE GROUP

1. In this section, we will formulate the congruence subgroup problem, which we will then study for split simple simply connected groups.

Let k be a finite degree extension of $\mathbf{Q}$ or an algebraic function field of dimension 1 over a finite field. A subring $\mathfrak{o}$ of k will be called a *Dedekind ring of arithmetic type*, if there exists a non-empty finite set S of places of k containing the archimedean places and such that $\mathfrak{o}$ is the set of elements x of k such that $v(x) \leq 1$ for every place $v \notin S$; where the valuation v is denoted multiplicatively. If this is the case, k is the field of fractions of $\mathfrak{o}$ and S is uniquely determined by $\mathfrak{o}$. The prime ideals of $\mathfrak{o}$ correspond canonically to the places $v \notin S$ of k.

Let G be a connected linear algebraic group defined over k and G_k the group of rational points of G. One chooses an embedding of G into a linear group GL_m and one considers the group $G_{\mathfrak{o}}$ of points of G

with values in $\mathfrak{o}$, where $\mathfrak{o} = \mathfrak{o}(S)$ is the Dedekind ring of arithmetic type defined from a set S of places of k. A subgroup Γ' of G_k is called an S-arithmetic subgroup if $\Gamma' \cap G_{\mathfrak{o}}$ is of finite index in Γ' and in $G_{\mathfrak{o}}$. An S-arithmetic subgroup Γ' of G_k is called an S-congruence subgroup if Γ' contains, for a non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$, the subgroup $\Gamma_{\mathfrak{q}}$ of $G_{\mathfrak{o}}$ formed by the elements g such that $g \equiv 1$ modulo $\mathfrak{q}$.

One sees easily that these definitions do not depend on the choice of the embedding of G into a linear group GL_m. Consequently, one can define on G_k a group topology $T_a(S)$ [resp. $T_c(S)$] by taking as a basis of neighborhoods of e the S-arithmetic subgroups [resp. S-congruence subgroups] of G_k. Let then $\widehat{G_k(S)}$ [resp. $\overline{G_k(S)}$] be the completion of G_k with respect to $T_a(S)$ [resp. $T_c(S)$]; the groups $\widehat{G_k(S)}$ and $\overline{G_k(S)}$ are locally compact and totally disconnected. Since $T_a(S)$ is finer than $T_c(S)$, there is a canonical morphism π from $\widehat{G_k(S)}$ into $\overline{G_k(S)}$.

The completion $\widehat{G_{\mathfrak{o}}}$ [resp. $\overline{G_{\mathfrak{o}}}$] of $G_{\mathfrak{o}}$ with respect to $T_a(S)$ [resp. $T_c(S)$] is an open compact subgroup of $\widehat{G_k(S)}$ [resp. $\overline{G_k(S)}$]. The group $\widehat{G_{\mathfrak{o}}}$ is none other than the completion of $G_{\mathfrak{o}}$ with respect to the topology of finite index subgroups. One sees easily that the restriction of π to $\widehat{G_{\mathfrak{o}}}$ has as image $\overline{G_{\mathfrak{o}}}$ and as kernel a profinite group $C(G_{\mathfrak{o}})$. Consequently, the morphism π maps $\widehat{G_k(S)}$ onto $\overline{G_k(S)}$, and the kernel $C^S(G_k)$ of π, which is canonically identified with $C(G_{\mathfrak{o}})$, is a profinite group.

We thus have an exact sequence of locally compact groups:

$$1 \to C^S(G_k) \to \widehat{G_k(S)} \xrightarrow{\pi} \overline{G_k(S)} \to 1. \tag{1}$$

The congruence subgroup problem for G_k relative to S consists in determining whether $C^S(G_k)$ reduces to the identity element, that is to say, if every finite index subgroup in $G_{\mathfrak{o}}$ contains $\Gamma_{\mathfrak{q}}$ for a non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$.

The usual case is the one where S is the set S_∞ of archimedean places of k, which is equivalent to saying that $\mathfrak{o}$ is the ring of integers of a number field k. One easily verifies that $C^{S_\infty}(G_k)$ reduces to $\{e\}$ if G is a unipotent group. According to C. Chevalley [12], this is still the case if G is a torus. On the other hand, $C^{S_\infty}(G_k)$ is generally infinite if G is a non-simply connected semi-simple group (J.-P. Serre [28]). Moreover, it has been known since F. Klein that $C^{S_\infty}(G_k)$ is infinite for $G = SL_2$ and $k = \mathbf{Q}$.

We will assume in the remainder of this section that G is a simply connected semi-simple group.

Let $A_k(S)$ be the ring of S-adeles of k, that is to say the restricted product of the completions of k for the places $v \notin S$ of k. The group $G_{A_k(S)}$ of points of G with values in $A_k(S)$, equipped with the topology defined from that

of $A_k(S)$, is a locally compact group. One sees that the topology of $G_{A_k(S)}$ induces on G_k exactly the topology $T_c(S)$, and therefore that $\overline{G_k(S)}$ is identified with the closure of G_k in $G_{A_k(S)}$. Consequently, if k is of characteristic 0 or if G satisfies strong approximation relative to S, $\overline{G_k(S)}$ is identified with the adelified group $G_{A_k(S)}$ (cf. M. Kneser [18]).

On the other hand, the exact sequence (1.1) is trivial over the group G_k, regarded as a discrete group, contained in $\overline{G_k(S)}$.

When $C^S(G_k)$ is finite and the quotient of G_k by its center Z_k has no non-trivial finite quotient, one sees immediately that $C^S(G_k)$ is central in $\overline{G_k(S)}$. In general, let us denote by $C_0^S(G_k)$ the closure of the subgroup of $C^S(G_k)$ generated by the commutators $xyx^{-1}y^{-1}$ for $x \in C^S(G_k)$ and $y \in \overline{G_k(S)}$. One then has, by passing to the quotient, a central extension $\widetilde{G_k(S)}$ of $\overline{G_k(S)}$:

$$1 \to C_u^S(G_k) \to \widetilde{G_k(S)} \xrightarrow{\tilde{\pi}} \overline{G_k(S)} \to 1;$$

where

$$\widetilde{G_k(S)} = \overline{G_k(S)}/C_0^S(G_k) \quad \text{and} \quad C_u^S(G_k) = C^S(G_k)/C_0^S(G_k).$$

This being so, the following theorem expresses a universal character of the extension $\widetilde{G_k(S)}$ of $\overline{G_k(S)}$ (Bass-Milnor-Serre [4]):

Theorem 0.1. *With the above notation, let an extension E of $\overline{G_k(S)}$ by a profinite group F be given such that its restriction to the group G_k, regarded as discrete, is trivial.*

(a) There then exists a morphism φ from $\overline{G_k(S)}$ into E such that $p \circ \varphi = \pi$, where p denotes the projection of E onto $\overline{G_k(S)}$.

(b) Suppose that G_k does not possess a non-trivial finite abelian quotient. Then, if in addition F is central in E, there exists one and only one morphism $\tilde{\varphi}$ from $\widetilde{G_k(S)}$ into E such that $p \circ \tilde{\varphi} = \tilde{\pi}$.

Proof. For statement (a), we reproduce the proof from [4]. There exists, by hypothesis, a morphism σ from G_k into E such that $p \circ \sigma(g) = g$ for all $g \in G_k$. The map $\varphi = \sigma \circ \pi$ is thus a morphism defined on the group G_k contained in $\overline{G_k(S)}$. However, the image of G_o by φ is contained in the profinite subgroup $p^{-1}(\overline{G}_o)$ of E. Consequently, given the definition of $\widehat{G}_o$, $\varphi|_{G_o}$ extends to a continuous morphism from $\widehat{G}_o$ into $p^{-1}(\overline{G}_o)$.

The morphism φ thus extends to a continuous morphism φ from $\overline{G_k(S)}$ into E. It is clear that $p \circ \varphi = \pi$. As for statement (b), one deduces from (a), by passing to the quotient, that there exists a morphism $\tilde{\varphi}$ from $\widetilde{G_k(S)}$ into E... □

such that $p \circ \tilde{\varphi} = \tilde{\pi}$. If $\tilde{\psi}$ is a morphism from $\widetilde{G_k(S)}$ into E having the same property, the map δ, defined by $\delta(x) = \tilde{\psi}(x)\tilde{\varphi}(x)^{-1}$, is a morphism from $\widetilde{G_k(S)}$ into F; since F is an abelian profinite group, our hypothesis on G_k implies that δ is trivial in G_k and therefore, by continuity, in the whole of $\widetilde{G_k(S)}$. This completes the proof of the theorem.

2. In this section, we will recall the construction of certain group schemes split over $\mathbf{Z}$ and some of their fundamental properties (cf. C. Chevalley [15] and M. Demazure [16]).

Let G be a connected semi-simple algebraic group over $\mathbf{C}$ and H a maximal torus of G. We denote by $\mathfrak{g}$ and $\mathfrak{h}$ the Lie algebras of G and H respectively. Let Φ be the root system of G relative to H. We choose a Chevalley lattice $\mathfrak{g}_{\mathbf{Z}}$ of $\mathfrak{g}$ adapted to $\mathfrak{h}$:

$$\mathfrak{g}_{\mathbf{Z}} = \mathfrak{h}_{\mathbf{Z}} + \sum_{\alpha} \mathbf{Z} e_\alpha$$

with a lattice $\mathfrak{h}_{\mathbf{Z}}$ of $\mathfrak{h}$ and a system of root elements e_α. For each $\alpha \in \Phi$, the element $h_\alpha = [e_\alpha, e_{-\alpha}]$ belongs to $\mathfrak{h}_{\mathbf{Z}}$ and we have $\alpha(h_\alpha) = 2$, the root α being identified with its differential; $\alpha(h)$ is in $\mathbf{Z}$ for any $h \in \mathfrak{h}_{\mathbf{Z}}$; if α, β and $\alpha + \beta$ are roots, we have $[e_\alpha, e_\beta] = N_{\alpha,\beta} e_{\alpha+\beta}$ with a non-zero integer $N_{\alpha,\beta}$; we have $N_{\alpha,\beta} = -N_{-\alpha,-\beta}$. Note that, if $\{\varepsilon_\alpha\}, \alpha \in \Phi$, is a set of integers ± 1 such that $\varepsilon_\alpha = \varepsilon_{-\alpha}$ for all α, then the elements $\varepsilon_\alpha e_\alpha$ also form a system of root elements for the Chevalley lattice $\mathfrak{g}_{\mathbf{Z}}$.

For $\alpha, \beta \in \Phi$, the integer $\alpha(h_\beta)$ will be called the Cartan integer relative to α and β, and will be denoted $\alpha\beta^*$.

Let ρ be a representation of G in a vector space V; we have $V = \sum_\lambda V^\lambda$, where λ ranges over the set $\Lambda(\rho)$ of weights of ρ relative to H and where V^λ is the subspace of V belonging to λ. Recall that a lattice $V_{\mathbf{Z}}$ of V is said to be *admissible* with respect to $\mathfrak{g}_{\mathbf{Z}}$ if $V_{\mathbf{Z}}$ satisfies the following conditions:

(i) $V_{\mathbf{Z}} = \sum_\lambda V_{\mathbf{Z}}^\lambda$, where $V_{\mathbf{Z}}^\lambda = V_{\mathbf{Z}} \cap V_\lambda$;

(ii) For all $\alpha \in \Phi$ and every integer $k \geq 0$, $\rho(e_\alpha)^k/k!$ leaves $V_{\mathbf{Z}}$ stable.

Consider a faithful representation ρ of G in a vector space V and an admissible lattice $V_{\mathbf{Z}}$ of V with respect to $\mathfrak{g}_{\mathbf{Z}}$. Any basis $\{v_1, \ldots, v_m\}$ of $V_{\mathbf{Z}}$ determines coordinates $t_{i,j}$ $(1 \leq i, j \leq m)$ on $GL(V)$, and the restrictions of these to G generate a subring $\mathbf{Z}[G]$ of the affine algebra $\mathbf{C}[G]$ of G. The ring $\mathbf{Z}[G]$, equipped with the comultiplication $\tilde{\pi}$ induced from...

...that of $\mathbb{C}[G]$, defines a group scheme $\mathfrak{G}$ over $\mathbb{Z}$, which we shall call the Chevalley-Demazure scheme associated to G. It is known that the group scheme $\mathfrak{G}$ does not depend, up to isomorphism, on the choice of H, of $\mathfrak{g}_{\mathbb{Z}}$, of ρ, or of $V_{\mathbb{Z}}$.

The algebra $\mathbb{Q}[G] = \mathbb{Z}[G] \otimes_{\mathbb{Z}} \mathbb{Q}$ makes G an algebraic group defined over $\mathbb{Q}$. If X is a subgroup of G defined over $\mathbb{Q}$, the canonical embedding ι of X into G induces a morphism $\tilde{\iota}$ from $\mathbb{C}[G]$ onto $\mathbb{C}[X]$; we will denote by $\mathbb{Z}[X]$ the subring $\tilde{\iota}(\mathbb{Z}[G])$ of $\mathbb{C}[X]$.

Let us introduce some subgroups of G. Let N be the normalizer of H in G; the quotient N/H is the Weyl group W of G relative to H and we have $N = HN_{\mathbb{Z}}$. When G is simply connected, the group $N_{\mathbb{Z}}$ is called the extended Weyl group associated to G. For each root α, U^{α} will denote the root subgroup $\exp(\mathbb{C}e_{\alpha})$, and G^{α} the simple subgroup of G generated by U^{α} and $U^{-\alpha}$. One defines the isomorphism x_{α} of the additive group onto U^{α} by $x_{\alpha}(t) = \exp(te_{\alpha})$ for $t \in \mathbb{C}$; there then exists a morphism ϕ_{α} from the group SL_2 onto G^{α} such that $\phi_{\alpha}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} = x_{\alpha}(t)$ and that $\phi_{-\alpha}\begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix} = x_{-\alpha}(t)$ for all $t \in \mathbb{C}$.

Let Δ be a system of simple roots of Φ, and Φ^+ the set of positive roots. The subgroup U^+ [resp. U^-] of G generated by the U^{α} [resp. $U^{-\alpha}$], $\alpha \in \Phi^+$, is a maximal unipotent subgroup of G.

Let Δ' be a subset of Δ and let us set $\Delta'' = \Delta - \Delta'$. Let us denote by $G(\Delta')$ [resp. $G(\Delta'')$] the subgroup of G generated by the $G^{\alpha}, \alpha \in \Delta'$ [resp. $\alpha \in \Delta''$]. Let $U^+(\Delta')$ [resp. $U^-(\Delta')$] further be the subgroup generated by the U^{α} [resp. $U^{-\alpha}$], $\alpha \in \Phi^+ - \Phi'^+$, where Φ'^+ is the set of positive roots of $G(\Delta')$ relative to $H(\Delta') = G(\Delta') \cap H$. If $H(\Delta'')$ denotes $G(\Delta'') \cap H$, we have $H = H(\Delta')H(\Delta'')$; it is known that $H(\Delta') \cap H(\Delta'') = \{e\}$ if G is simply connected. The reductive group $S(\Delta') = G(\Delta')H(\Delta'')$ normalizes $U^+(\Delta')$, and $U^+(\Delta')S(\Delta')$ is a parabolic subgroup of G. These subgroups of G are all defined over $\mathbb{Q}$.

Let us recall the following result on the affine algebras of these groups:

a. When the elements α of Φ^+ are suitably ordered, the map ε from an affine space $\mathbb{C}^p$ ($p = \dim U^+$) onto U^+, defined by

$$\varepsilon(t_1, \ldots, t_p) = \prod_i \exp(t_i e_{\alpha_i}),$$

is an isomorphism of algebraic varieties. Furthermore, $\mathbb{Z}[U^+]$ is the polynomial algebra over $\mathbb{Z}$ in the p variables $t_i \circ \varepsilon^{-1}$. We have an analogous result for the groups U^-, $U^+(\Delta')$ and $U^-(\Delta')$.

b. The algebra $\mathbb{Z}[H]$ is generated over $\mathbb{Z}$ by $\chi_1, \chi_1^{-1}, \ldots, \chi_r, \chi_r^{-1}$ if $\{\chi_1, \ldots, \chi_r\}$ is a generating set for the character group of H. We have an analogous result for $H(\Delta')$ and $H(\Delta'')$.

c. Let $\mathfrak{g}'$ be the Lie algebra of $G' = G(\Delta')$. Then $\mathfrak{g}'_{\mathbb{Z}} = \mathfrak{g}' \cap \mathfrak{g}_{\mathbb{Z}}$ is written in the form $\mathfrak{h}_{\mathbb{Z}} \cap \mathfrak{g}' + \sum_{\alpha} \mathbb{Z}e_{\alpha}$, where α runs through the set of roots of G' relative to $H(\Delta')$; thus $\mathfrak{g}'_{\mathbb{Z}}$ constitutes a Chevalley lattice of $\mathfrak{g}'$ adapted to $\mathfrak{g}' \cap \mathfrak{h}$. For the restriction of ρ to G', $V_{\mathbb{Z}}$ is an admissible lattice of V with respect to $\mathfrak{g}'_{\mathbb{Z}}$. Consequently, the algebra $\mathbb{Z}[G']$ that we have defined for the subgroup G' of G is identical to the one that can be defined, in the same way as for G, for the semisimple group G'.

We will now recall a decomposition theorem. Let μ be a character of H, dominant with respect to the system of simple roots Δ, such that μ vanishes on $H(\Delta')$ and does not vanish on $H \cap G^{\alpha}$ for any $\alpha \in \Delta''$. We choose an irreducible representation ρ of G in a vector space V, of highest weight μ; we have $V = \sum_{\lambda} V^{\lambda}$, where λ runs through the set of weights of ρ relative to H and where V^{λ} is the subspace of V belonging to the weight λ. Then, the subgroup $U^+(\Delta')S(\Delta')$ is the stabilizer in G of V^{μ}, and the function t_{μ} on G, defined by $\rho(g)v^{\mu} \equiv t_{\mu}(g)v^{\mu}$ modulo $\sum_{\lambda \neq \mu} V^{\lambda}$ with a non-zero vector v^{μ} of V^{μ}, belongs to $\mathbb{Z}[G]$. The map $\tilde{\pi} \otimes \pi$ from $U^-(\Delta') \times S(\Delta') \times U^+(\Delta')$ into G defined by $\tilde{\pi} \otimes \pi(u', s, u) = u'su$ is an isomorphism of algebraic varieties from $U^-(\Delta') \times S(\Delta') \times U^+(\Delta')$ onto an affine open set Ω_{μ} of G, whose affine algebra is precisely $\mathbb{Z}[G](t_{\mu}^{-1})$. The same is obviously true for the map, defined analogously, from $U^-(\Delta') \times U^+(\Delta') \times S(\Delta')$ onto Ω_{μ}. If $H(\Delta'') \cap G(\Delta') = \{e\}$, the map π from $H(\Delta'') \times G(\Delta')$ onto $S(\Delta')$ is also an isomorphism of algebraic varieties.

More precisely, according to [15], we thus have the following algebra isomorphisms:

$$\tilde{\pi} \otimes \bar{\pi} : \mathbb{Z}[G](t_{\mu}^{-1}) \to \mathbb{Z}[U^-(\Delta')] \otimes \mathbb{Z}[S(\Delta')] \otimes \mathbb{Z}[U^+(\Delta')],$$

$$\tilde{\pi} \otimes \bar{\pi} : \mathbb{Z}[G](t_{\mu}^{-1}) \to \mathbb{Z}[U^-(\Delta')] \otimes \mathbb{Z}[U^+(\Delta')] \otimes \mathbb{Z}[S(\Delta')]$$

and, if $H(\Delta'') \cap G(\Delta') = \{e\}$,

$$\tilde{\pi} : \mathbb{Z}[S(\Delta')] \to \mathbb{Z}[H(\Delta'')] \otimes \mathbb{Z}[G(\Delta')];$$

where one has respectively

$$\tilde{\pi} \otimes \bar{\pi}(t_{\mu}) = 1 \otimes (t_{\mu}|S(\Delta')) \otimes 1,$$
$$\tilde{\pi} \otimes \bar{\pi}(t_{\mu}) = 1 \otimes 1 \otimes (t_{\mu}|S(\Delta'))$$

and

$$\tilde{\pi}(t_{\mu}|S(\Delta')) = (\mu|H(\Delta'')) \otimes 1.$$

In particular, for $\Delta' = \varnothing$, this gives a morphism from $U^- \times H \times U^+$ onto an affine open set Ω_{μ_0} of G, and an algebra isomorphism

$$\tilde{\pi} \otimes \bar{\pi} : \mathbb{Z}[G](t_{\mu_0}^{-1}) \to \mathbb{Z}[U^-] \otimes \mathbb{Z}[H] \otimes \mathbb{Z}[U^+].$$

We will then study the admissible lattices of certain representations of G.

Let ρ be a representation of G in a vector space V. If $V_{\mathbf{Z}}$ is an admissible lattice of V with respect to $\mathfrak{g}_{\mathbf{Z}}$, the coefficients of ρ relative to a basis of $V_{\mathbf{Z}}$ belong to $\mathbf{Z}[G]$; the action of G on V thus induces an algebra morphism $\tilde{\rho}$ from $\mathbf{Z}[V]$ to $\mathbf{Z}[G] \otimes \mathbf{Z}[V]$, where $\mathbf{Z}[V]$ denotes the subring of the affine algebra $\mathbf{C}[V]$ of V generated by the coordinates of V relative to a basis of $V_{\mathbf{Z}}$.

To avoid an unnecessary complication, we will assume in the remainder of this section that G is simple.

It is easily seen that for an irreducible representation ρ of G the following conditions are equivalent:

(C) All non-zero weights of ρ are transformed into one another by the Weyl group W of G;

(C') If λ and μ are non-zero weights of ρ whose difference is a root α, then λ is the transform of μ by the reflection $\sigma_\alpha \in W$ associated with α.

We will call a basic representation any non-trivial irreducible representation of G satisfying these conditions.

We first have the following lemma ([14], exposé 20):

Lemma 0.2. *Let ρ be a basic representation of G.*

(a) Every non-zero weight of ρ is then of multiplicity 1 and the multiplicity of the zero weight is the number of simple roots appearing among the non-zero weights of ρ.

(b) Let $\Delta(\rho)$ be the set of these latter simple roots, and for each $\alpha \in \Delta(\rho)$, let v^α be a non-zero vector of V^α. Then, V^0 possesses a basis consisting of elements $v^0_\beta, \beta \in \Delta(\rho)$, such that $\rho(e_\alpha)v^0_\beta = \delta_{\alpha,\beta}v^\alpha$, where $\delta_{\alpha,\beta} = 1$ if $\alpha = \beta$ and $\delta_{\alpha,\beta} = 0$ if $\alpha \neq \beta$.

Proof. Indeed, the first assertion of (a) is well known. Consider next the subalgebra $\mathfrak{m}$ of $\mathfrak{g}$ generated by the e_α and $e_{-\alpha}, \alpha \in \Delta(\rho)$, and the representation of $\mathfrak{m}$ in the smallest subspace V' of V containing V^0 and stable under $\mathfrak{m}$. According to (C'), we have $V' = V^0 + \sum_\lambda V^\lambda$, where λ ranges over the set of transforms of the $\alpha \in \Delta(\rho)$ by the Weyl group of $\mathfrak{m}$ relative to $\mathfrak{m} \cap \mathfrak{h}$. The representation of $\mathfrak{m}$ in V' is thus equivalent to the adjoint representation of $\mathfrak{m}$. Whence the assertions of our lemma. □

Recall that G is simple. A root α of G is said to be long [resp. short], if no root of G has a length greater [resp. smaller] than that of α.

It is known that there exists a basic representation of G whose non-zero weights are the short roots of G; it is, moreover, up to an equivalence[1]

...up to [a homothety], the only basic representation of G to effectively have weight zero. On the other hand, every basic representation of G is a fundamental representation, with the exception of the adjoint representation of a group of type A_n. If G is simply connected, there exist at least c non-equivalent basic representations of G where c is the order of the center of G ([14], loc. cit.). Note also that for the classical groups, their usual realizations (SL, Sp, SO and $Spin$) satisfy condition (C).

We point out the following lemma:

Lemma 0.3. *2.2. — If G is simply connected and of rank ≥ 2, G admits a fundamental representation satisfying (C) and the following condition:*

(D) It is of degree ≥ 3 and its highest weight is not the sum of two simple roots.

Indeed, if G is of rank ≥ 3, every non-trivial irreducible representation of G satisfies this last condition, because every dominant weight $\neq 0$ is a linear combination of all the simple roots with strictly positive coefficients. If G is of type A_2 [resp. C_2], G possesses a fundamental representation of degree 3 [resp. 4] satisfying (C) and not having zero as a weight. For the group of type G_2, the fundamental representation of degree 7 satisfies our conditions.

[1] [[ILLEGIBLE — REASON: text ends mid-sentence at the bottom of the page]]

Now let ρ be a basic representation of G in a vector space V. We will choose an admissible lattice of V uniquely determined up to a homothety. Let $\Lambda(\rho)$ denote the set of weights of ρ, $\Lambda(\rho)^*$ the set of non-zero weights, and $\Delta(\rho)$ the set of simple roots appearing in $\Lambda(\rho)$. We take an admissible lattice $V_{\mathbf{Z}}$ of V with respect to $\mathfrak{g}_{\mathbf{Z}}$: we have $V_{\mathbf{Z}} = \sum_\lambda V_{\mathbf{Z}}^\lambda$, where $V_{\mathbf{Z}}^\lambda = V_{\mathbf{Z}} \cap V^\lambda$ for $\lambda \in \Lambda(\rho)$. We then choose, for each $\lambda \in \Lambda(\rho)^*$, a primitive vector v^λ of $V_{\mathbf{Z}}^\lambda$. There then exists, according to Lemma 2.1, a basis $\{v_\alpha^0\}$ of V^0 indexed by $\Delta(\rho)$ and such that $\rho(e_\alpha)v_\beta^0 = \delta_{\alpha,\beta}v^\alpha$, for all $\alpha, \beta \in \Delta(\rho)$. This also implies that

$$\rho(e_{-\alpha})v^\beta = \pm\delta_{\alpha,\beta}v_\alpha^0$$

for $\alpha, \beta \in \Delta(\rho)$. We deduce from this that the $\mathbf{Z}$-module generated by the elements v^λ and the v_α^0 is an admissible lattice of V with respect to $\mathfrak{g}_{\mathbf{Z}}$.

Suppose that the lattice $V_{\mathbf{Z}}$ is thus normalized; $V_{\mathbf{Z}}$ has a basis consisting of the $v^\lambda, \lambda \in \Lambda(\rho)^*$, and the $v_\alpha^0, \alpha \in \Delta(\rho)$. This basis allows us to see explicitly how the root subgroups U^α operate on V.

Lemma 0.4. *2.3. — The action of the root subgroups $U^\alpha(\alpha \in \Phi)$ on V is written, in terms of the normalized basis, by the following formulas [where it is agreed to write $x_\alpha(t)$ instead of $\rho(x_\alpha(t))$ and where ± 1 signifies a constant ± 1 independent of the variable t]:*

(a) If $\lambda \in \Lambda(\rho)^*$ and $\lambda + \alpha \notin \Lambda(\rho)$; $x_\alpha(t)v^\lambda = v^\lambda$;

(b) If $\lambda, \lambda + \alpha \in \Lambda(\rho)^*$; $x_\alpha(t)v^\lambda = v^\lambda \pm tv^{\lambda+\alpha}$;

(c) If $\alpha \notin \Lambda(\rho)$; $x_\alpha(t)v^0 = v^0$ for all $v^0 \in V^0$;

(d) If $\alpha \in \Lambda(\rho)$; $x_\alpha(t)v^0 = v^0 + t\alpha_*(v^0)v^\alpha$ for all $v^0 \in V^0$, and $x_\alpha(t)v^{-\alpha} = v^{-\alpha} + tv^0(\alpha) \pm t^2v^\alpha$, where α_* is a non-zero element of the dual $(V^0)^*$ of V^0 and $v^0(\alpha)$ a non-zero element of $V_{\mathbf{Z}}^0$;

(e) For all $\alpha \in \Lambda(\rho)$, $v^0(\alpha) = \pm v^0(-\alpha)$ and $\alpha_* = \pm(-\alpha)_*$. The α_*, $\alpha \in \Lambda(\rho)$, constitute in $(V^0)^*$ a basis of the lattice $(V_{\mathbf{Z}}^0)_*$ dual to $V_{\mathbf{Z}}^0$.

This follows immediately from what precedes the lemma and from an elementary result on the representations of a simple group of rank 1.

3. In this section we will study the problem of congruence groups for split simply connected simple groups.

Let G be a simply connected simple group over $\mathbf{C}$. We associate to G the group scheme $\mathfrak{G}$ over $\mathbf{Z}$, by choosing a maximal torus H of G and a Chevalley lattice $\mathfrak{g}_{\mathbf{Z}}$ with a system of root elements e_α (see no. 2); Φ will denote the root system of G relative to H and Δ a system of simple roots in Φ.

Let $\mathfrak{o}$ be a commutative ring with unit. One deduces from $\mathfrak{G}$, by extension of scalars from $\mathbf{Z}$ to $\mathfrak{o}$, an affine group scheme ${}_{\mathfrak{o}}G$ over $\mathfrak{o}$; the group $G_{\mathfrak{o}}$ of points of ${}_{\mathfrak{o}}G$ with values in $\mathfrak{o}$ is the set of morphisms of the algebra $\mathbf{Z}[G] \otimes_{\mathbf{Z}} \mathfrak{o}$ into $\mathfrak{o}$. If $\mathfrak{o}$ is a field k, ${}_kG$ defines a simply connected simple group of the same type as G. Every split simply connected simple group over k is isomorphic over k to a group defined in this manner ([16], [6]).

The group $G_{\mathfrak{o}}$ possesses the subgroups corresponding to the various group sub-schemes of $\mathfrak{G}$ defined in no. 2: in particular, the torus $H_{\mathfrak{o}}$, the unipotent groups $U_{\mathfrak{o}}^+$ and $U_{\mathfrak{o}}^-$, the root subgroups $U_{\mathfrak{o}}^\alpha$ $(\alpha \in \Phi)$. For each $\alpha \in \Phi$, x_α will denote the isomorphism of the additive group onto $U_{\mathfrak{o}}^\alpha$, determined by the choice of the root element e_α (see no. 2).

Let $\mathfrak{o}^\times$ be the multiplicative group of invertible elements of $\mathfrak{o}$. For all $\alpha \in \Phi$ and $t \in \mathfrak{o}^\times$, let $w_\alpha(t) = x_\alpha(t)x_{-\alpha}(-t^{-1})x_\alpha(t)$; this element belongs to the algebraic normalizer $N_{\mathfrak{o}}$ of $H_{\mathfrak{o}}$. The map h_α, defined by $h_\alpha(t) = w_\alpha(t)w_\alpha(1)^{-1}$, is an isomorphism of $\mathfrak{o}^\times$ onto a subgroup $H_{\mathfrak{o}}^\alpha$ of $H_{\mathfrak{o}}$. Furthermore, $H_{\mathfrak{o}}$ is the direct product of the subgroups $H_{\mathfrak{o}}^\alpha$ as α ranges over the system of simple roots Δ. On the other hand, the elements $w_\alpha(1)$, $\alpha \in \Phi$, generate a finite subgroup $\mathfrak{N}_{\mathfrak{o}}$ of $N_{\mathfrak{o}}$, and one has $N_{\mathfrak{o}} = H_{\mathfrak{o}}\mathfrak{N}_{\mathfrak{o}}$.

In the case of a field k, the group G_k is generated by the root subgroups U_k^α, and one has $G_k = U_k^- N_k U_k^+ = U_k^+ N_k U_k^+$. When k is an infinite field, it is known that G_k is equal to its commutator subgroup and that the quotient of G_k by its center Z_k is simple as an abstract group [13].

To simplify the subsequent presentation, we agree to say, if G is isomorphic to one of the symplectic groups, that the simply connected simple group G or its Dynkin diagram Δ is *of symplectic type*; otherwise, we will say that G or Δ is *of non-symplectic type*.

We now come to the problem of congruence groups for G_k, where k is the fraction field of a Dedekind ring $\mathfrak{o}$ of arithmetic type; the ring $\mathfrak{o}$ is defined in k from a set S of places of k. Consider the exact sequence (1.1):

$$1 \to C^S(G_k) \to \widehat{G}_k(S) \xrightarrow{\pi} \overline{G_k(S)} \to 1. \tag{2}$$

On one hand, according to what precedes, the commutator group of $\widehat{G}_k(S)$ [resp. of $\overline{G_k(S)}$] is dense in $\widehat{G}_k(S)$ [resp. $\overline{G_k(S)}$]. On the other hand, we see that G_k is dense in the adelified group $G_{\mathbb{A}_k(S)}$, according to the strong approximation theorem or according to the fact that G_k is generated by its root subgroups $U_\alpha^{k'}$ for each completion k' of k.

In order to study the exact sequence restricted to $G_\mathfrak{o}$, we will define some subgroups of $G_\mathfrak{o}$ where $\mathfrak{o}$ is a commutative ring with unit. When the ring $\mathfrak{o}$ has been fixed, Γ will denote $G_\mathfrak{o}$, and the subgroup of $G_\mathfrak{o}$ generated by the root subgroups $U_\alpha^\mathfrak{o}$ will be denoted $E_\mathfrak{o}$. If $\mathfrak{q}$ is an ideal of $\mathfrak{o}$, we will denote by $\Gamma_\mathfrak{q}$ the kernel of the reduction morphism from $G_\mathfrak{o}$ into $G_{\mathfrak{o}/\mathfrak{q}}$, and $E_\mathfrak{q}$ the smallest normal subgroup of $E_\mathfrak{o}$ containing the $\Gamma_\mathfrak{q} \cap U_\alpha^\mathfrak{o}$ ($\alpha \in \Phi$). We have seen above that $E_\mathfrak{o}$ contains $N_\mathfrak{o}$.

Lemma 0.5. *3.1. — Suppose that G is of rank ≥ 2 and that $\mathfrak{o}$ is an infinite integral domain. Then, every subgroup of finite index in $G_\mathfrak{o}$ contains the group $E_\mathfrak{q}$ for a non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$.*

Proof. Let Γ' be a subgroup of finite index in $G_\mathfrak{o}$: to prove the lemma, we can suppose that Γ' is normal in $G_\mathfrak{o}$. Let us first note that, if $\mathfrak{o}$ is of characteristic 0, our assertion is trivial and also valid for SL_2; indeed, if $G_\mathfrak{o}/\Gamma'$ is of order n, Γ' contains all the elements of the form $x_\alpha(nt)$ ($\alpha \in \Phi, t \in \mathfrak{o}$), and therefore the group $E_\mathfrak{q}$ for $\mathfrak{q} = n\mathfrak{o}$. Even in arbitrary characteristic, it suffices to consider the cases of rank 2, since E contains $N_\mathfrak{o}$ and every root subgroup $U_\alpha^\mathfrak{o}$ is transformed by an element of $N_\mathfrak{o}$ into $U_\beta^\mathfrak{o}$ for a simple root β. We distinguish the three cases of rank 2.

In the case of type A_2, let us denote by α, β and $\alpha+\beta$ the positive roots. We have the commutation relation ([13], [16]):

$$x_\alpha(u)x_\beta(v)x_\alpha(u)^{-1}x_\beta(v)^{-1} = x_{\alpha+\beta}(\pm uv);$$

the constant ± 1 obviously depends on the choice of isomorphisms x_γ, and we can suppose that it is equal to 1, possibly by replacing $x_\gamma(u)$ □

by $x_\gamma(\varepsilon_\gamma u)$ ($\gamma \in \Phi$), where $\{\varepsilon_\gamma\}$ is a set of integers ± 1 such that $\varepsilon_\gamma = \varepsilon_{-\gamma}$ for all γ; this last remark will also be applied to types C_2 and G_2. First, since $\mathfrak{o}$ is an infinite additive group, Γ' contains $x_\alpha(s)$ for an element $s \neq 0$ of $\mathfrak{o}$. Then, the commutation relation shows that Γ' contains $x_{\alpha+\beta}(q)$ for $q = s\mathfrak{o}$. The group Γ' contains $x_\gamma(q)$ for all $\gamma \in \Phi$, and thus $E_\mathfrak{q}$.

For type C_2, let α and β be simple roots with $\alpha\beta^* = -1$ and $\beta\alpha^* = -2$; the positive roots are α, β, $\alpha+\beta$, and $2\alpha+\beta$. One can suppose that the commutation relation for U^α and U^β is given by

$$x_\alpha(u)x_\beta(v)x_\alpha(u)^{-1}x_\beta(v)^{-1} = x_{\alpha+\beta}(uv)x_{2\alpha+\beta}(u^2v).$$

Let us take an element $x_\alpha(s)$ of Γ' such that $s(s-1) \neq 0$. Then, according to the commutation relation, Γ' contains, for all $t \in \mathfrak{o}$,

$$g_1 = x_{\alpha+\beta}(st)x_{2\alpha+\beta}(s^2t)$$

as well as its transform g_2 by $w_\alpha(1)$; we have

$$g_2 = x_{\alpha+\beta}(-st)x_\beta(\pm s^2t) = g_1^{-1}g_3, \quad \text{where } g_3 = x_{\alpha+\beta}(s^2t)x_\beta(\pm s^2t).$$

The element

$$g_4 = x_\alpha(1)g_3x_\alpha(1)^{-1}g_3^{-1} = x_{\alpha+\beta}(\pm s^2t)x_{2\alpha+\beta}(\pm s^2t)$$

belongs to Γ'. Consequently, Γ' contains

$$g_4g_1^{-1} = x_{\alpha+\beta}(s(1-s)t).$$

For $\mathfrak{q}' = s(1-s)\mathfrak{o}$, Γ' therefore contains $x_{\alpha+\beta}(\mathfrak{q}')$ and also $x_\alpha(\mathfrak{q}')$. The commutation relation then shows that $x_{2\alpha+\beta}(q)$ is contained in Γ' for $q = \mathfrak{q}'^2$. Consequently, Γ' contains $E_\mathfrak{q}$.

For type G_2, let α and β be simple roots with $\alpha\beta^* = -1$ and $\beta\alpha^* = -3$; the positive roots are α, β, $\alpha+\beta$, $2\alpha+\beta$, $3\alpha+\beta$, and $3\alpha+2\beta$. The commutation relation for U^α and U^β can be written as

$$x_\alpha(u)x_\beta(v)x_\alpha(u)^{-1}x_\beta(v)^{-1} = x_{\alpha+\beta}(uv)x_{2\alpha+\beta}(u^2v)x_{3\alpha+\beta}(u^3v)x_{3\alpha+2\beta}(u^3v^2).$$

We see first, as the long roots $\pm\beta, \pm(3\alpha+\beta)$, and $\pm(3\alpha+2\beta)$ form a root system of type A_2, that Γ' contains $x_\gamma(\mathfrak{q}')$ for every long root γ, where $\mathfrak{q}'$ is a non-zero ideal of $\mathfrak{o}$. Let us take an element $x_\alpha(s)$ of Γ' such that $s \in \mathfrak{q}'$ and that $s(1-s) \neq 0$. By calculating

$$x_\alpha(s)x_\beta(t)x_\alpha(s)^{-1}x_\beta(t)^{-1} \quad \text{and} \quad x_\alpha(1)x_\beta(s^2t)x_\alpha(1)^{-1}x_\beta(s^2t)^{-1},$$

one sees that Γ' contains

$$x_{\alpha+\beta}(st)x_{2\alpha+\beta}(s^2t) \quad \text{and} \quad x_{\alpha+\beta}(s^2t)x_{2\alpha+\beta}(s^2t)$$

for all $t \in \mathfrak{o}$; Γ' therefore contains $x_{\alpha+\beta}(q)$ for $q = s(1-s)\mathfrak{o}$. One concludes from this that Γ' contains $E_\mathfrak{q}$. This completes the proof of Lemma 3.1.

The following lemma will be proven in no. 4:

Lemma 0.6. *3.2. — Suppose that G is of rank ≥ 2 and that the quotient $\mathfrak{o}/\mathfrak{q}$ is semi-local for every non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$. Then:*

(a) $E_\mathfrak{q}$ is a normal subgroup of $G_\mathfrak{o}$ for every ideal $\mathfrak{q}$ of $\mathfrak{o}$;

(b) If $\mathfrak{q}$ and $\mathfrak{q}'$ are non-zero ideals of $\mathfrak{o}$ such that $\mathfrak{q}$ contains $\mathfrak{q}'$, we have $\Gamma_\mathfrak{q} = E_\mathfrak{q}\Gamma_{\mathfrak{q}'}$;

(c) If $\mathfrak{q}$ is an ideal of $\mathfrak{o}$, the commutator $xyx^{-1}y^{-1}$ belongs to $E_\mathfrak{q}$ for all $x \in E_\mathfrak{o}$ and $y \in \Gamma_\mathfrak{q}$.

We are going to apply these lemmas to the problem of congruence groups. Suppose that G is of rank ≥ 2 and that $\mathfrak{o}$ is a Dedekind ring of arithmetic type.

Let $\mathfrak{q}$ be a non-zero ideal of $\mathfrak{o}$. According to Lemma 3.2, (a), $E_\mathfrak{q}$ is a normal subgroup of Γ; we therefore have an exact sequence:

$$1 \to C_\mathfrak{q} \to \Gamma/E_\mathfrak{q} \to \Gamma/\Gamma_\mathfrak{q} \to 1,$$

where $C_\mathfrak{q} = \Gamma_\mathfrak{q}/E_\mathfrak{q}$. Let us equip $\Gamma/E_\mathfrak{q}$ and therefore $C_\mathfrak{q}$ with the topology of subgroups of finite index; by denoting by $\widehat{\Gamma/E_\mathfrak{q}}$ and $\widehat{C}_\mathfrak{q}$ the separated completions of $\Gamma/E_\mathfrak{q}$ and $C_\mathfrak{q}$, we have an exact sequence of profinite groups:

$$1 \to \widehat{C}_\mathfrak{q} \to \widehat{\Gamma/E_\mathfrak{q}} \to \Gamma/\Gamma_\mathfrak{q} \to 1.$$

If $\mathfrak{q}'$ is a non-zero ideal of $\mathfrak{o}$ contained in $\mathfrak{q}$, we have a commutative diagram:

$$\begin{array}{ccccccccc} 1 & \to & C_{\mathfrak{q}'} & \to & \Gamma/E_{\mathfrak{q}'} & \to & \Gamma/\Gamma_{\mathfrak{q}'} & \to & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \to & C_\mathfrak{q} & \to & \Gamma/E_\mathfrak{q} & \to & \Gamma/\Gamma_\mathfrak{q} & \to & 1. \end{array}$$

The morphism from $C_{\mathfrak{q}'}$ into $C_\mathfrak{q}$ is surjective, according to Lemma 3.2, (b). From this we deduce an analogous diagram for the separated completions where the morphism from $\widehat{C}_{\mathfrak{q}'}$ into $\widehat{C}_\mathfrak{q}$ is surjective.

Let us pass to the projective limit over the set of non-zero ideals of $\mathfrak{o}$. The limit of the $\Gamma/\Gamma_\mathfrak{q}$ identifies with the group $\overline{G}_\mathfrak{o}$ and, according to Lemma 3.1, the limit of the $\Gamma/E_\mathfrak{q}$ identifies with $\hat{G}_\mathfrak{o}$. Furthermore, the limit morphism of $\hat{G}_\mathfrak{o}$ onto $\overline{G}_\mathfrak{o}$ is none other than the one denoted by π in no. 1. We therefore have the exact sequence:

$$1 \to \varprojlim \widehat{C}_\mathfrak{q} \to \hat{G}_\mathfrak{o} \xrightarrow{\pi} \overline{G}_\mathfrak{o} \to 1,$$

where $\varprojlim \widehat{C}_\mathfrak{q}$ identifies canonically with $C(G_\mathfrak{o})$.

Let us recall that $\hat{G}_\sigma$ is the completion of G_σ for the topology of subgroups of finite index. Lemma 3.2, (c) asserts that every element of E_σ commutes with the elements of $\varprojlim \hat{C}_q$. Let us return to the exact sequence (1.1):

$$1 \to C^S(G_k) \to \hat{G}_k(S) \xrightarrow{\pi} G_{A_k(S)} \to 1. \tag{3}$$

We recall that $\hat{G}_k(S)$ is the completion of G_k for the topology $T_a(S)$. The centralizer of $C^S(G_k)$ in G_k is a normal subgroup of G_k, and we have just seen that it contains E_σ. Since the quotient of G_k by its center is simple as an abstract group, this implies that every element of G_k commutes with the elements of $C^S(G_k)$. Since G_k is dense in $\hat{G}_k(S)$, we conclude that $C^S(G_k)$ is central in $\hat{G}_k(S)$.

Combining this result with Theorem 1.1, we have the following theorem:

Theorem 0.7. *Let k be the field of fractions of a Dedekind ring $\mathfrak{o}(S)$ of arithmetic type. Let G be a simple simply connected split group over k of rank ≥ 2. The exact sequence (1.1)*

$$1 \to C^S(G_k) \to \hat{G}_k(S) \xrightarrow{\pi} G_{A_k(S)} \to 1,$$

associated with G_k relative to $\mathfrak{o}(S)$, enjoys the following properties:

(a) The kernel $C^S(G_k)$ is central in $\hat{G}_k(S)$ and is a profinite group;

(b) The exact sequence is trivial over the discrete group G_k contained in $G_{A_k(S)}$;

(c) The commutator subgroup of $\hat{G}_k(S)$ is dense in $\hat{G}_k(S)$;

(d) If E is a central extension of $G_{A_k(S)}$ by a profinite abelian group, which is trivial over G_k, then there exists a morphism ϕ from $\hat{G}_k(S)$ into E such that $p \circ \phi = \pi$, where p denotes the projection of E onto $G_{A_k(S)}$.

Note that these properties characterize $\hat{G}_k(S)$ among all extensions of $G_{A_k(S)}$; that is, if an extension of $G_{A_k(S)}$ by a locally compact group B possesses these properties, it is isomorphic to $\hat{G}_k(S)$ as an extension of $G_{A_k(S)}$, B being isomorphic to $C^S(G_k)$. One could immediately deduce from this that, if S' contains S, the group $C^{S'}(G_k)$ is isomorphic to a quotient of $C^S(G_k)$.

4. In this section, we will study the subgroups Γ_q and E_q of G_σ introduced in No. 3 and will give a proof of Lemma 3.2.

We resume the notations of No. 3. Furthermore, we will always assume that $\mathfrak{o}$ is a commutative ring with identity such that $\mathfrak{o}/\mathfrak{q}$ is semi-local for every non-zero ideal $\mathfrak{q}$ of $\mathfrak{o}$.

We shall need the following lemma:

Lemma 0.8. *Let $\mathfrak{q}$ be an ideal of $\mathfrak{o}$ and let a_0, a_1 and a_2 be elements of $\mathfrak{o}$ such that $a_0\mathfrak{o} + \mathfrak{q} = \mathfrak{o}$ and that $\mathfrak{o}/a_0\mathfrak{o}$ is semi-local. Then:*

(a) If $\mathfrak{m}$ is a maximal ideal of $\mathfrak{o}$ containing a_0, there exists an element t of $\mathfrak{q}$ such that $t \notin \mathfrak{m}$ and that $\mathfrak{o}/(a_0 + a_1 t)\mathfrak{o}$ is semi-local;

(b) If $\mathfrak{m}$ is a maximal ideal of $\mathfrak{o}$ containing a_0 and a_1, there exists an element t of $\mathfrak{q}$ such that $t \notin \mathfrak{m}$ and that every maximal ideal of $\mathfrak{o}$ containing a_0 and $(a_1 + a_2 t)$ also contains a_1.

Proof. Let us first prove (a). Since we have $\mathfrak{m} + \mathfrak{q} = \mathfrak{o}$, there exists an element t of $\mathfrak{q}$ such that $t \notin \mathfrak{m}$. If $\mathfrak{q} \neq \mathfrak{o}$, then $a_0' = a_0 + a_1 t$ is different from $\mathfrak{o}$ and, by our hypothesis on $\mathfrak{o}$, $\mathfrak{o}/a_0'\mathfrak{o}$ is semi-local. If $\mathfrak{o}$ is itself semi-local, $\mathfrak{o}/a_0'\mathfrak{o}$ is obviously so. Suppose therefore that $\mathfrak{q} = \mathfrak{o}$ and that there exists an infinity of maximal ideals of $\mathfrak{o}$. Let $\mathfrak{m}'$ then be a maximal ideal of $\mathfrak{o}$ such that $a_0 \notin \mathfrak{m}'$. We have $\mathfrak{m} + \mathfrak{m}' = \mathfrak{o}$, and there exists an element t of $\mathfrak{m}'$ such that $t \notin \mathfrak{m}$; then, $a_0' = a_0 + a_1 t$ is different from $\mathfrak{o}$ and $\mathfrak{o}/a_0'\mathfrak{o}$ is semi-local.

As for statement (b), let us consider the maximal ideals of $\mathfrak{o}$ containing a_0 and not a_1. These ideals $\mathfrak{m}_i (1 \leq i \leq k)$ as well as $\mathfrak{m}$ and $\mathfrak{q}$ form a finite system of pairwise comaximal distinct ideals. There exists therefore an element t of $\mathfrak{q}$ such that $t \notin \mathfrak{m}$ and $t \in \mathfrak{m}_i$ for all i. Then, $a_1 + a_2 t$ is in no $\mathfrak{m}_i$ for any i and, consequently, every maximal ideal of $\mathfrak{o}$ containing a_0 and $a_1 + a_2 t$ necessarily contains a_1. This completes the proof of the lemma. □

We shall consider certain representations of $G_\mathfrak{o}$ in $\mathfrak{o}$-modules. Let ρ be a representation of G in a vector space V and $V_\mathfrak{o}$ an admissible lattice of V with respect to $\mathfrak{g}_\mathbb{Z}$ (see no. 2). The algebra morphism from $\mathbb{Z}[V]$ to $\mathbb{Z}[G] \otimes \mathbb{Z}[V]$ defines an action ρ of $G_\mathfrak{o}$ on $V_\mathfrak{o}$, where $V_\mathfrak{o} = V_\mathbb{Z} \otimes \mathfrak{o} = \mathrm{Hom}(\mathbb{Z}[V], \mathfrak{o})$.

Suppose G has rank ≥ 2 and let us take a fundamental representation ρ of G satisfying conditions (C) and (D) of no. 2. We then have a normalized admissible lattice $V_\mathfrak{o}$ of V, a basis of which is composed of the $v^\lambda (\lambda \in \Lambda(\rho)^+)$ and the $v_\alpha (\alpha \in \Delta(\rho))$. These elements therefore form a basis of $V_\mathfrak{o}$ over $\mathfrak{o}$.

That being said, Lemma 2.3 will allow us to prove

Lemma 0.9. *Let $\mathfrak{q}$ be an ideal of $\mathfrak{o}$ and v_0 be a unimodular element of $V_\mathfrak{o}$ such that $v_0 \equiv uv^\lambda \pmod{\mathfrak{q}V_\mathfrak{o}}$ with a non-zero weight λ and an invertible element u of $\mathfrak{o}$. There then exists an element g of $E_\mathfrak{q}$ such that $\rho(g)v_0 \equiv uv^\lambda \pmod{\sum_\nu \mathfrak{q}V_\mathfrak{o}^\nu}$, where ν runs through the set of weights of ρ different from λ.*

Our proof of this lemma is directly inspired by that which [2] and [3] give in the case of the natural representation of SL_n.

in $\mathfrak{o}^n$ $(n \geq 3)$. We will write gv below instead of $\rho(g)v$ for $g \in G_{\mathfrak{o}}$ and $v \in V_{\mathfrak{o}}$.

We first note that $N_{\mathfrak{o}}$ contains an element n such that $n(uv^{\lambda}) = v^{\mu}$, where μ is the highest weight of ρ. Now $N_{\mathfrak{o}}$ normalizes $E_{\mathfrak{q}}$ and every element of $N_{\mathfrak{o}}$ permutes the submodules $V_{\mathfrak{o}}^{\lambda}$ of $V_{\mathfrak{o}}$ $(\lambda \in \Lambda(\rho))$ among themselves. Consequently, it is sufficient to prove the lemma in the case where $\lambda = \mu$ and $u = 1$.

Suppose then that $v_{\mathfrak{o}} \equiv v^{\mu}$ (mod $\mathfrak{q}V_{\mathfrak{o}}$). We will say that a weight λ is of height $-\sum m_i$ if $\mu - \lambda = \sum m_i \alpha_i$ with respect to the simple roots α_i; μ is the only one of height 0, there is one weight μ_1 and only one of height -1, and, according to condition (D), there is at least one weight of height -2 and zero does not figure among these weights. On the other hand, every element v of $V_{\mathfrak{o}}$ decomposes into λ-components $(\lambda \in \Lambda(\rho))$ and we set

$$v = \sum_{\lambda} a_{\lambda}(v) v^{\lambda} + \sum_{\alpha} b^{\alpha}(v) v_{\alpha}^{0}$$

with respect to the normalized basis of $V_{\mathfrak{o}}$. We will establish in this step of the proof that there exists an element g of $E_{\mathfrak{q}}$ tel que

$$a_{\mu}(gv_{\mathfrak{o}})\mathfrak{o} + a_{\mu_1}(gv_{\mathfrak{o}})\mathfrak{o} = \mathfrak{o}.$$

Let us first note that we can assume $a_{\mu}(v_{\mathfrak{o}}) \neq 0$; it is evident if $\mathfrak{q} \neq 0$. In the case where $\mathfrak{q} = 0$, if we have $a_{\lambda}(v_{\mathfrak{o}}) \neq 0$ for some $\lambda \in \Lambda(\rho)^*$, there exists an element g of $N_{\mathfrak{o}}$ such that $a_{\mu}(gv_{\mathfrak{o}}) = a_{\lambda}(v_{\mathfrak{o}})$; if $a_{\lambda}(v_{\mathfrak{o}}) = 0$ for all $\lambda \in \Lambda(\rho)^*$, we find an element $g = x_{\alpha}(\pm 1)$, $\alpha \in \Delta(\rho)$, such that $a_{\alpha}(gv_{\mathfrak{o}}) = b^{\alpha}(v_{\mathfrak{o}}) \neq 0$, which brings us back to the previous case. Suppose then that $\mathfrak{o}/a_{\mu}(v_{\mathfrak{o}})\mathfrak{o}$ is semi-local. To prove our assertion above, it now suffices to verify the following statement, denoting by $\mathfrak{I}(v)$, $v \in V_{\mathfrak{o}}$, the set of maximal ideals of $\mathfrak{o}$ containing $a_{\mu}(v)$ and $a_{\mu_1}(v)$:

Let $\mathfrak{m}$ be an element of $\mathfrak{I}(v_{\mathfrak{o}})$ and λ be a weight of ρ having the highest height among the weights ν such that the ν-component of $v_{\mathfrak{o}}$ is not zero mod $\mathfrak{m}V_{\mathfrak{o}}$. Then there exists an element g of $E_{\mathfrak{q}}$ such that $\mathfrak{o}/a_{\mu}(gv_{\mathfrak{o}})\mathfrak{o}$ is semi-local, that $\mathfrak{I}(v_{\mathfrak{o}})$ contains $\mathfrak{I}(gv_{\mathfrak{o}})$ and that $gv_{\mathfrak{o}}$ has a λ'-component not zero mod $\mathfrak{m}V_{\mathfrak{o}}$ for a weight λ' of height strictly higher than that of λ.

Indeed, this allows us to eliminate one after the other the elements of $\mathfrak{I}(v_{\mathfrak{o}})$ by replacing $v_{\mathfrak{o}}$ with $gv_{\mathfrak{o}}$ with an element g of $E_{\mathfrak{q}}$. Let us prove the statement above; we treat two cases separately, depending on whether λ is equal to 0 or not.

In the case where $\lambda \neq 0$, let α be a simple root such that $\lambda + \alpha$ is a weight of ρ. If $\alpha = \mu_1 - \mu_2$ for a weight μ_2 of height -2, we take an element t of $\mathfrak{q}$ which satisfies Lemma 4.1, (b) for

$$(a_0, a_1, a_2) = (a_{\mu}(v_{\mathfrak{o}}), a_{\mu_1}(v_{\mathfrak{o}}), a_{\mu_2}(v_{\mathfrak{o}}));$$

otherwise, one takes an element t of $\mathfrak{q}$ which satisfies Lemma 4.1, (a) for

$$(a_0, a_1) = (a_\mu(v_0), a_{\mu_1}(v_0)).$$

By setting $g = x_\alpha(\pm t)$, one then verifies, according to Lemmas 2.3 and 4.1, that $\mathfrak{o}/a_\mu(gv_0)\mathfrak{o}$ is semi-local, that $\mathfrak{J}(gv_0)$ is contained in $\mathfrak{J}(v_0)$ and that the λ'-component of gv_0 is not zero modulo $\mathfrak{m}V_\mathfrak{o}$ for $\lambda' = \lambda + \alpha$ or $\lambda + 2\alpha$, according to whether $\lambda + \alpha$ is equal to 0 or not.

In the case where $\lambda = 0$, let α be a simple root such that $b^\alpha(v_0)$ is not in $\mathfrak{m}$. One first finds, according to Lemma 2.3, an element $g' = x_{-\alpha}(s)$, $s \in \mathfrak{q}$, such that $a_{-\alpha}(g'v_0)$ is in $\mathfrak{m}$. One then chooses an element t of $\mathfrak{q}$ for α and $g'v_0$, as was done for α and v_0 in the previous case. By setting $g = x_\alpha(\pm t)g'$, one sees by Lemmas 2.3 and 4.1 that $\mathfrak{o}/a_\mu(gv_0)\mathfrak{o}$ is semi-local, that $\mathfrak{J}(gv_0)$ is contained in $\mathfrak{J}(v_0)$ and that the α-component of gv_0 is not zero modulo $\mathfrak{m}V_\mathfrak{o}$. This completes the proof of our statement and establishes that there exists an element g of $E_\mathfrak{q}$ such that

$$a_\mu(gv_0)\mathfrak{o} + a_{\mu_1}(gv_0)\mathfrak{o} = \mathfrak{o}.$$

Suppose then that $v_0 \equiv v^\mu \pmod{\mathfrak{q}V_\mathfrak{o}}$ and that $a_0 = a_\mu(v_0)$ and $a_1 = a_{\mu_1}(v_0)$ are relatively prime. Let μ_2 be a weight of ρ of height -2; one has $\mu_1 = \mu - \alpha_1$ and $\mu_2 = \mu_1 - \alpha_2 \neq 0$, where α_1 and α_2 are two simple roots and $\beta = \alpha_1 + \alpha_2$ a root. Let s and s' be two elements of $\mathfrak{o}$ such that $sa_0 + s'a_1 = 1$. Then, setting

$$g_1 = x_{-\alpha_2}(\pm s't)x_{-\beta}(\pm st), \quad \text{where } t = (a_0 - 1) - a_{\mu_2}(v_0),$$

one has, for $v_1 = g_1v_0$, $a_\mu(v_1) = a_0$ and $a_{\mu_1}(v_1) = a_0 - 1$, by virtue of Lemma 2.3. We then have

$$a_\mu(v_2) = 1 \quad \text{and} \quad a_{\mu_2}(v_2) = a_0 - 1 \quad \text{for } v_2 = g_2v_1,$$

with $g_2 = x_\beta(\pm 1)$; then

$$a_\mu(v_3) = 1 \quad \text{and} \quad a_{\mu_2}(v_3) = 0 \quad \text{for } v_3 = g_3v_2,$$

with $g_3 = x_{-\beta}(\pm(a_0 - 1))$. Consequently, one still has $a_\mu(v_4) = 1$ for $v_4 = g_2^{-1}v_3$. One thus has $a_\mu(gv_0) = 1$ for $g = g_2^{-1}g_3g_2g_1$, and g belongs to $E_\mathfrak{q}$ since g_1 and g_3 are in $E_\mathfrak{q}$.

This finally completes the proof of Lemma 4.2.

We will now proceed to a reduction theorem which will have Lemma 3.2 as a corollary. Since ρ is a fundamental representation, there exists a unique simple root α_1 such that $\mu - \alpha_1$ is a weight of ρ. By setting $\Delta'' = \{\alpha_1\}$ and $\Delta' = \Delta - \Delta''$, we define, as in no. 2, subgroups of G: $G' = G(\Delta')$, $H'' = H(\Delta'')$, $U^+(\Delta')$ and $U^-(\Delta')$. We recall that G' is a simply connected semi-simple group, H'' a torus,

$U^+(\Delta')$ and $U^-(\Delta')$ of unipotent groups. One has, in $G_\mathfrak{o}$, the corresponding subgroups: $G'_\mathfrak{o}, H''_\mathfrak{o}, U^+_\mathfrak{o}(\Delta')$ and $U^-_\mathfrak{o}(\Delta')$.

Theorem 0.10. *We maintain the hypotheses and notations above: in particular, G is of rank ≥ 2. One has $\Gamma_\mathfrak{q} = E_\mathfrak{q}(\Gamma_\mathfrak{q} \cap G'_\mathfrak{o})$ for every ideal $\mathfrak{q}$ of $\mathfrak{o}$.*

Proof. In order to prove this theorem, let us first recall the decomposition theorem stated in section 2. The algebra $\mathbf{Z}[G]$ possesses an element t_μ such that the algebra morphism, dual to the multiplication in G, from $\mathbf{Z}[G](t_\mu^{-1})$ into $\mathbf{Z}[U^-(\Delta')] \otimes \mathbf{Z}[U^+(\Delta')] \otimes \mathbf{Z}[H''] \otimes \mathbf{Z}[G']$, is an isomorphism between these two algebras. The image of t_μ is $1 \otimes 1 \otimes \mu'' \otimes 1$, where μ'' is the restriction to H'' of t_μ or of the weight μ. Consequently, the map π from $U^-_\mathfrak{o}(\Delta') \times U^+_\mathfrak{o}(\Delta') \times H''_\mathfrak{o} \times G'_\mathfrak{o}$ into $G_\mathfrak{o}$, defined by

$$\pi(u^-, u^+, h'', g') = u^-u^+h''g',$$

is injective and has as image the subset of $G_\mathfrak{o}$ formed by elements g such that $t_\mu(g)$ is invertible in $\mathfrak{o}$. One also knows that $\mathbf{Z}[H'']$ is generated over $\mathbf{Z}$ by μ'' and $(\mu'')^{-1}$.

To prove our theorem, let g be an element of $\Gamma_\mathfrak{q}$. Let us consider the representation ρ of $G_\mathfrak{o}$ in $V_\mathfrak{o}$. One easily sees that

$$\rho(g)v^\mu \equiv v^\mu \pmod{\mathfrak{q}V_\mathfrak{o}}.$$

Consequently, according to Lemma 4.2, there exists an element g_1 of $E_\mathfrak{q}$ such that

$$\rho(g_1g)v^\mu \equiv v^\mu \pmod{\sum_{\nu \neq \mu} \mathfrak{q}V_\mathfrak{o}^\nu},$$

which signifies that $t_\mu(g_1g) = 1$. By the decomposition theorem, g_1g can thus be written in the form $u^-u^+h''g'$, where $u^- \in U^-_\mathfrak{o}(\Delta')$, $u^+ \in U^+_\mathfrak{o}(\Delta')$, $h'' \in H''_\mathfrak{o}$ and $g' \in G'_\mathfrak{o}$. One has $\mu(h'') = t_\mu(g_1g) = 1$;

hence $h'' = e$. One also sees that u^-, u^+ and g' are in $\Gamma_{\mathfrak{q}}$, by virtue of the uniqueness of the analogous decomposition in $G_{\mathfrak{o}/\mathfrak{q}}$. Now, one easily verifies, according to the remark in section 2 on the algebras $\mathbf{Z}[U^-(\Delta')]$ and $\mathbf{Z}[U^+(\Delta')]$, that $U^-_{\mathfrak{o}}(\Delta') \cap \Gamma_{\mathfrak{q}}$ and $U^+_{\mathfrak{o}}(\Delta') \cap \Gamma_{\mathfrak{q}}$ are contained in $E_{\mathfrak{q}}$. Consequently, g belongs to $E_{\mathfrak{q}}(\Gamma_{\mathfrak{q}} \cap G'_{\mathfrak{o}})$. The theorem is thus proven. □

The following corollary contains Lemma 3.2:

Corollary 0.11. *(a) For every $x \in E_{\mathfrak{o}}$ and every $y \in \Gamma_{\mathfrak{q}}$, the commutator $xyx^{-1}y^{-1}$ belongs to $E_{\mathfrak{q}}$.*

(b) If $\mathfrak{o}$ is a semi-local ring, $\Gamma_{\mathfrak{q}}$ is equal to $E_{\mathfrak{q}}$ even for G of rank 1.

(c) $E_{\mathfrak{q}}$ is a normal subgroup of $G_{\mathfrak{o}}$.

(d) If $\mathfrak{q}'$ is a non-zero ideal of $\mathfrak{o}$ contained in $\mathfrak{q}$, $\Gamma_{\mathfrak{q}}$ is equal to $E_{\mathfrak{q}}\Gamma_{\mathfrak{q}'}$.

Let us first recall that $G'_{\mathfrak{o}}$ is the group of points with values in $\mathfrak{o}$ of the group scheme over $\mathfrak{o}$ deduced from the group scheme over $\mathbf{Z}$ associated to the simply connected semi-simple group G' and that the root subgroups of $G'_{\mathfrak{o}}$ are also root subgroups of $G_{\mathfrak{o}}$; this will sometimes allow us to reason by induction on the rank of G. Let us prove statement (a). Let x be an element of $U^+_\alpha(\Delta')$; if y is an element of $\Gamma_{\mathfrak{q}} \cap G'_{\mathfrak{o}}$, the commutator $xyx^{-1}y^{-1}$ is in $U^+_\alpha(\Delta') \cap \Gamma_{\mathfrak{q}}$, thus in $E_{\mathfrak{q}}$; taking into account Theorem 4.3, we deduce that $xyx^{-1}y^{-1}$ is in $E_{\mathfrak{q}}$ for all $y \in \Gamma_{\mathfrak{q}}$. Consequently, as each root subgroup $U^\pm_\delta$ of $G_{\mathfrak{o}}$ has in $U^+_\alpha(\Delta')$ a conjugate by an element of $\mathfrak{N}_{\mathfrak{o}}$, statement (a) is verified for every element x of $U^\pm_\delta$. Since $E_{\mathfrak{o}}$ is generated by the root subgroups, this immediately entails our assertion for every element x of $E_{\mathfrak{o}}$. As for statement (b), one sees easily that it is true for the group of rank 1, i.e., for SL_2. Consequently, we conclude from Theorem 4.3, by induction on the rank of G, that it is true for G. To prove statement (c), let us suppose that $\mathfrak{q}$ is a non-zero ideal (if $\mathfrak{q} = \{0\}$, it is trivial). Then the group $G_{\mathfrak{o}/\mathfrak{q}}$ is generated by root subgroups; whence $G_{\mathfrak{o}} = \Gamma_{\mathfrak{q}} E_{\mathfrak{o}}$. Consequently, $E_{\mathfrak{q}}$ is a normal subgroup of $G_{\mathfrak{o}}$, because, according to (a), $\Gamma_{\mathfrak{q}}$ normalizes $E_{\mathfrak{q}}$. Statement (d) is an immediate consequence of (b).

Corollary 0.12. *4.5. — Let α be a long root in Δ and G^α the simple subgroup of G generated by U^α and $U^{-\alpha}$. If $G^\alpha_{\mathfrak{o}}$ denotes the subgroup of $G_{\mathfrak{o}}$ corresponding to G^α, we have $\Gamma_{\mathfrak{q}} = E_{\mathfrak{q}}(\Gamma_{\mathfrak{q}} \cap G^\alpha_{\mathfrak{o}})$ for every ideal $\mathfrak{q}$ of $\mathfrak{o}$.*

Proof. Indeed, according to Theorem 4.3, we have $\Gamma_{\mathfrak{q}} = E_{\mathfrak{q}}(\Gamma_{\mathfrak{q}} \cap G'_{\mathfrak{o}})$ for a semi-simple subgroup G' of rank $r-1$, where r is the rank of G. Reasoning by induction on the rank of G, we thus have $\Gamma_{\mathfrak{q}} = E_{\mathfrak{q}} \prod_\beta (\Gamma_{\mathfrak{q}} \cap G^\beta_{\mathfrak{o}})$, the elements β of Δ being suitably ordered. Now Δ possesses one or two elements β and γ such that every element of Δ can be transformed into β or γ by an element of the Weyl group W and such that β and γ are not orthogonal. Let $G^{\beta\gamma}$ then be the subgroup of G generated by G^β and G^γ. For each δ of Δ, there exists an element n of $\mathfrak{N}_{\mathfrak{o}}$ such that $nG^\delta_{\mathfrak{o}}n^{-1} \subset G^{\beta\gamma}_{\mathfrak{o}}$; we thus have $\Gamma_{\mathfrak{q}} = E_{\mathfrak{q}}(\Gamma_{\mathfrak{q}} \cap G^{\beta\gamma}_{\mathfrak{o}})$ according to Corollary 4.4. By applying Theorem 4.3 to the group $G^{\beta\gamma}$, we deduce from there that $\Gamma_{\mathfrak{q}} = E_{\mathfrak{q}}(\Gamma_{\mathfrak{q}} \cap G^\delta_{\mathfrak{o}})$, δ being one of the roots β and γ; according to the proof of Lemma 2.2, δ must be a long root in the cases of rank 2. This immediately entails our corollary. □

The following remarks on groups of various types will also be observed.

a. If G is simple of rank ≥ 2, one can choose the representation ρ verifying conditions (C) and (D) such that G' is, itself also, simple.

b. If G is of non-symplectic type of rank ≥ 3, Δ contains two non-orthogonal long roots α, β; according to Corollary 4.5, we have $\Gamma_q = E_q(\Gamma_q \cap G^{\alpha\beta})$, where $G^{\alpha\beta}$ is a simple group of type A_2. On the other hand, if G is of symplectic type, two non-opposite long roots are always orthogonal. For the group of type G_2, the root subgroups U^α corresponding to the long roots generate a simply connected simple group $G^{\alpha\beta}$ of type A_2; we therefore have $\Gamma_q = E_q(\Gamma_q \cap G^{\alpha\beta})$.

c. If G is of rank ≥ 2, the commutator $xyx^{-1}y^{-1}$ belongs to E_q for all $x \in G_\sigma$ and $y \in \Gamma_q$; in other words, Γ_q/E_q is central in G_σ/E. Given Corollary 4.4, this is equivalent to saying that Γ_q/E_q is abelian. For groups of rank ≥ 3 and the group of type C_2, this follows easily from Corollaries 4.4 and 4.5, because the root system Φ possesses, in these cases, two long roots orthogonal to each other. For groups of types A_2 and G_2, our assertion follows, in view of the previous remark, from a theorem of Mennicke, which implies in particular that Γ_q/E_q is abelian for $G = SL_3$ (see [4], theor. 5.4, lemma 2.9).

The question of evaluating the groups Γ_q/E_q is thus reduced, to a certain extent, to the cases of the groups SL_2, SL_3 and Sp_4.

Corollary 0.13. *4.6.* — *The group Γ_q/E_q for G of rank ≥ 2 is isomorphic to a quotient of the corresponding group for either SL_3 or Sp_4. In particular, we have $G_\sigma = E_\sigma$ for G of rank ≥ 2 if this is true, for either SL_2, or for SL_3 and Sp_4.*

CHAPTER II.
Central Extensions of a Split Simple Group.

5. We shall study in this chapter, according to the method of R. Steinberg [30] and C. Moore [26], the central extensions of the group G_k, where G is a split simply connected simple group over a field k.

We resume the notation of No. 3; k is a field and G_k is the group of rational points of the split simply connected simple group over k defined from a scheme $\mathfrak{G}$ in groups split over $\mathbf{Z}$.

It is known ([13], [16]) that the group G_k is generated by the root subgroups U_α^k ($\alpha \in \Phi$) and that these generators satisfy the following relations (R 1), (R 2) and (R 3), x_α denoting the isomorphism chosen from k to U_α^k:

(R1) $x_\alpha(u)\, x_\alpha(v) = x_\alpha(u+v)$ for all $\alpha \in \Phi$ and $u, v \in k$;

(R2) (i) $x_\alpha(u)\, x_\beta(v)\, x_\alpha(u)^{-1}\, x_\beta(v)^{-1} = \prod_{i,j\in\mathbf{N}^*} x_{i\alpha+j\beta}\big(N_{\alpha,\beta;i,j}\, u^i v^j\big)$ for all $\alpha, \beta \in \Phi$[2] and $u, v \in k$.

$\beta \in \Phi$ such that $\alpha + \beta \neq 0$ and for all $u, v \in k$; where the $N_{\alpha,\beta;i,j}$ are certain rational integers;

(ii) $w_\alpha(u) x_\alpha(v) w_\alpha(u)^{-1} = x_{-\alpha}(-u^{-2}v)$ for all $\alpha \in \Phi, u \in k^\times$ and $v \in k$, with $w_\alpha(u) = x_\alpha(u) x_{-\alpha}(-u^{-1}) x_\alpha(u)$;

(R 3) $h_\alpha(u) h_\alpha(v) = h_\alpha(uv)$ for all $\alpha \in \Phi$ and $u, v \in k^\times$, with $h_\alpha(u) = w_\alpha(u) w_\alpha(1)^{-1}$.

R. Steinberg [30] has demonstrated the following results:

a. (R 1), (R 2) and (R 3) form an exhaustive family of relations for the generators $x_\alpha(u)$ of the abstract group G_k.

b. Let $E(G_k)$ be the group generated by the same generators subject only to the relations (R 1) and (R 2). Then, the kernel $\pi_1(G_k)$ of the natural morphism p from $E(G_k)$ onto G_k is central in $E(G_k)$. If k is a finite field, $\pi_1(G_k)$ reduces to $\{e\}$.

c. $E(G_k)$ is equal to its commutator group except if $k = \mathbf{F}_2, G = \mathrm{SL}_2, \mathrm{Sp}_4, \mathrm{G}_2$ or if $k = \mathbf{F}_3, G = \mathrm{SL}_2$.

d. If k possesses at least 11 elements, every central extension of $E(G_k)$ is trivial.

These results affirm, in homological language, that if A is a trivial G_k-module, the cohomology group $H^2(G_k, A)$ is isomorphic to the group $\mathrm{Hom}(\pi_1(G_k), A)$ of morphisms from $\pi_1(G_k)$ into A, when k has enough elements.

In this section and the following ones, we shall assume unless otherwise mentioned that k is an infinite field, and we will write simply X instead of X_k for a subgroup X of G.

Let us begin by examining the result of Steinberg more closely.

Let A be an abelian group and let E be a central extension of G_k by A:

$$1 \to A \to E \xrightarrow{\pi} G_k \to 1.$$

According to [30], there exist lifts $r(U^\alpha)$ in E of the root groups U^α such that the elements $\tilde{x}_\alpha(u) = r(x_\alpha(u))$ satisfy the relations (R 1) and (R 2). We then set, for all $\alpha \in \Phi$ and $t \in k^\times$,

$$\tilde{w}_\alpha(t) = \tilde{x}_\alpha(t)\tilde{x}_{-\alpha}(-t^{-1})\tilde{x}_\alpha(t) \quad \text{and} \quad \tilde{h}_\alpha(t) = \tilde{w}_\alpha(t)\tilde{w}_\alpha(1)^{-1}.$$

We first have the following lemma ([30], [16]):

Lemma 5.1. — *(a) For all $\alpha \in \Phi$, x_α defines an isomorphism of k onto $\tilde{U}^\alpha$. Let $\tilde{U}^+$ [resp. $\tilde{U}^-$] be the subgroup of E generated by the $\tilde{U}^\beta$ [resp. the $\tilde{U}^{-\beta}$], $\beta \in \Phi^+$. Then π induces an isomorphism of $\tilde{U}^+$ [resp. of $\tilde{U}^-$] onto U^+ [resp. U^-].*

[2] [[UNCERTAIN: $\alpha + \beta \neq 0$ — CONFIDENCE: 70% — REASON: The line is cut off at the bottom of the page. Based on the standard Chevalley relations, the missing condition is likely $\alpha + \beta \neq 0$.]]

(b) Let α and β be elements of Φ. We have

$$\tilde{w}_\alpha(t)\tilde{x}_\beta(u)\tilde{w}_\alpha(t)^{-1} = \tilde{x}_{\sigma_\alpha\beta}(\eta t^{-c}u)$$

for all $t \in k^\times$ and $u \in k$; where $\sigma_\alpha \in W$ is the reflection associated with α, c is the Cartan integer $\beta\alpha^*$ relative to β and α, and η is a constant $\eta(\alpha,\beta)$ equal to ± 1.

(c) We have

$$\begin{gathered}\eta(\alpha,\beta) = \eta(\alpha,-\beta); \quad \eta(\alpha,\alpha) = \eta(\alpha,-\alpha) = -1;\\ \eta(\alpha,\beta) = 1 \quad \text{if } \alpha \pm \beta \neq 0, \alpha \pm \beta \notin \Phi;\\ \eta(\alpha,\beta)\eta(\beta,\alpha) = -1 \quad \text{if } \alpha\beta^* = \beta\alpha^* = -1;\\ \eta(\alpha,\beta) = -1 \quad \text{if } \alpha\beta^* = 0, \alpha \pm \beta \in \Phi.\end{gathered}$$

This lemma allows for the deduction of various relations between the elements $\tilde{x}_\alpha(u)$. We collect some of them here [30]:

Lemma 0.14. *5.2. — We have the following relations:*

(a) $\tilde{w}_\alpha(t)\tilde{x}_\beta(u)\tilde{w}_\alpha(t)^{-1} = \tilde{x}_{\sigma_\alpha\beta}(\eta t^{-c}u)$;

(b) $\tilde{w}_\alpha(t)\tilde{h}_\beta(u)\tilde{w}_\alpha(t)^{-1} = \tilde{h}_{\sigma_\alpha\beta}(ut^{-c})\tilde{h}_{\sigma_\alpha\beta}(t^{-c})^{-1}$;

(c) $\tilde{h}_\alpha(t)\tilde{x}_\beta(u)\tilde{h}_\alpha(t)^{-1} = \tilde{x}_\beta(t^c u)$,
$\tilde{h}_\alpha(t)\tilde{x}_\alpha(u)\tilde{h}_\alpha(t)^{-1} = \tilde{x}_\alpha(t^2 u)$;

(d) $\tilde{h}_\alpha(t)\tilde{w}_\beta(u)\tilde{h}_\alpha(t)^{-1} = \tilde{w}_\beta(t^c u)$;

(e) $\tilde{h}_\alpha(t)\tilde{h}_\beta(u)\tilde{h}_\alpha(t)^{-1} = \tilde{h}_\beta(t^c u)\tilde{h}_\beta(t^c)^{-1}, \quad c = \beta\alpha^*$;

(f) $\tilde{w}_\alpha(1)\tilde{h}_\beta(t)\tilde{w}_\alpha(1)^{-1} = \tilde{h}_\beta(t)\tilde{h}_\alpha(t^{-d}), \quad d = \alpha\beta^*$;

(g) $\tilde{w}_\alpha(t) = \tilde{w}_{-\alpha}(-t^{-1}), \quad \tilde{h}_\alpha(t) = \tilde{h}_{-\alpha}(t)^{-1}$,
$\tilde{w}_\alpha(1)\tilde{h}_\alpha(t)\tilde{w}_\alpha(1)^{-1} = \tilde{h}_\alpha(t^{-1})$;

(h) $\tilde{w}_\alpha(1)^{-1}\tilde{x}_\alpha(t)\tilde{w}_\alpha(1) = \tilde{x}_{-\alpha}(-t) = \tilde{x}_\alpha(-t^{-1})\tilde{w}_\alpha(t^{-1})\tilde{x}_\alpha(-t^{-1}), \quad t \neq 0$.

Set $\tilde{H} = \pi^{-1}(H)$, $\tilde{N} = \pi^{-1}(N)$, and recall the Bruhat decomposition of G_k: $G_k = U^+NU^- = U^+NU^+$. Lifting it to E, we have $E = \tilde{U}^+\tilde{N}\tilde{U}^- = \tilde{U}^+\tilde{N}\tilde{U}^+$. The following lemma specifies this decomposition of E:

Lemma 0.15. *5.3. — (a) For every element $\tilde{n}$ of $\tilde{N}$, $\tilde{U}^+\tilde{n}\tilde{U}^+ \cap \tilde{N}$ reduces to $\{\tilde{n}\}$.*

(b) We define maps ν and $\tilde{\nu}$ as follows:

$$\begin{gathered}\nu : G_k \to N, \quad \nu(unu') = n \quad \textit{for } u, u' \in U^+ \textit{ and } n \in N;\\ \tilde{\nu} : E \to \tilde{N}, \quad \tilde{\nu}(\tilde{u}\tilde{n}\tilde{u}') = \tilde{n} \quad \textit{for } \tilde{u}, \tilde{u}' \in \tilde{U}^+ \textit{ and } \tilde{n} \in \tilde{N}.\end{gathered}$$

Then we have $\nu \circ \pi = \pi \circ \tilde{\nu}$.

(c) $\tilde{\nu}(\tilde{u}\tilde{g}) = \tilde{\nu}(\tilde{g})$ for all $\tilde{u} \in \tilde{U}^+$ and $\tilde{g} \in E$;
$\tilde{\nu}(\tilde{h}\tilde{g}) = \tilde{h}\tilde{\nu}(\tilde{g})$ for all $\tilde{h} \in \tilde{H}$ and $\tilde{g} \in E$.

(d) Let α be a simple root. Then, for an element $\tilde{g}$ of E, $\tilde{\nu}(\tilde{w}_\alpha(-1)\tilde{g})$ is equal either to $\tilde{w}_\alpha(-1)\tilde{\nu}(\tilde{g})$ or to $\tilde{h}_\alpha(t^{-1})\tilde{\nu}(\tilde{g})$ with an element t of $k^\times$.

Proof. Let us first prove statement (a), which justifies the definition of $\tilde{\nu}$. The analogous statement for G_k is well known, and if n is an element of N, $U_n^+ = U^+ \cap nU^+n^{-1}$ is generated by root groups U^β. Suppose that $\tilde{n}' = \tilde{u}\tilde{n}\tilde{u}'$, with $\tilde{n}, \tilde{n}'$ in $\tilde{N}$ and $\tilde{u}, \tilde{u}'$ in $\tilde{U}^+$. Descending to G_k, we first see that $\tilde{n}' = \tilde{n}a$ with an element a of A. We thus have $a = (\tilde{n}^{-1}\tilde{u}\tilde{n})\tilde{u}'$. This shows that $\pi(\tilde{u}) \in U_n^+$ for $n = \pi(\tilde{n})$, and thus that $\tilde{u}$ belongs to the lifting $\tilde{U}_n^+$ of U_n^+ into $\tilde{U}^+$. Now, since $\tilde{U}_n^+$ is generated by root groups $\tilde{U}^\beta$, we have $\tilde{n}^{-1}\tilde{U}_n^+\tilde{n} \subset \tilde{U}^+$, by virtue of lemma 5.1, (b), which implies that every element of $\tilde{N}$ permutes the root groups $\tilde{U}^\gamma$ among themselves. It follows that $a \in \tilde{U}^+$; whence $a = e$. This proves statement (a). Assertions (b) and (c) are now obvious. Let us examine (d). Let us first note

that $\tilde{U}_\alpha^+ = \tilde{U}^+ \cap \tilde{w}_\alpha(1)\tilde{U}^+\tilde{w}_\alpha(1)^{-1}$ is the subgroup of $\tilde{U}^+$ generated by the root groups $\tilde{U}^\beta$ for $\beta \in \Phi^+$, $\beta \neq \alpha$, and that we have

$$\tilde{U}^+ = \tilde{U}_\alpha^+\tilde{U}_\alpha, \quad \tilde{U}_\alpha^+ \cap \tilde{U}_\alpha = \{e\}.$$

Let us set $\tilde{g} = \tilde{u}\tilde{n}\tilde{u}'$, with $\tilde{n} = \nu(\tilde{g})$ and $\tilde{u}, \tilde{u}'$ in $\tilde{U}^+$, and then $\tilde{u} = \tilde{u}_1\tilde{x}_\alpha(t)$, with $\tilde{u}_1 \in \tilde{U}_\alpha^+$ and $t \in k$. We have $\tilde{n}^{-1}\tilde{U}_\alpha^+\tilde{n} = \tilde{U}_\alpha^{+\sigma^{-1}}$, with $\sigma = \omega(\pi(\tilde{n}))$, where ω is the canonical morphism from N to W. We now distinguish two cases:

(i) If $\tilde{n}^{-1}\tilde{x}_\alpha(t)\tilde{n}$ is in $\tilde{U}^+$, in other words if $\sigma^{-1}\alpha > 0$ or if $t = 0$, then we have

$$\tilde{w}_\alpha(-1)\tilde{g} = \tilde{w}_\alpha(-1)\tilde{u}_1\tilde{w}_\alpha(1) \cdot \tilde{w}_\alpha(-1)\tilde{n} \cdot \tilde{n}^{-1}\tilde{x}_\alpha(t)\tilde{n} \cdot \tilde{u}' = \tilde{u}_2\tilde{w}_\alpha(-1)\tilde{n}\tilde{u}_2',$$

with $\tilde{u}_2 \in \tilde{U}_\alpha^+$ and $\tilde{u}_2' \in \tilde{U}^+$. Whence

$$\tilde{\nu}(\tilde{w}_\alpha(-1)\tilde{g}) = \tilde{w}_\alpha(-1)\tilde{\nu}(\tilde{g}).$$

(ii) If $\sigma^{-1}\alpha < 0$ and $t \neq 0$, then we have

$$\begin{aligned}\tilde{w}_\alpha(-1)\tilde{g} &= \tilde{w}_\alpha(-1)\tilde{u}_1\tilde{w}_\alpha(1) \cdot \tilde{w}_\alpha(-1)\tilde{x}_\alpha(t)\tilde{w}_\alpha(1) \cdot \tilde{w}_\alpha(-1)\tilde{n}\tilde{u}' \\ &= \tilde{u}_2\tilde{x}_\alpha(-t^{-1})\tilde{w}_\alpha(t^{-1})\tilde{x}_\alpha(-t^{-1}) \cdot \tilde{w}_\alpha(-1)\tilde{n}\tilde{u}', \quad \text{with } \tilde{u}_2 \in \tilde{U}_\alpha^+ \\ &= \tilde{u}_2\tilde{x}_\alpha(-t^{-1})\tilde{w}_\alpha(t^{-1})\tilde{w}_\alpha(-1)\tilde{n}\tilde{u}_2', \quad \text{with } \tilde{u}_2' \in \tilde{U}^+.\end{aligned}$$

Whence

$$\tilde{\nu}(\tilde{w}_\alpha(-1)\tilde{g}) = \tilde{w}_\alpha(t^{-1})\tilde{w}_\alpha(-1)\tilde{n} = \tilde{h}_\alpha(t^{-1})\tilde{n}.$$

Assertion (d) is therefore proved. □

We will henceforth be interested in certain cocycles on $k^\times$ with values in A.

For each $\alpha \in \Phi$, one defines a map c_α from $k^\times \times k^\times$ into A by

$$c_\alpha(s,t) = \tilde{h}_\alpha(s)\tilde{h}_\alpha(t)\tilde{h}_\alpha(st)^{-1};$$

it is a normalized cocycle on $k^\times$ with values in A. The cocycle c_α will be called the *Steinberg cocycle* in α of E relative to the isomorphisms x_α. Recall that we have chosen once and for all the isomorphisms x_α of the additive group onto the root subgroups U^α of G_k; we will see at the end of this section that in general the map $\tilde{h}_\alpha$ and the cocycle c_α depend on the choice of x_α and $x_{-\alpha}$, while the monomorphism h_α of the multiplicative group into H does not depend on it. Nevertheless, when one has chosen the isomorphisms x_α, the cocycle c_α is uniquely determined by E or even by the class of extensions of E, because the lifted isomorphisms $\tilde{x}_\alpha$ are, according to Lemma 5.2, (c), uniquely determined by E.

We are going to study the properties of the cocycles c_α.

Lemma 0.16. *(a) $c_\alpha(s,t) = c_{-\alpha}(t,s)^{-1}$ for all $\alpha \in \Phi$.*

(b) If there exists an element σ of the Weyl group W such that $\beta = \sigma\alpha$, then c_β is identical to c_α or to $c_{-\alpha}$.

(c) For all $\alpha, \beta \in \Phi$, we have

$$\tilde{h}_\alpha(s)\tilde{h}_\beta(t)\tilde{h}_\alpha(s)^{-1}\tilde{h}_\beta(t)^{-1} = c_\alpha(s,t^d) = c_\beta(t,s^c)^{-1},$$

where $c = \beta\alpha^*$ and $d = \alpha\beta^*$.

(d) The cocycles c_α are bilinear maps from $k^\times \times k^\times$ into A except possibly for the long roots α of a symplectic group G.

Proof. Statement (a) is an immediate consequence of Lemma 5.2, (g). Assertion (b) results from Lemma 5.2, (b), for the latter asserts that, if $\beta = \sigma_\alpha\gamma$, one has either

$$\tilde{w}_\alpha(1)\tilde{h}_\gamma(u)\tilde{w}_\alpha(1)^{-1} = \tilde{h}_\beta(u)$$

or

$$\tilde{w}_\beta(1)\tilde{w}_\alpha(1)\tilde{h}_\gamma(u)\tilde{w}_\alpha(1)^{-1}\tilde{w}_\beta(1)^{-1} = \tilde{h}_{-\beta}(u).$$

Statement (c) is easily deduced from Lemma 5.2, (e), and it implies that $c_\alpha(s,t^d)$, where $d = \alpha\beta^*$, is a bilinear function of (s,t). From this results the last assertion, considering the fact that for a root α there exists another β such that $\alpha\beta^* = 1$, unless α is a long root in the root system of symplectic type. □

We then have the following proposition, considering the case of the group G of rank 1:

Proposition 0.17. *The cocycles c_α satisfy the following equations:*

(S1) $c(x,y)c(xy,z) = c(x,yz)c(y,z)$;

(S2) $c(1,1) = 1; \quad c(x,y) = c(x^{-1},y^{-1})$;

(S3) $c(x,y) = c(x,(1-x)y) \quad$ *if* $x \neq 1$.

Indeed, (S 1) is nothing but the cocycle identity, and (S 2) follows immediately from Lemma 5.2, (g). It remains to verify the identity (S 3). We first see that $c_\alpha(t,-t^{-1}) = 1$ for all $t \in k^\times$; indeed, according to Lemma 5.2, (a), (g),

$$\begin{aligned} c_\alpha(t,-t^{-1}) &= \tilde{h}_\alpha(t)\tilde{h}_\alpha(-t^{-1})\tilde{h}_\alpha(-1)^{-1} \\ &= \tilde{w}_\alpha(t)\tilde{w}_\alpha(-1)\tilde{w}_\alpha(-t^{-1})\tilde{w}_\alpha(-1)\tilde{w}_\alpha(-1)^{-2} \\ &= \tilde{w}_\alpha(t)\tilde{w}_{-\alpha}(t^{-1}) = \tilde{w}_\alpha(t)\tilde{w}_\alpha(-t) = 1. \end{aligned}$$

This identity, combined with (S 1), implies that

$$c_\alpha(s,t) = c_\alpha(s,t)c_\alpha(st,-s^{-1}t^{-1}) = c_\alpha(s,-s^{-1})c_\alpha(t,-s^{-1}t^{-1}) = c_\alpha(t,-s^{-1}t^{-1});$$

which, by repetition, gives the identity

$$c_\alpha(s,t) = c_\alpha(-s^{-1}t^{-1},s).$$

Now let us calculate

$$\tilde{\nu}(\tilde{w}_\alpha(-1)\tilde{g}\tilde{w}_\alpha(-1)) \quad \text{for } \tilde{g} = \tilde{x}_\alpha(u)\tilde{w}_\alpha(1)\tilde{x}_\alpha(v) \quad \text{with } uv(1-uv) \neq 0.$$

On the one hand, according to Lemma 5.2, (g),

$$\begin{aligned} \tilde{w}_\alpha(-1)\tilde{g}\tilde{w}_\alpha(-1) &= \tilde{w}_\alpha(-1)\tilde{x}_\alpha(u)\tilde{w}_\alpha(1)\tilde{x}_\alpha(v)\tilde{w}_\alpha(-1) \\ &= \tilde{w}_\alpha(-1)\tilde{x}_\alpha(u-v^{-1})\tilde{w}_\alpha(v^{-1})\tilde{x}_\alpha(-v^{-1}); \end{aligned}$$

whence, according to Lemma 5.3, (d),

$$\begin{aligned} \tilde{\nu}(\tilde{w}_\alpha(-1)\tilde{g}\tilde{w}_\alpha(-1)) &= \tilde{h}_\alpha((u-v^{-1})^{-1})\tilde{w}_\alpha(v^{-1}) \\ &= \tilde{h}_\alpha((u-v^{-1})^{-1})\tilde{h}_\alpha(v^{-1})\tilde{w}_\alpha(1). \end{aligned}$$

On the other hand,

$$\begin{aligned} \tilde{w}_\alpha(-1)\tilde{g}\tilde{w}_\alpha(-1) &= \tilde{x}_\alpha(-u^{-1})\tilde{h}_\alpha(u^{-1})\tilde{w}_\alpha(1)\tilde{x}_\alpha(v-u^{-1})\tilde{w}_\alpha(-1) \\ &= \tilde{x}_\alpha(-u^{-1})\tilde{h}_\alpha(u^{-1})\tilde{x}_\alpha(-(v-u^{-1})^{-1})\tilde{w}_\alpha((v-u^{-1})^{-1})\tilde{x}_\alpha(-(v-u^{-1})^{-1}); \end{aligned}$$

whence

$$\begin{aligned} \tilde{\nu}(\tilde{w}_\alpha(-1)\tilde{g}\tilde{w}_\alpha(-1)) &= \tilde{h}_\alpha(u^{-1})\tilde{w}_\alpha((v-u^{-1})^{-1}) \\ &= \tilde{h}_\alpha(u^{-1})\tilde{h}_\alpha((v-u^{-1})^{-1})\tilde{w}_\alpha(1). \end{aligned}$$

Comparing these two expressions for $\tilde{\nu}(\tilde{w}_\alpha(-1)\tilde{g}\tilde{w}_\alpha(-1))$, we have

$$c_\alpha(u^{-1},(v-u^{-1})^{-1}) = c_\alpha((u-v^{-1})^{-1},v^{-1});$$

hence, according to (S 2),

$$c_\alpha(u,v-u^{-1}) = c_\alpha(u-v^{-1},v).$$

Whence, by applying the identity

$$c_\alpha(s,t) = c_\alpha(-s^{-1}t^{-1},s)$$

to both sides, we deduce that

$$c_\alpha((1-uv)^{-1},u) = c_\alpha((1-uv)^{-1},u-v^{-1}).$$

By substituting t for $(1-u\theta)^{-1}$, one concludes that

$$c_z(t,u) = c_z(t,u(1-t)^{-1}).$$

Hence identity (S 3). Our proposition is thus proved. The following lemma is easy to prove:

Lemma 0.18. *Equations (S 1), (S 2) and (S 3) are satisfied by any map c from $k^\times \times k^\times$ into A satisfying the following:*

($S^0$1) $c(x, yz) = c(x, y)c(x, z)$;

($S^0$2) $c(xy, z) = c(x, z)c(y, z)$;

($S^0$3) $c(x, 1 - x) = 1$ *if* $x \neq 1$.

A map c from $k^\times \times k^\times$ into A satisfying equations (S 1), (S 2) and (S 3) will be called a *Steinberg cocycle* on $k^\times$ with values in A. Such a cocycle c will be called *bilinear* if c satisfies equations (S^0 1), (S^0 2) and (S^0 3). It is known [30] that, if k is a finite field, any map from $k^\times \times k^\times$ into A satisfying (S 1), (S 2) and (S 3) is trivial for any A.

Let $S(k^\times, A)$ denote the multiplicative group of Steinberg cocycles and $S^b(k^\times, A)$ the subgroup of bilinear Steinberg cocycles on $k^\times$ with values in A.

The following proposition gives fundamental properties of a Steinberg cocycle:

Proposition 0.19. *Let c be a Steinberg cocycle on $k^\times$ with values in A. Then:*

(a) We have
$$c(x, y) = c(x, -xy) = c(-xy, y) = c(y^{-1}, x)$$
and
$$c(x^2, y) = c(x, y^2)$$
for all $x, y \in k^\times$.

(b) The map $c^\sharp$ defined by $c^\sharp(x, y) = c(x, y^2)$ is a bilinear Steinberg cocycle. We have $c^\sharp(x, -1) = 1$ for all $x \in k^\times$.

(c) The map $\check{c}$ defined by $\check{c}(x, y) = c(y, x)^{-1}$ is a Steinberg cocycle. We have $c^\sharp = (\check{c})^\sharp = c\check{c}$. If c is bilinear, we have $\check{c} = c$ and $c^\sharp = c^2$.

(d) The central extension of $k^\times$ by A defined by the cocycle c is an abelian group if and only if $c^\sharp$ is trivial.

First, let us note that c is a normalized cocycle. According to conditions (S 2) and (S 3), we have
$$\begin{aligned} c(x, y) &= c(x^{-1}, y^{-1}) = c(x^{-1}, (1 - x^{-1})y^{-1}) \\ &= c(x, (1 - x^{-1})^{-1}y) = c(x, -(1 - x)^{-1}xy) = c(x, -xy); \end{aligned}$$

which gives a part of (a) and implies that
$$c(x, -x^{-1}) = c(x, 1) = 1$$
for all $x \in k^\times$. As a result, we have
$$c(y, -1) = c(y, -1)c(x, -x^{-1}) = c(y, x)c(yx, -x^{-1});$$
whence
$$c(y, y)^{-1}c(xy, y) = c(y, -1)^{-1}c(xy, -x^{-1}) = c(y, x)^{-1}.$$
We thus see that
$$c(x, y^2) = c(x, y)c(xy, y)c(y, y)^{-1} = c(x, y)c(y, x)^{-1};$$
which means $c^\sharp = c\check{c}$ in our notation.

Consider the extension ε of $k^\times$ defined from the cocycle c. The identity $c^\sharp = c\check{c}$ shows that we have
$$c^\sharp(x, y) = s(x)s(y)s(x)^{-1}s(y)^{-1}$$
for a section s of ε; this proves statement (d), and implies that $c^\sharp$ is a bilinear map from $k^\times \times k^\times$ into A. Since the condition (S^0 3) for $c^\sharp$ is an immediate consequence of (S3) for c, we conclude that $c^\sharp$ is a bilinear Steinberg cocycle. Then, $\check{c} = c^\sharp c^{-1}$ is also a Steinberg cocycle, and the other assertions of (c) are obvious.

It remains to prove statement (a). We have already seen that $c(x,y) = c(x,-xy)$, and, since $\check{c}$ is also a Steinberg cocycle, we also verify that $c(x,y) = c(-xy,y)$. This implies that

$$c(x,y) = c(-xy^{-1},y) = c(-xy^{-1},x) = c(y^{-1},x).$$

Finally, we have

$$c(x^2,y)^{-1} = \check{c}^\natural(y,x) = c^\natural(y,x) = c^\natural(x,y)^{-1} = c(x,y^2)^{-1}.$$

We have thus proved all the assertions of the proposition.

Let us return to the extension E of G_k. According to Lemma 5.4 and Proposition 5.7, if G is of non-symplectic type, the Steinberg cocycles c_α are identical for all roots α of the same length. If G is of symplectic type, the simple root system Δ possesses only one long root.

We therefore give the following definition: if α is a long root in Δ, the Steinberg cocycle c_α at α will be called the Steinberg cocycle of E relative to Δ and to the isomorphisms x_α.

The following lemma is a consequence of Lemma 5.4:

Lemma 0.20. *5.8. — Let c be the Steinberg cocycle of the extension E.*

(a) For each $\alpha \in \Phi$, the Steinberg cocycle c_α at α of E is equal to one of the following: $c, \check{c}, c^\natural, c^\sharp$.

(b) The subgroup $\tilde{H}$ of E is abelian if and only if one of the following conditions is satisfied: (i) c is trivial; (ii) G is of symplectic type and $c^\natural$ is trivial.

Proof. Indeed, if α is a long root in Φ, according to Lemma 5.4, (b), we have $c_\alpha = c$ or $\check{c}$. If β is a non-long root in Φ, there exists another root α such that $\beta\alpha^* = -1$ and $\alpha\beta^* = -3$ or -2, according to whether G is of type G_2 or not; we then have $c_\beta = c_\alpha^\natural$ or $\check{c}_\alpha^\natural$, by virtue of Lemma 5.4, (c). This proves statement (a). Assertion (b) follows easily from Lemma 5.4, (c). □

We will later need the following lemma:

Lemma 0.21. *5.9. — Let α be an element of Φ. If u and v are elements of $k^\times$ such that $1 + uv \neq 0$, the element $\tilde{x}_{-\alpha}(u)\tilde{x}_\alpha(v)$ can be written in the form*

$$\tilde{x}_\alpha(v(1+uv)^{-1})\tilde{h}_\alpha((1+uv)^{-1})c_\alpha(-u,1+uv)c_\alpha(1+uv,-1)^{-1}\tilde{x}_{-\alpha}(u(1+uv)^{-1}).$$

Proof. Indeed, if we consider the case of rank 1, it is a matter of putting $\tilde{x}_{-\alpha}(u)\tilde{x}_\alpha(v)$ in the "big cell" $\tilde{U}^+\tilde{H}\tilde{U}^-$ of $E = \tilde{U}^+\tilde{N}\tilde{U}^-$. This is done by an easy calculation, using notably Lemma 5.2, (h). □

Recall that the classes of central extensions of G_k by A form, with Baer multiplication, the cohomology group $H^2(G_k, A)$. It is already known that the Steinberg cocycle c_E of E relative to Δ and to the isomorphisms x_α depends only on the class of extensions of E. According to Steinberg's theorem on G_k, the extension E is trivial if and only if c_E is. Moreover, it is easily seen that, if c_E and $c_{E'}$ are thus associated with extensions E and E', the product $c_E c_{E'}$ is the Steinberg cocycle of the Baer product of E and of E'.

We can now state one of the main theorems of this chapter; recall that k is an infinite field.

Theorem 0.22. *5.10. — When one associates to a central extension E of G_k by A its Steinberg cocycle relative to Δ and to the isomorphisms x_α, one obtains an isomorphism from the group $H^2(G_k, A)$ of classes of central extensions of G_k by A onto the group $S(k^\times, A)$ [resp. $S^0(k^\times, A)$] of Steinberg cocycles [resp. bilinear Steinberg cocycles] on $k^\times$ with values in A, if G is of symplectic type [resp. non-symplectic].*

To prove this theorem, it remains for us to see that the map $E \mapsto c_E$ has for its image the whole of $S(k^\times, A)$ or $S^0(k^\times, A)$. We will verify this in the following two sections.

This theorem allows representing by generators and relations the kernel $\pi_1(G_k)$ of the covering $E(G_k)$ of G_k considered by Steinberg [30] and taken up again at the beginning of this section.

Corollary 0.23. *5.11. — If G is of symplectic type [resp. non-symplectic], $\pi_1(G_k)$ is isomorphic to the abelian group $S(k)$ [resp. $S^0(k)$] generated...*

by the symbols $c(x,y)$ $(x, y \in k^\times)$ subject to the relations (S 1), (S 2) and (S 3) [resp. (S° 1), (S° 2) and (S° 3)].

For $G = \mathrm{SL}_2$, Theorem 5.10 and its corollary were obtained by C. Moore [26].

In the case of SL_2, one can also give a nice formula which expresses a cocycle on G_k determining E in terms of c_E.

Let us define a map χ from G_k to $k^\times$ as follows:

$$\chi(g) = \chi(\nu(g)) \quad \text{for all } g \in G_k;$$
$$\chi(h_\alpha(t)) = t, \quad \chi(w_\alpha(t)) = -t \quad \text{for } t \in k^\times, \quad \alpha \in \Delta.$$

On the other hand, we choose a section $\mathfrak{s}$ of E as follows:

$$\tilde{\mathfrak{s}}(g) = \mathfrak{s}(\nu(g)) \quad \text{for all } g \in G_k;$$
$$\mathfrak{s}(h_\alpha(t)) = c_E(t,t)^{-1}\tilde{h}_\alpha(t), \quad \mathfrak{s}(w_\alpha(t)) = \tilde{w}_\alpha(t) \quad \text{for } t \in k^\times, \quad \alpha \in \Delta.$$

Then, the cocycle z_E on G_k with respect to $\mathfrak{s}$ is given by

$$z_E(g,g') = c_E(\chi(g),\chi(g'))^{-1}c_E(\chi(gg'), -\chi(g)^{-1}\chi(g')).$$

The following corollary is therefore a consequence of Theorem 5.10:

Corollary 0.24. *5.12. — Let $G = SL_2$. Let χ be the map from G_k into $k^\times$ defined as follows:*

$$\chi\begin{pmatrix} s & t \\ u & v \end{pmatrix} = v \text{ or } u, \text{ depending on whether } u = 0 \text{ or not.}$$

Then:

(a) If c is a Steinberg cocycle on $k^\times$ with values in A, the formula

$$z_c(g,g') = c(\chi(g),\chi(g'))^{-1}c(\chi(gg'), -\chi(g)^{-1}\chi(g'))$$

gives a cocycle on G_k with values in A.

(b) The map $c \mapsto z_c$ establishes an isomorphism from $S(k^\times, A)$ onto $H^2(G_k, A)$.

T. Kubota [19] had directly proved Corollary 5.12, (a) in the particular case where k is a local field and c is the norm residue symbol of k (cf. n° 9).

Let us return to the general case and examine how the algebraic automorphisms of G_k act on $H^2(G_k, A)$.

Recall that G_k is the group of rational points of the group scheme ${}_kG$ over k derived from a split group scheme $\mathfrak{G}$ over $\mathbb{Z}$. We will denote by $\mathrm{Aut}({}_kG)_k$ the group of automorphisms of ${}_kG$ defined over k.

We have chosen a maximal split torus ${}_kH$ of ${}_kG$ and, for each root α of ${}_kG$ relative to ${}_kH$, an isomorphism x_α from the additive group onto the root subgroup ${}_kU^\alpha$; x_α and $x_{-\alpha}$ are related in such a way that

$x_\alpha(t)x_{-\alpha}(-t^{-1})x_\alpha(t)$ be in the normalizer ${}_kN$ of ${}_kH$ for all $t \neq 0$. If x'_α and $x'_{-\alpha}$ are other isomorphisms of the additive group onto ${}_kU^\alpha$ and ${}_kU^{-\alpha}$ defined over k and possessing this last property, there exists an element r of $k^\times$ such that $x'_\alpha(u) = x_\alpha(ru)$ and $x'_{-\alpha}(u) = x_{-\alpha}(r^{-1}u)$ for all u.

On the other hand, $\mathrm{Aut}({}_kG)_k$ contains the group $\mathrm{Ad}(G_k)$ of automorphisms of ${}_kG$ defined by elements of G_k. By virtue of the conjugation theorem of maximal split tori [6], if α is a root of ${}_kG$ relative to ${}_kH$ and if σ is an element of $\mathrm{Aut}({}_kG)_k$, there exists an element τ of $\mathrm{Ad}(G_k)$ such that $\sigma\tau$ leaves stable ${}_kH, {}_kU^\alpha$ and ${}_kU^{-\alpha}$. If $\{r_\alpha\}(\alpha \in \Delta)$ is a set of elements of $k^\times$, there exists an element σ of $\mathrm{Aut}({}_kG)_k$ such that σ leaves fixed the elements of ${}_kH$ and that $x_\alpha(u)^\sigma = x_\alpha(r_\alpha u)$ for all $\alpha \in \Delta$.

An element σ of $\mathrm{Aut}({}_kG)_k$ acts on the cocycles z on G_k with values in A, by $z^\sigma(g_1, g_2) = z(g_1^\sigma, g_2^\sigma)$; σ thus defines an automorphism of $H^2(G_k, A)$. The group $\mathrm{Aut}({}_kG)_k$ thus operates on $H^2(G_k, A)$, and it is evident that $\mathrm{Ad}(G_k)$ acts trivially on it.

Suppose that an element σ of $\mathrm{Aut}({}_kG)_k$ leaves stable ${}_kH, {}_kU^\alpha$ and ${}_kU^{-\alpha}$ for an element α of Φ; we have

$$x'_\alpha(u) = x_\alpha(u)^\sigma = x_\alpha(ru) \quad \text{and} \quad x'_{-\alpha}(u) = x_{-\alpha}(u)^\sigma = x_{-\alpha}(r^{-1}u)$$

with an element r of $k^\times$. Let E be a central extension of G_k by A, and let us calculate in terms of c_α the Steinberg cocycle c'_α at α of E^σ. Let us set in E

$$\tilde{x}'_\alpha(u) = \tilde{x}_\alpha(ru) \quad \text{and} \quad \tilde{x}'_{-\alpha}(u) = \tilde{x}_{-\alpha}(r^{-1}u);$$

then

$$\tilde{w}'_\alpha(t) = \tilde{x}'_\alpha(t)\tilde{x}'_{-\alpha}(-t^{-1})\tilde{x}'_\alpha(t) = \tilde{w}_\alpha(rt);$$

$$\tilde{h}'_\alpha(t) = \tilde{w}'_\alpha(t)\tilde{w}'_\alpha(-1) = \tilde{h}_\alpha(rt)\tilde{h}_\alpha(r)^{-1} = c_\alpha(t,r)^{-1}\tilde{h}_\alpha(t);$$

whence

$$c'_\alpha(s,t) = c_\alpha(s,t)c_\alpha(st,r)c_\alpha(s,r)^{-1}c_\alpha(t,r)^{-1}.$$

We are thus led to the following result:

Proposition 0.25. *5.13. — (a) The group $k^\times$ operates on $S(k^\times, A)$ by the following formula:*

$$(\hat{\delta}(t)c)(x,y) = c(x,y)c(xy,t)c(x,t)^{-1}c(y,t)^{-1}.$$

The elements of $(k^\times)^2$ act trivially on it, and $\hat{\delta}(-1)c = \check{c}$ for all $c \in S(k^\times, A)$.

(b) $Aut({}_kG)_k$ operates on $H^2(G_k, A)$ and therefore on $S(k^\times, A)$ [resp. $S^0(k^\times, A)$] according to the isomorphism of Theorem 5.10. The action of an element of $Aut({}_kG)_k$ is identical to $\hat{\delta}(t)$ for an element t of $k^\times$.

(c) The invariants of $Aut({}_kG)_k$ in $H^2(G_k, A)$ are the elements that correspond to those of $S^0(k^\times, A)$.

(*d*) The restricted isomorphism from $S^o(k^\times, A)$ to $H^2(G_k, A)^o$ depends neither on the choice of the isomorphisms x_α nor on the choice of the system of simple roots Δ nor on the choice of the maximal torus ${}_kH$ split over k.

One could therefore say that the isomorphism of $S(k^\times, A)$ [resp. of $S^o(k^\times, A)$] onto $H^2(G_k, A)$ is canonical after restriction to $S^o(k^\times, A)$.

6. This section and the following one are devoted to the proof of Theorem 5.10.

We will first study a central extension of the extended Weyl group $\mathfrak{N} = N_Z$ associated with the simply connected simple group G. The group $\mathfrak{N}$ is an extension of the Weyl group W by the abelian group $\mathfrak{H} = H_Z$ of order 2^r, where r is the rank of G. Recall that Φ denotes the root system of G relative to the maximal torus H and that we have chosen a system of simple roots Δ of Φ.

We have the following lemma (Demazure [16], Tits [31]):

Lemma 0.26. *The group $\mathfrak{N}$ is generated by the elements $w_\alpha = w_\alpha(-1)$ for $\alpha \in \Delta$ and defined by the following relations (W 1), (W 2) and (W 3):*

(W 1) $h_\alpha w_\beta h_\alpha^{-1} = w_\beta h_\alpha^{c'}$ *with* $h_\alpha = w_\alpha^2$ *and* $c' = (-1)^{\beta\alpha^*}$, *for all* $\alpha, \beta \in \Delta$;

(W 2) $$\begin{cases} w_\alpha w_\beta = w_\beta w_\alpha & \text{if } \beta\alpha^* = 0; \\ w_\alpha w_\beta w_\alpha = w_\beta w_\alpha w_\beta & \text{if } \alpha\beta^* = \beta\alpha^* = -1; \\ w_\beta w_\alpha w_\beta w_\alpha^{-1} = w_\alpha w_\beta w_\alpha^{-1} w_\beta & \text{if } \beta\alpha^* = -2; \\ w_\beta w_\gamma w_\beta^{-1} = w_\gamma w_\beta^{-1} w_\gamma^{-1} \text{ with } w_\gamma = w_\alpha w_\beta w_\alpha^{-1}, & \text{if } \beta\alpha^* = -3; \end{cases}$$

(W 3) $h_\alpha^2 = e$ *for all* $\alpha \in \Delta$.

Proof. Indeed, applying Lemmas 5.1 and 5.2 to G, we easily see that the generators w_α satisfy these relations. Let $\mathfrak{N}'$ be the group generated by the symbols w_α subject to the relations (W 1), (W 2) and (W 3); it is naturally an extension of $\mathfrak{N}$. According to (W 1), we have

$$w_\beta h_\alpha w_\beta^{-1} = h_\alpha h_\beta^{c''} \quad \text{et} \quad h_\beta h_\alpha h_\beta^{-1} = h_\alpha h_\beta^{2c''} \quad \text{avec } 2c'' = (-1)^{\beta\alpha^*} - 1;$$

the elements h_α therefore generate a normal subgroup $\mathfrak{H}'$ of $\mathfrak{N}'$, and according to (W 3), $\mathfrak{H}'$ is abelian of order 2^r. The quotient group $\mathfrak{N}'/\mathfrak{H}'$ is generated by the involutive elements $\bar{w}_\alpha$ satisfying the relations (W 2), which form an exhaustive family of relations for the generators σ_α of the Weyl group W (see N. Bourbaki [8]). Consequently, $\mathfrak{N}'$ is isomorphic to $\mathfrak{N}$. □

On this occasion, we will recall other properties of the Weyl group W. We know that W is generated by the reflections σ_α for $\alpha \in \Delta$ and defined by the relations (W 2) with the w_α replaced by the σ_α. We define the length function l on W with respect to Δ as follows: for $\sigma \in W$, $l(\sigma)$ denotes the smallest non-negative integer m such that σ can be written as...

product $\prod_{i=1}^m \sigma_{\alpha_i}$, with $\alpha_i \in \Delta$. We will also denote by l the function on N defined by $l(n) = l(\omega(n))$, where ω denotes the canonical morphism from N onto W.

The following results are then known about the Weyl group W, its action on Φ and the map ν from lemma 5.3, (b) (cf. [8], [20], [31]).

Lemma 0.27. *6.2. — (a) Let Δ' be a subset of Δ and W' the subgroup of W generated by the elements σ_α for $\alpha \in \Delta'$. Then, each class $W'\sigma$ of W modulo W' possesses an element σ_0 such that $l(\sigma'\sigma_0) = l(\sigma') + l(\sigma_0)$ for all $\sigma' \in W'$.*

(b) Let β be a positive root, and let σ, σ' be elements of W such that

$$l(\sigma'\sigma) = l(\sigma') + l(\sigma).$$

Then, if $\sigma'\sigma\beta$ is positive, the same is true of $\sigma\beta$.

(c) If g and g' are elements of G_k such that

$$l(\nu(g)\nu(g')) = l(\nu(g)) + l(\nu(g')),$$

then $\nu(gg') = \nu(g)\nu(g')$.

(d) Suppose that elements d_α, indexed by Δ, of a monoid with identity D satisfy the following relations:

$$(W\ 2') \quad \begin{cases} d_\alpha d_\beta = d_\beta d_\alpha & \text{if } \beta\alpha^* = 0; \\ d_\alpha d_\beta d_\alpha = d_\beta d_\alpha d_\beta & \text{if } \alpha\beta^* = \beta\alpha^* = -1; \\ (d_\alpha d_\beta)^2 = (d_\beta d_\alpha)^2 & \text{if } \beta\alpha^* = -2; \\ (d_\alpha d_\beta)^3 = (d_\beta d_\alpha)^3 & \text{if } \beta\alpha^* = -3. \end{cases}$$

If an element σ of W is expressed in two minimal words $\prod_{i=1}^m \sigma_{\alpha_i}$ and $\prod_{j=1}^m \sigma_{\alpha'_j}$, where $m = l(\sigma)$ and $\alpha_i, \alpha'_j \in \Delta$, then we have, in D,

$$\prod_{i=1}^m d_{\alpha_i} = \prod_{j=1}^m d_{\alpha'_j}.$$

Recall first that $l(\sigma_\alpha\sigma) = l(\sigma) + 1$ for $\sigma \in W$ and $\alpha \in \Delta$ if and only if $\sigma^{-1}\alpha$ is positive. From this, one easily deduces statements (b) and (c), by induction on $l(\sigma')$ or $l(\nu(g))$. For assertions (a) and (d), a proof can be found in [8].

The following theorem will play an important role in the proof of theorem 5.10.

Theorem 0.28. *6.3. — Let $\tilde{\mathfrak{N}}$ be the group generated by the symbols $w_\alpha (\alpha \in \Delta)$ subject only to the relations (W 1) and (W 2), and π the natural morphism from $\tilde{\mathfrak{N}}$ onto $\mathfrak{N}$. Then:*

(a) The kernel $\mathfrak{Z}$ of π is central in $\tilde{\mathfrak{N}}$ and generated by w_α^2, where α is a long root in Δ. The group $\mathfrak{Z}$ is cyclic of infinite order [resp. of order 2] if Δ is of symplectic type [resp. non-symplectic].

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(b) *The group $\tilde{\mathfrak{N}}$ can be defined by the following relations:*

(W 1') $w_\beta^{-1} h_\alpha w_\beta = h_\alpha w_\alpha^{c''}$ with $h_\alpha = w_\alpha^2$ and $c'' = 1 - (-1)^{\beta\alpha^*}$, for all $\alpha, \beta \in \Delta$;

$$\text{(W 2')} \quad \begin{cases} w_\alpha w_\beta = w_\beta w_\alpha & \text{if } \beta\alpha^* = 0; \\ w_\alpha w_\beta w_\alpha = w_\beta w_\alpha w_\beta & \text{if } \alpha\beta^* = \beta\alpha^* = -1; \\ (w_\alpha w_\beta)^2 = (w_\beta w_\alpha)^2 & \text{if } \beta\alpha^* = -2; \\ (w_\alpha w_\beta)^3 = (w_\beta w_\alpha)^3 & \text{if } \beta\alpha^* = -3. \end{cases}$$

(c) *The subgroup $\mathfrak{K}$ of $\tilde{\mathfrak{N}}$ generated by the elements $h_\alpha (\alpha \in \Delta)$ is defined by the following relations:*

$$h_\beta^{-1} h_\alpha h_\beta = h_\alpha h_\alpha^{c''} \text{ with } c'' = 1 - (-1)^{\beta\alpha^*}, \text{ for all } \alpha, \beta \in \Delta.$$

First, the relations (W 1) imply that the elements $z_\alpha = w_\alpha^4$ generate in $\tilde{\mathfrak{N}}$ a central subgroup, which, by Lemma 6.1, is equal to the kernel $\mathfrak{Z}$ of π. It is also easily seen that the families of relations (W 1) and (W 1') are equivalent and that, under their assumption, the same holds for the relations (W 2) and (W 2'). On the other hand, the relations (W 1') immediately entail those of statement (c); in other words,

$$h_\beta^{-1} h_\alpha^{-1} h_\beta h_\alpha = h_\beta^{d''} = h_\alpha^{-c''}, \quad \text{with } c'' = 1 - (-1)^{\beta\alpha^*},\ d'' = 1 - (-1)^{\alpha\beta^*}.$$

From this we deduce that $z_\beta^2 = e$ if $\beta\alpha^*$ is odd for an element α of Δ, that $z_\beta = e$ if $\alpha\beta^* = -2$, and that $z_\alpha = z_\beta$ if $\alpha\beta^*$ and $\beta\alpha^*$ are odd. Now, the roots of Δ have, at most, two different lengths, and those which are long form an irreducible subsystem. We conclude that $\mathfrak{Z}$ is generated by z_α, where α is any long root in Δ, and that $\mathfrak{Z}$ is of order 1 or 2 if Δ is of non-symplectic type.

It remains for us to determine the order of $\mathfrak{Z}$. First, from the relations (W 1') and (W 2'), one obtains a presentation of the quotient group of $\tilde{\mathfrak{N}}$ by its commutator subgroup; if Δ is of symplectic type, one immediately sees there that the element w_α is of infinite order in this abelian group, α being the long root in Δ. Consequently, $\mathfrak{Z}$ is an infinite group in the symplectic case. To determine the order of $\mathfrak{Z}$ in the non-symplectic case, we shall construct an extension of $\mathfrak{N}$ by a group of order 2, which will be isomorphic to $\tilde{\mathfrak{N}}$ in the non-symplectic case. This construction is valid for all types, but it suffices, to complete the proof of the theorem, to treat the cases where Δ consists of roots of the same length. Indeed, the cases of types $B_n (n \geq 3)$, F_4, and G_2 can easily be concluded by letting an "exterior" automorphism of order 2 or 3 (cf. Lemma 6.4) act on the groups $\mathfrak{N}$ of types D_{n+1}, E_6, and D_4.

Let $\tilde{\mathfrak{H}}'$ be the group generated by the symbols $t_\alpha (\alpha \in \Delta)$ subject to the following relations:

$$t_\alpha^2 = e, \quad t_\beta^{-1} t_\alpha t_\beta = t_\alpha t_\beta^{c''} \text{ with } c'' = 1 - (-1)^{\beta\alpha^*}, \text{ for all } \alpha, \beta \in \Delta.$$

As seen above, $\tilde{\mathfrak{H}}'$ is a central extension of $\mathfrak{H}$ by the kernel of order 2. Then a group of automorphisms of $\tilde{\mathfrak{H}}'$ is defined; for each $\alpha \in \Delta$, the map $t_\beta \mapsto t_\beta t_\alpha^{d'} (\beta \in \Delta, 2d' = 1 - (-1)^{\alpha\beta^*})$ extends to an automorphism θ_α of $\tilde{\mathfrak{H}}'$. These automorphisms $\theta_\alpha (\alpha \in \Delta)$ satisfy relations (W 1), (W 2) and (W 3) with the w_α replaced by the θ_α; we also note that $\theta_\alpha^2 = \mathrm{Ad}(t_\alpha)$ for all $\alpha \in \Delta$.

One then defines, for each $\alpha \in \Delta$, a transformation λ_α of the product set $\tilde{\mathfrak{H}} \times W$:

$$\lambda_\alpha(\iota, \sigma) = (\theta_\alpha(\iota), \sigma_\alpha \sigma) \quad \text{if } l(\sigma_\alpha \sigma) > l(\sigma);$$
$$\lambda_\alpha(\iota, \sigma) = (\theta_\alpha(\iota t_\alpha), \sigma_\alpha \sigma) \quad \text{if } l(\sigma_\alpha \sigma) < l(\sigma).$$

Let Λ be the group of transformations of $\tilde{\mathfrak{H}}' \times W$ generated by the λ_α. It is easily seen that $\lambda_\alpha^2(\iota, \sigma) = (t_\alpha \iota, \sigma)$ for all α, and that $\lambda_\alpha^2 \lambda_\beta \lambda_\alpha^{-2} = \lambda_\beta^{c'}$ with $c' = (-1)^{\beta\alpha^*}$ for all $\alpha, \beta \in \Delta$; in other words, the generators λ_α satisfy relations (W 1') and the elements λ_α^2 generate a normal subgroup of Λ which is isomorphic to $\tilde{\mathfrak{H}}'$. Consequently, when relations (W 2') have been verified for Λ, it will be concluded, in the non-symplectic case, that Λ is isomorphic to $\tilde{\mathfrak{N}}$ and that $\mathfrak{Z}$ is of order 2.

Proof. Let α, β be two elements of Δ, and W' the subgroup of W generated by σ_α and σ_β. Recall that θ_α and θ_β satisfy relations (W 1') and (W 2'). Then, by virtue of lemma 6.2, (a) and of the definition of λ_α and λ_β, we see that the relation (W 2') for λ_α and λ_β is satisfied if it is so on the subset $\{e\} \times W'$ of $\tilde{\mathfrak{H}}' \times W$. We are thus brought back to the cases of rank 2. The case where $\alpha\beta^* = 0$ is trivial. In the case of type A_2, a direct calculation shows that the relation (W 2') for λ_α and λ_β is true on the elements of $\{e\} \times W'$. Theorem 6.3 is thus proved. □

Remark 1. It will be noted that the construction of Λ is motivated by lemma 6.2, (d) and that our arguments closely resemble those of Tits [31] which allow one to establish, starting from W, the existence of a group isomorphic to $\mathfrak{N}$.

Furthermore, it is known that the universal covering $\tilde{G}_{\mathbf{R}}$ of the connected Lie group $G_{\mathbf{R}}$ possesses a subgroup isomorphic to $\tilde{\mathfrak{N}}$; it suffices to take $\pi^{-1}(N_{\mathbf{Z}})$, where π denotes the projection of $\tilde{G}_{\mathbf{R}}$ onto $G_{\mathbf{R}}$ (cf. no. 9). This provides another proof of the non-triviality of $\mathfrak{Z}$.

We shall now begin the proof of theorem 5.10. Let c be a given Steinberg cocycle (bilinear if G is of non-symplectic type) on $k^\times$ with values in A. The task then is to show the existence...

...tence of a central extension of G_k by A whose Steinberg cocycle at α is equal to c for a long root α in Δ.

Let us first note the following lemma, which will allow us subsequently to restrict ourselves to the cases of certain types.

Lemma 0.29. *6.4. — There exists an embedding of the group G' of type B_n $(n \geq 3)$ [resp. F_4 or G_2] into the group G of type D_{n+1} [resp. E_6 or D_4] such that the conclusion of Theorem 5.10 for G' is deduced by restriction from that for G.*

Proof. Indeed, the diagrams Δ of types A_{2n-1} $(n \geq 2)$, D_{n+1} $(n \geq 3)$ and E_6 [resp. of type D_4] admit an automorphism ρ of order 2 [resp. 3] which leaves fixed at least one vertex. This automorphism ρ lifts to an automorphism ρ of G such that $\rho \circ x_\alpha = x_{\rho\alpha}$ and $\rho \circ x_{-\alpha} = x_{-\rho\alpha}$ for all $\alpha \in \Delta$. The fixed points of ρ in G form a simply connected simple group G' of type C_n $(n \geq 2)$, B_n $(n \geq 3)$, F_4 or G_2. The

group $H' = G' \cap H$ is a maximal torus of G', the roots of G' relative to H' are the restrictions to H' of the roots α of G, and the long roots of G' are thus identified with the roots of G which are fixed by ρ. The automorphism ρ of G carries over to an automorphism of the group scheme $\mathcal{G}$ over $\mathbb{Z}$ and then into an automorphism of G_k, also denoted by ρ. The group G'_k of fixed points of ρ in G_k is the set of rational points of a simply connected simple group split over k and of the same type as G'. If E is a central extension of G_k by A, we deduce from it by restriction a central extension E' of G'_k by A. Then, a long root α of G' being identified with a root of G, one sees immediately that the Steinberg cocycles at α of E and E' are identical. Consequently, if G' is of non-symplectic type, the conclusion of Theorem 5.10 for G' results from that for G. □

From now on, we shall only treat, on the one hand, the case where G is of symplectic type and, on the other hand, the case where the roots of G have the same length and where the given Steinberg cocycle c is bilinear. We will freely use the properties of the Steinberg cocycles stated in Proposition 5.7.

The first step of the proof consists of defining a central extension of H by A. Let us set $c^0 = c^{1/2}$ in the symplectic case and $c^0 = c$ in the non-symplectic case; c^0 is a bilinear Steinberg cocycle. We define as follows the cocycles c_α ($\alpha \in \Delta$) and $c_{\alpha\beta}$ ($\alpha, \beta \in \Delta, \alpha \neq \beta$): $c_\alpha = c$ for the long roots α and $c_\beta = c^0$ for the short roots β in the symplectic case; $c_{\alpha\beta} = 1$ if $\beta\alpha^* = 0$ and $c_{\alpha\beta} = (c^0)^{-1}$ if $\beta\alpha^* \neq 0$. Let us then set

$$f\left(\prod_\alpha h_\alpha(u_\alpha), \prod_\beta h_\beta(v_\beta)\right) = \prod_\alpha c_\alpha(u_\alpha, v_\alpha) \prod_{\beta<\alpha} c_{\alpha\beta}(u_\alpha, v_\beta),$$

Δ being equipped with any total order; c is a cocycle on H with values in A. Let $\tilde{H}$ be the central extension of H by A defined from c:

$$1 \to A \to \tilde{H} \to H \to 1.$$

One can lift into $\tilde{H}$ the elements $h_\alpha(t)$ ($\alpha \in \Delta, t \in k^\times$) in such a way that the lifted elements $\tilde{h}_\alpha(t)$ satisfy the following relations:

$$\tilde{h}_\beta(u)\tilde{h}_\beta(v) = c_\beta(u, v)\tilde{h}_\beta(uv)$$

and

$$\tilde{h}_\beta(u)\tilde{h}_\gamma(v)\tilde{h}_\beta(u)^{-1}\tilde{h}_\gamma(v)^{-1} = c_\beta(u, v^{\beta\gamma^*}) = c_\gamma(u^{\gamma\beta^*}, v)$$

for all $\beta, \gamma \in \Delta$.

Then we are going to define a group of automorphisms of $\tilde{H}$.

Lemma 0.30. *(a) For each $\alpha \in \Delta$, there exists an automorphism θ_α of $\tilde{H}$ such that $\theta_\alpha^{-1}(a) = a$ for $a \in A$ and that $\theta_\alpha^{-1}(\tilde{h}_\beta(t)) = \tilde{h}_\beta(t)\tilde{h}_\alpha(t^{-\alpha\beta^*})$ for all $\beta \in \Delta$.*

(b) θ_α induces, by passing to the quotient, the automorphism of H defined by the element $w_\alpha(-1)$ of N.

(c) $\theta_\alpha^2 = \theta_{-\alpha}^2 = \mathrm{Ad}(\tilde{h}_\alpha(-1))$ for all $\alpha \in \Delta$. The automorphisms θ_α satisfy the relations $(W1)$, $(W2)$ and $(W3)$.

Let us first note that $c_\alpha(u, v^{\beta^*})$ is bilinear in (u, v) for any $\alpha, \beta \in \Delta$. To verify (a), it suffices to show that the elements $\theta_\alpha^{-1}(\tilde{h}_\beta(t))$ satisfy the same relations as the $\tilde{h}_\beta(t)$. We have

$$\begin{aligned}
&\theta_\alpha^{-1}(\tilde{h}_\beta(u))\theta_\alpha^{-1}(\tilde{h}_\beta(v)) \\
&\quad = \tilde{h}_\beta(u)\tilde{h}_\alpha(u^{-\alpha\beta^*})\tilde{h}_\beta(v)\tilde{h}_\alpha(v^{-\alpha\beta^*}) \\
&\quad = \tilde{h}_\beta(u)\tilde{h}_\beta(v)\tilde{h}_\alpha((uv)^{-\alpha\beta^*})c_\alpha(u^{-\alpha\beta^*}, v^{\alpha\beta^*})c_\alpha(u^{-\alpha\beta^*}, v^{-\alpha\beta^*}) \\
&\quad = c_\beta(u, v)\tilde{h}_\beta(uv)\tilde{h}_\alpha((uv)^{-\alpha\beta^*}) \\
&\quad = c_\beta(u, v)\theta_\alpha^{-1}(\tilde{h}_\beta(uv));
\end{aligned}$$

$$\begin{aligned}
&\theta_\alpha^{-1}(\tilde{h}_\beta(u))\theta_\alpha^{-1}(\tilde{h}_\gamma(v))(\theta_\alpha^{-1}(\tilde{h}_\beta(u)))^{-1}(\theta_\alpha^{-1}(\tilde{h}_\gamma(v)))^{-1} \\
&\quad = c_\beta(u, v^{\beta\gamma^*})c_\alpha(u^{-\alpha\beta^*}, v^{\alpha\gamma^*})c_\alpha(u^{\alpha\beta^*}, v^{-\alpha\gamma^*})c_\alpha(u^{-\alpha\beta^*}, v^{-\alpha\gamma^*})^2 \\
&\quad = c_\beta(u, v^{\beta\gamma^*}).
\end{aligned}$$

Statement (a) is thus proven. Assertion (b) follows immediately from Lemma 5.2, (f) applied to G_k. Let us verify (c). We first have

$$\begin{aligned}\theta_\alpha^{-1}(\tilde{h}_\alpha(t)) &= \tilde{h}_\alpha(t)\tilde{h}_\alpha(t^{-2}) = \tilde{h}_\alpha(t^{-1});\\ \theta_\alpha^{-2}(\tilde{h}_\beta(u)) &= \tilde{h}_\beta(u)\tilde{h}_\alpha(u^{-\alpha\beta^*})\tilde{h}_\alpha(u^{\alpha\beta^*})\\ &= \tilde{h}_\beta(u)c_\alpha(u^{-\alpha\beta^*}, u^{\alpha\beta^*}) = \tilde{h}_\beta(u)c_\alpha(u^{-\alpha\beta^*}, -1)\\ &= \tilde{h}_\alpha(-1)\tilde{h}_\beta(u)\tilde{h}_\alpha(-1)^{-1}.\end{aligned}$$

This proves that $\theta_\alpha^{-2} = \theta_\alpha^2 = \mathrm{Ad}(\tilde{h}_\alpha(-1))$ and easily implies that the automorphisms θ_α satisfy the relations (W 1) and (W 3). The relation (W 2') for θ_α and θ_β is obvious if $\beta\alpha^* = 0$. If $\alpha\beta^* = \beta\alpha^* = -1$, we have

$$\begin{aligned}(\theta_\alpha\theta_\beta\theta_\alpha)^{-1}(\tilde{h}_\gamma(t)) &= (\theta_\beta\theta_\alpha)^{-1}(\tilde{h}_\gamma(t)\tilde{h}_\alpha(t^{-\alpha\gamma^*}))\\ &= \theta_\alpha^{-1}(\tilde{h}_\gamma(t)\tilde{h}_\beta(t^{-\beta\gamma^*})\tilde{h}_\alpha(t^{-\alpha\gamma^*})\tilde{h}_\beta(t^{-\beta\gamma^*}))\\ &= \tilde{h}_\gamma(t)\tilde{h}_\alpha(t^{-\alpha\gamma^*})\tilde{h}_\beta(t^{-\beta\gamma^*})\tilde{h}_\alpha(t^{-\beta\gamma^*})\tilde{h}_\alpha(t^{\alpha\gamma^*})\tilde{h}_\beta(t^{-\alpha\gamma^*})\tilde{h}_\alpha(t^{-\gamma\alpha^*})\\ &= \tilde{h}_\gamma(t)\tilde{h}_\alpha(t^{-\alpha\gamma^*-\beta\gamma^*})\tilde{h}_\beta(t^{-\alpha\gamma^*-\beta\gamma^*})c_\beta(-1, t^{\beta\gamma^*})\\ &= \tilde{h}_\gamma(t)\tilde{h}_\beta(t^{-\alpha\gamma^*-\beta\gamma^*})\tilde{h}_\alpha(t^{-\alpha\gamma^*-\beta\gamma^*})c_\alpha(-1, t^{\alpha\gamma^*})\\ &= (\theta_\beta\theta_\alpha\theta_\beta)^{-1}(\tilde{h}_\gamma(t));\end{aligned}$$

from which we get the relation (W 2') for θ_α and θ_β. Finally, let us treat the case where $\beta\alpha^* = -2$. We will compare $(\theta_\alpha\theta_\beta)^2(\tilde{h}_\gamma(t))$ and $(\theta_\beta\theta_\alpha)^2(\tilde{h}_\gamma(t))$. Let us first note that the elements of the form $\tilde{h}_\alpha(t^i)\tilde{h}_\beta(t^{2j})$ $(i, j \in \mathbf{Z})$ form an abelian subgroup $\tilde{H}'(t)$ of $\tilde{H}$. The automorphisms θ_α and θ_β leave $\tilde{H}'(t)$ and $\tilde{h}_\gamma(t)\tilde{H}'(t)$ stable. We have therefore

$$(\theta_\alpha\theta_\beta)^2(\tilde{h}_\gamma(t)) = \tilde{h}_\gamma(t)\tilde{h}'_1 \quad \text{and} \quad (\theta_\beta\theta_\alpha)^2(\tilde{h}_\gamma(t)) = \tilde{h}_\gamma(t)\tilde{h}'_2,$$

with $\tilde{h}'_1, \tilde{h}'_2 \in \tilde{H}'(t)$. From this, we deduce by statement (b) and Lemma 6.1 that $\varphi(\tilde{h}'_1) = \varphi(\tilde{h}'_2)$. Hence $\tilde{h}'_1 = \tilde{h}'_2$, since the restriction of φ to $\tilde{H}'(t)$ is injective. This completes the proof of (c).

Let $\tilde{\mathfrak{N}}$ be the group generated by the symbols $\tilde{w}_\alpha$ $(\alpha \in \Delta)$ subject to the relations (W 1) and (W 2), and $\tilde{\mathfrak{H}}$ its subgroup generated by the elements $\tilde{h}_\alpha = \tilde{w}_\alpha^2$. According to Lemma 6.5, (c), we can then form a group N^* which is a semi-direct product of $\tilde{H}$ by $\tilde{\mathfrak{N}}$, where $\tilde{\mathfrak{N}}$ acts on $\tilde{H}$ such that $\tilde{w}_\alpha\tilde{h}\tilde{w}_\alpha^{-1} = \theta_\alpha(\tilde{h})$ for $\alpha \in \Delta$ and $\tilde{h} \in \tilde{H}$. We then have the following lemma:

Lemma 0.31. *6.6. — (a) There exists a morphism j from $\tilde{\mathfrak{H}}$ into $\tilde{H}$ such that $j(\tilde{h}_\alpha) = \tilde{h}_\alpha(-1)$ for all $\alpha \in \Delta$. The elements of the form $\tilde{h}j(\tilde{h})^{-1}$ $(\tilde{h} \in \tilde{\mathfrak{H}})$ constitute a normal subgroup J of N^*.*

(b) Let $\tilde{N}$ be the quotient group of N^* by J. Then $\tilde{H}$ is identified with a normal subgroup of $\tilde{N}$. The group $\tilde{N}$ is a central extension of N by A:

$$1 \to A \to \tilde{N} \xrightarrow{p} N \to 1;$$

where $\varphi(\tilde{h}_\alpha(t)) = h_\alpha(t)$ and $\varphi(p^*(\tilde{w}_\alpha)) = w_\alpha(-1)$ for $\alpha \in \Delta$, p^* denoting the natural morphism from N^* onto $\tilde{N}$. Let

$$\tilde{w}_\alpha(t) = \tilde{h}_\alpha(t)p^*(\tilde{w}_\alpha)^{-1} \text{ for all } \alpha \in \Delta \text{ and } t \in k^\times;$$

then

$$p^*(\tilde{w}_\alpha) = \tilde{w}_\alpha(-1) \quad \text{and} \quad \tilde{w}_\alpha(t)^{-1} = \tilde{w}_\alpha(-t).$$

The existence of the morphism j results from Theorem 6.3, (c) and from the fact that $c^\sigma(-1,-1)^2 = c^\sharp(-1,-1) = 1$. Lemma 6.5, (c) implies that the elements $\tilde{h}j(\tilde{h})^{-1}$ form a normal subgroup of N^*. Statement (b) is then easy to verify.

We are now going to define an extension E of G_k. Let S be the subset of $G_k \times \tilde{N}$ formed of the pairs $[g, \tilde{n}]$ such that $\nu(g) = \varphi(\tilde{n})$, where ν is the map from G_k into N defined in Lemma 5.3, (b). We have a commutative diagram:

$$\begin{array}{ccc} S & \xrightarrow{\tilde{\nu}} & \tilde{N} \\ p\downarrow & & \downarrow\varphi \\ G_k & \xrightarrow{\nu} & N \end{array}$$

where $p([g, \tilde{n}]) = g$ and $\tilde{\nu}([g, \tilde{n}]) = \tilde{n}$. Let us define transformations of S as follows, $\lambda(\tilde{h}), \lambda(u)$ and λ_α $(\tilde{h} \in \tilde{H}, u \in U^+, \alpha \in \Delta)$:

(i) For all $\tilde{h} \in \tilde{H}$, $\lambda(\tilde{h})[g, \tilde{n}] = [\varphi(\tilde{h})g, \tilde{h}\,\tilde{n}]$;

(ii) For all $u \in U^+$, $\lambda(u)[g, \tilde{n}] = [ug, \tilde{n}]$;

(iii) For all $\alpha \in \Delta$, $\lambda_\alpha[g, \tilde{n}]$ is equal to

$$[w_\alpha(-1)g, \tilde{w}_\alpha(-1)\tilde{n}] \quad \text{or to} \quad [w_\alpha(-1)g, \tilde{h}_\alpha(t^{-1})\tilde{n}],$$

according to whether $\nu(w_\alpha(-1)g)$ is equal to $w_\alpha(-1)\nu(g)$ or to $h_\alpha(t^{-1})\nu(g)$.

Let E be the group of transformations of S generated by $\lambda(\tilde{H}), \lambda(U^+)$ and the λ_α ($\alpha \in \Delta$). The map λ gives an isomorphism of $\tilde{H}$ [resp. U^+] onto $\lambda(\tilde{H})$ [resp. $\lambda(U^+)$], and $\lambda(A)$ is a central subgroup of E. Consequently, E operates on the set of orbits of $\lambda(A)$, which identifies with G_k via p. The elements of E thus act on G_k as left translations; hence a canonical morphism π from E onto G_k. Our task henceforth will consist of establishing that the kernel of π indeed reduces to $\lambda(A)$.

We are going to see the relations existing between various elements of the group E.

Lemma 0.32. *(a)* $\lambda(\tilde{h})\lambda(u)\lambda(\tilde{h})^{-1} = \lambda(\varphi(\tilde{h})u\varphi(\tilde{h})^{-1})$ *for all* $\tilde{h} \in \tilde{H}$ *and* $u \in U^+$.

(b) Let us set

$$U_\alpha^\sharp = U^+ \cap w_\alpha(1)U^+w_\alpha(1)^{-1} \quad \textit{for} \quad \alpha \in \Delta.$$

Then we have

$$U_\alpha^\sharp = w_\alpha(1)U_\alpha^\sharp w_\alpha(1)^{-1} \quad \textit{and} \quad \lambda_\alpha^{-1}\lambda(u)\lambda_\alpha = \lambda(w_\alpha(1)uw_\alpha(1)^{-1})$$

for all $u \in U_\alpha^\sharp$.

(c) $\lambda_\alpha^2 = \lambda(\tilde{h}_\alpha(-1))$ *for all* $\alpha \in \Delta$.

(d) $\lambda_\alpha^{-1}\lambda(\tilde{h})\lambda_\alpha = \lambda(\tilde{w}_\alpha(1)\tilde{h}\tilde{w}_\alpha(1)^{-1})$ *for all* $\alpha \in \Delta$ *and* $\tilde{h} \in \tilde{H}$.

The first two assertions are immediate. Let us verify (c). If $\nu(w_\alpha(-1)g) = w_\alpha(-1)\nu(g)$, then

$$\nu(w_\alpha(-1)^2g) = w_\alpha(-1)\nu(w_\alpha(-1)g);$$

consequently,

$$\lambda_\alpha^2[g, \tilde{n}] = \lambda_\alpha[\tilde{w}_\alpha(-1)g, \tilde{w}_\alpha(-1)\tilde{n}] = [h_\alpha(-1)g, \tilde{w}_\alpha(-1)^2\tilde{n}].$$

If $\nu(w_\alpha(-1)g) = h_\alpha(t^{-1})\nu(g)$, then

$$\nu(w_\alpha(-1)^2g) = h_\alpha(-t)\nu(w_\alpha(-1)g);$$

consequently,

$$\lambda_\alpha^2[g, \tilde{n}] = \lambda_\alpha[w_\alpha(-1)g, \tilde{h}_\alpha(t^{-1})\tilde{n}] = [h_\alpha(-1)g, \tilde{h}_\alpha(-t)\tilde{h}_\alpha(t^{-1})\tilde{n}].$$

We have therefore, in both cases, $\lambda_\alpha^2[g, \tilde{n}] = [h_\alpha(-1)g, \tilde{h}_\alpha(-1)\tilde{n}]$. Let us add that statement (c) allows us to write $\lambda_\alpha^{-1} : \lambda_\alpha^{-1}[g, \tilde{n}]$ is equal to $[w_\alpha(1)g, \tilde{w}_\alpha(1)\tilde{n}]$ or to $[w_\alpha(1)g, \tilde{h}_\alpha(-t)^{-1}\tilde{n}]$, according to whether $\nu(w_\alpha(1)g)$ is equal to $w_\alpha(1)\nu(g)$ or to $h_\alpha(-t)^{-1}\nu(g)$.

As for the last assertion, we can assume that $\tilde{h}$ is of the form $\tilde{h}_\beta(s)$ ($\beta \in \Delta, s \in k^\times$).

If $\nu(w_\alpha(-1)g) = w_\alpha(-1)\nu(g)$, then we have

$$\lambda_\alpha^{-1}\lambda(\tilde{h})\lambda_\alpha[g, \tilde{n}] = [w_\alpha(1)h_\beta(s)w_\alpha(-1)g, \tilde{w}_\alpha(1)\tilde{h}_\beta(s)\tilde{w}_\alpha(-1)\tilde{n}].$$

If $\nu(w_\alpha(-1)g) = \tilde{h}_\alpha(t^{-1})\nu(g)$, then we have

$$\nu(w_\alpha(1)h_\beta(s)w_\alpha(-1)g) = h_\alpha(t^{-1}s^{\alpha\beta^*})^{-1}\nu(h_\beta(s)w_\alpha(-1)g);$$

consequently,

$$\lambda_\alpha^{-1}\lambda(h)\lambda_\alpha[g, \tilde{n}] = [w_\alpha(1)h_\beta(s)w_\alpha(-1)g, \tilde{h}_\alpha(t^{-1}s^{\alpha\beta^*})^{-1}\tilde{h}_\beta(s)\tilde{h}_\alpha(t^{-1})\tilde{n}].$$

However, we have in $\tilde{H}$

$$\begin{aligned}\tilde{h}_\alpha(t^{-1}s^{\alpha\beta^*})^{-1}\tilde{h}_\beta(s)\tilde{h}_\alpha(t^{-1}) \\ &= \tilde{h}_\beta(s)\tilde{h}_\alpha(s^{-\alpha\beta^*})c_\alpha(ts^{-\alpha\beta^*}, s^{\alpha\beta^*})c_\alpha(t^{-1}s^{\alpha\beta^*}, s^{\alpha\beta^*}) \\ &= \tilde{h}_\beta(s)\tilde{h}_\alpha(s^{-\alpha\beta^*}) = \tilde{w}(1)\tilde{h}_\beta(s)\tilde{w}_\alpha(1)^{-1}.\end{aligned}$$

We therefore conclude that

$$\lambda_\alpha^{-1}\lambda(\tilde{h}_\beta(s))\lambda_\alpha = \lambda(\tilde{w}_\alpha(1)\tilde{h}_\beta(s)\tilde{w}_\alpha(1)^{-1}).$$

We then have the following lemma:

Lemma 0.33. *6.8. — (a) If G is of rank 1, E is a central extension of G_k by $\lambda(A)$. The Steinberg cocycle c_E of E is equal to $\lambda \circ c$, where c is the Steinberg cocycle initially given.*

(b) If G is of rank 2, the two elements λ_α and λ_β satisfy the relations (W 1) and (W 2) of Lemma 6.1.

This lemma will be proven in the following section. Let us note however here that it is during this proof that the non-multiplicative condition (S 3) of Steinberg cocycles truly intervenes.

Now let Δ' be a subset of Δ and G'_k be the subgroup of G_k generated by the root groups U^β and $U^{-\beta}$ ($\beta \in \Delta'$). Let us set

$$U'^+ = G'_k \cap U^+, \quad H' = G'_k \cap H, \quad N' = G'_k \cap N \quad \text{and} \quad \tilde{H}' = \varphi^{-1}(H'),$$

$$\tilde{N}' = \varphi^{-1}(N'), \quad S' = p^{-1}(G'_k).$$

We then have the following lemma:

Lemma 0.34. *6.9. — Let E' be the subgroup of E generated by $\lambda(\tilde{H}')$, $\lambda(U'^+)$ and the elements λ_β ($\beta \in \Delta'$). Then E' leaves S' stable, and, if an element of E' acts trivially on S', it is the identity element.*

Proof. It is clear that E' leaves S' stable. Consider an arbitrary coset $G'_k g$ of G_k modulo G'_k. According to Lemma 6.2, (a) and (c), there exists in $G'_k g$ an element g_1 such that $\nu(g'g_1) = \nu(g')\nu(g_1)$ for all $g' \in G'_k$. We then choose an element $\tilde{n}_1$ of $\tilde{N}$ such that $\varphi(\tilde{n}_1) = \nu(g_1)$ and define a map $\tau = \tau(g_1, \tilde{n}_1)$ from S' to S by $\tau([g', \tilde{n}']) = [g'g_1, \tilde{n}'\tilde{n}_1]$; the image of τ, which is the orbit of E' passing through $[g_1, \tilde{n}_1]$, is stable under E'. One then sees from the very definition of the initial generators of E, that $\epsilon \circ \tau = \tau \circ \epsilon$ on S' for the generators ϵ of E', namely $\lambda(\tilde{H}')$, $\lambda(U'^+)$ and the λ_β ($\beta \in \Delta'$). The equality $\epsilon \circ \tau = \tau \circ \epsilon$ thus holds for every element ϵ of E'. Consequently, if an element of E' acts trivially on S', it also acts trivially on $\tau(S')$. Since $\tau(S')$ is an arbitrary orbit of E' in S, this proves our assertion. □

The following lemma is a consequence of the two preceding ones:

Lemma 0.35. *6.10. — (a) If we take for Δ' an element α of Δ, the subgroup E' of E is a central extension of G'_k by $\lambda(A)$ and $\lambda \circ c_\alpha$ is the Steinberg cocycle at α of E'.*

(b) The elements λ_α ($\alpha \in \Delta$) of E satisfy the relations (W 1) and (W 2).

Indeed, if we take for Δ' one or two elements of Δ, the action of the group E' on S' can be regarded either as one of the cases of rank ≤ 2, or as a direct product of two cases of rank 1, cases which we have studied in Lemma 6.8. Our assertions thus follow from Lemmas 6.8 and 6.9.

On arrive finalement au résultat suivant :

Lemma 0.36. *6.11. — (a) The isomorphism λ from $\tilde{H}$ onto $\lambda(\tilde{H})$ extends to a monomorphism λ from $\tilde{N}$ into E such that $\lambda(\tilde{w}_\alpha(-1)) = \lambda_\alpha$ for every $\alpha \in \Delta$. The quotient of $\lambda(\tilde{N})$ by $\lambda(\tilde{H})$ is isomorphic to W.*

Theorem 0.37. *(b) Let λ_σ be a representative in $\lambda(\tilde{N})$ of an element σ of W. Then, if α and $\sigma\alpha$ are positive roots, we have in E*

$$\lambda_\sigma\lambda(U^\alpha)\lambda_\sigma^{-1} = \lambda(U^{\sigma\alpha}).$$

(c) The group E is the union of the subsets $\lambda(U^+)\lambda(\tilde{H})\lambda_\sigma\lambda(U^+)$, where $\{\lambda_\sigma\}$ is a system of representatives of W in $\lambda(\tilde{N})$.

(d) The kernel of the canonical morphism π of E onto G_k reduces to $\lambda(A)$. For each $\alpha \in \Delta$, the Steinberg cocycle at α of the central extension E is equal to $\lambda \circ c_\alpha$.

Proof. Let us denote by Λ the subgroup of E generated by the elements λ_α for $\alpha \in \Delta$. The quotient of Λ by $\Lambda \cap \lambda(\tilde{H})$ is isomorphic to W, by virtue of Theorem 6.3 and Lemma 6.10, (b). Our first statement then follows from Lemma 6.7, (d). As for statement (b), if $l(\sigma) = 1$, it follows immediately from Lemma 6.7, (a) and (b). The general case follows from there, according to Lemma 6.2, (b), by induction on $l(\sigma)$. One can now verify assertion (c) by a standard procedure. Indeed, it suffices to show that this union of subsets of E is stable under left multiplication by the initial generators of E. This is evident for $\lambda(\tilde{h})$ and $\lambda(u)$ ($\tilde{h} \in \tilde{H}$, $u \in U^+$). Let α be an element of Δ and σ an element of W. First, the set $\lambda_\alpha\lambda(U^+)\lambda(\tilde{H})\lambda_\sigma\lambda(U^+)$ is identical to $\lambda(U^*_\alpha)\lambda(\tilde{H})\lambda_\alpha\lambda(U^\alpha)\lambda_\sigma\lambda(U^+)$ [3]

$$\lambda(U^\alpha)\lambda_\sigma = \lambda_\sigma\lambda(U^\beta), \quad \text{where } \beta = \sigma^{-1}\alpha;$$

$\lambda_\alpha\lambda(U^+)\lambda(\tilde{H})\lambda_\sigma\lambda(U^+)$ is then contained in $\lambda(U^+)\lambda(\tilde{H})\lambda_{\sigma'}\lambda(U^+)$ for $\sigma' = \sigma_\alpha\sigma$. If β is not positive, then $\sigma'^{-1}\alpha$ is positive; on the other hand, $\lambda_\alpha\lambda(U^\alpha)\lambda_\alpha^{-1}$ is contained in the union of $\lambda(U^\alpha)\lambda(\tilde{H}^\alpha)$ and $\lambda(U^\alpha)\lambda(\tilde{H}^\alpha)\lambda_\alpha\lambda(U^\alpha)$, by virtue of Lemma 6.10, (a); it follows that $\lambda_\alpha\lambda(U^+)\lambda(\tilde{H})\lambda_\sigma\lambda(U^+)$ is contained in the union of $\lambda(U^+)\lambda(\tilde{H})\lambda_\sigma\lambda(U^+)$ and $\lambda(U^+)\lambda(\tilde{H})\lambda_{\sigma'}\lambda(U^+)$. Statement (c) is thus proved. One deduces from it immediately that the kernel of π is contained in $\lambda(\tilde{H})$ and thus that it reduces to $\lambda(A)$. Lemma 6.10, (a) leads to the last assertion of (d).

One thus concludes, by identifying via λ the kernel $\lambda(A)$ with A, that E is a central extension of G_k by A and that the Steinberg cocycle c_E of E is the initially given cocycle c.

We have thus completed the proof of Theorem 5.10. □

7.

We will give in this section a proof of Lemma 6.8. We will first introduce another transformation group of S. Let ι be the transformation of S of order 2 defined as follows:

$$^\iota[g, \tilde{n}] = [g^{-1}, \tilde{n}^{-1}] \quad \text{for all } [g, \tilde{n}] \in S.$$

Let E^* denote the transform of E by ι; $E^* = \iota E \iota$. The group E^* operates, like E, transitively on S.

We propose to show that the elements of E commute with those of E^* in the group of transformations of S. Let us set

$$\rho(\tilde{h}) = \iota\lambda(\tilde{h})\iota, \quad \rho(u) = \iota\lambda(u)\iota, \quad \rho_\alpha = \iota\lambda_\alpha\iota \quad \text{for } \tilde{h} \in \tilde{H}, \quad u \in U^+, \quad \alpha \in \Delta.$$

The following lemma will simplify our work.

Lemma 0.38. *7.1. — (a) The elements of $\lambda(\tilde{H})$ or $\lambda(U^+)$ [resp. of $\rho(\tilde{H})$ or $\rho(U^+)$] commute with all elements of E^* [resp. of E].*

(b) Let α, β be elements of Δ. If λ_α commutes with ρ_β, then ρ_α commutes with λ_β.

(c) One has $(\lambda_\alpha^{-1}\rho_\beta\lambda_\alpha\rho_\beta^{-1})\varepsilon = \varepsilon(\lambda_\alpha^{-1}\rho_\beta\lambda_\alpha\rho_\beta^{-1})$ for every element ε of $\lambda(\tilde{H})$, $\lambda(U^+_\alpha)$, $\rho(\tilde{H})$ or $\rho(U^+_\beta)$.

(d) If $\lambda_\alpha^{-1}\rho_\beta\lambda_\alpha\rho_\beta^{-1}$ leaves fixed an element s of S, the same holds for $\lambda_\alpha s$ and $\rho_\beta^{-1}s$.

Proof. One first sees easily that the elements of $\rho(\tilde{H})$ or $\rho(U^+)$ commute with the initial generators of E. Statement (b) is obvious. Regarding statement (c), if ε is an element of $\lambda(\tilde{H})$ or of $\lambda(U^+_\alpha)$, then

$$(\lambda_\alpha\rho_\beta^{-1})\varepsilon(\lambda_\alpha\rho_\beta^{-1})^{-1} = \lambda_\alpha\varepsilon\lambda_\alpha^{-1} = (\rho_\beta^{-1}\lambda_\alpha)\varepsilon(\rho_\beta^{-1}\lambda_\alpha)^{-1},$$

taking into account statement (a). Finally, if $\lambda_\alpha\rho_\beta^{-1}s = \rho_\beta^{-1}\lambda_\alpha s$, then

$$\lambda_\alpha\rho_\beta^{-1}(\lambda_\alpha s) = \lambda_\alpha(\rho_\beta^{-1}\lambda_\alpha s) = \lambda_\alpha^2\rho_\beta^{-1}s = \rho_\beta^{-1}\lambda_\alpha^2 s,$$

according to lemmas 6.7, (c) and 7.1, (a). The same holds for $\rho_\beta^{-1}s$. The lemma is thus proved. □

[3] [[UNCERTAIN: U^*_α — CONFIDENCE: 85

To see that $\lambda_\alpha \rho_\beta^{-1} = \rho_\beta^{-1}\lambda_\alpha$, it is sufficient, according to lemma 7.1, (c), to verify that $\lambda_\alpha \rho_\beta^{-1} s = \rho_\beta^{-1}\lambda_\alpha s$ for every element s of S_1, if S_1 is a subset of S that meets all the orbits of the group $\lambda(U_\alpha^+)\lambda(\tilde{H})\rho(\tilde{H})\rho(U_\beta^+)$. Using the Bruhat decomposition of G_k, one can easily determine the orbits in S of this last group.

It is known, according to lemmas 6.1 and 6.6, that the elements $w_\alpha(1)$ [resp. $\tilde{w}_\alpha(1)$] ($\alpha \in \Delta$) satisfy relations (W 1) and (W 2). Let us then define a system of representatives $\{w(\sigma)\}$ [resp. $\{\tilde{w}(\sigma)\}$] of W in N [resp. $\tilde{N}$] if σ is written as a minimal word $\prod_{i=1}^m \sigma_{\alpha_i}$, where $m = l(\sigma)$ and $\alpha_i \in \Delta$, then

$$w(\sigma) = \prod_{i=1}^m w_{\alpha_i}(1) \qquad \left[\text{resp. } \tilde{w}(\sigma) = \prod_{i=1}^m \tilde{w}_{\alpha_i}(1)\right].$$

This definition is justified by lemma 6.2, (d).

Then, it is easily seen that every orbit in S of the group $\lambda(U_\alpha^+)\lambda(\tilde{H})\rho(\tilde{H})\rho(U_\beta^+)$ contains an element $\mathbf{s}$ of the form

$$[x_\alpha(u)w(\sigma)x_\beta(v), \tilde{w}(\sigma)], \quad \text{where } \sigma \in W \text{ and } u, v \in k.$$

Recall that we only discuss the cases of type A_1, A_2 or C_2.

Lemma 0.39. *(a) If $\sigma\beta \neq \pm\alpha$, then*

$$l(\sigma_\alpha\sigma) - l(\sigma) = l(\sigma_\alpha\sigma\sigma_\beta) - l(\sigma\sigma_\beta) \quad \textit{and} \quad l(\sigma\sigma_\beta) - l(\sigma) = l(\sigma_\alpha\sigma\sigma_\beta) - l(\sigma_\alpha\sigma).$$

(b) If $\sigma\beta = \alpha$, then

$$\sigma\sigma_\beta = \sigma_\alpha\sigma, \quad w(\sigma)x_\beta(v)w(\sigma)^{-1} = x_\alpha(v) \quad \textit{and} \quad \tilde{w}(\sigma)\tilde{h}_\beta(t)\tilde{w}(\sigma)^{-1} = \tilde{h}_\alpha(t)$$

for all $v \in k$, $t \in k^\times$.

(c) If $\sigma\beta = -\alpha$, then

$$\sigma\sigma_\beta = \sigma_\alpha\sigma \quad \textit{and} \quad l(\sigma\sigma_\beta) = l(\sigma) - 1.$$

(d) If c is a Steinberg cocycle on $k^\times$ with values in A, then

$$c(u^{-1}, (v - u^{-1})^{-1}) = c((u - v^{-1})^{-1}, v^{-1})$$

for all $u, v \in k^\times$ such that $1 - uv \neq 0$.

Proof. Lemma 6.2 easily implies statements (a) and (c), which are valid regardless of the type of Δ. Assertion (d) is verified by the same reasoning as that of Proposition 5.5. As for statement (b), it is obvious that $\sigma\sigma_\beta = \sigma_\alpha\sigma$. To see the other assertions of (b), we distinguish several cases; the first case where $\alpha = \beta$ and $\sigma = e$, the second where Δ is of type C_2, $\alpha = \beta$ and $\sigma = \sigma_\gamma\sigma_\alpha\sigma_\gamma$ with the other element γ of Δ, and the third where Δ is of type A_2, $\alpha \neq \beta$ and $\sigma = \sigma_\beta\sigma_\alpha$. In the first case, our assertions are trivial. In the other cases, one easily deduces, by applying Lemma 5.1 to G_k, that $w(\sigma)x_\beta(v)w(\sigma)^{-1} = x_\alpha(v)$. Then, in the second case, we have

$$\begin{aligned}
\tilde{w}(\sigma)\tilde{h}_\alpha(t)\tilde{w}(\sigma)^{-1} &= (\tilde{w}_\gamma(1)\tilde{w}_\alpha(1))\tilde{h}_\alpha(t)\tilde{h}_\gamma(t^{-\gamma\alpha^*})(\tilde{w}_\gamma(1)\tilde{w}_\alpha(1))^{-1} \\
&= \tilde{w}_\gamma(1)\tilde{h}_\alpha(t^{-1})\tilde{h}_\gamma(t^{-\gamma\alpha^*})\tilde{h}_\alpha(t^2)\tilde{w}_\gamma(1)^{-1} \\
&= \tilde{w}_\gamma(1)\tilde{h}_\alpha(t)\tilde{h}_\gamma(t^{-\gamma\alpha^*})\tilde{w}_\gamma(1)^{-1} \\
&= \tilde{h}_\alpha(t)\tilde{h}_\gamma(t^{-\gamma\alpha^*})\tilde{h}_\gamma(t^{\gamma\alpha^*}) \\
&= \tilde{h}_\alpha(t).
\end{aligned}$$

Finally, in the third case, we have

$$\begin{aligned}
\tilde{w}(\sigma)\,\tilde{h}_\beta(t)\,\tilde{w}(\sigma)^{-1} &= \tilde{w}_\beta(1)\,\tilde{h}_\beta(t)\,\tilde{h}_\alpha(1)^4\,\tilde{w}_\beta(1)^{-1} \\
&= \tilde{h}_\beta(t^{-1})\,\tilde{h}_\alpha(t)\,\tilde{h}_\beta(t) \\
&= c_\beta(t,t)\,c_\beta(t^{-1},t)\,\tilde{h}_\alpha(t) \\
&= \tilde{h}_\alpha(t).
\end{aligned}$$

The assertions of statement (b) are thus proved. □

We will now verify that $\lambda_\alpha\rho_\beta^{-1}s = \rho_\beta^{-1}\lambda_\alpha s$ for every element s of the form $[x_\alpha(u)w(\sigma)x_\beta(v), \tilde{w}(\sigma)]$ where $\sigma \in W$ and $u, v \in k$.

If $\sigma\beta \neq \pm\alpha$, then we verify, according to lemma 7.2, (a), that

$$\nu(g)^{-1}\nu(gw_\beta(-1)) = \nu(w_\alpha(-1)g)^{-1}\nu(w_\alpha(-1)gw_\beta(-1))$$

and

$$\nu(w_\alpha(-1)g)\nu(g)^{-1} = \nu(w_\alpha(-1)gw_\beta(-1))\nu(gw_\beta(-1))^{-1}$$

for $g = x_\alpha(u)w(\sigma)x_\beta(v)$; which implies, given the definition of λ_α and ρ_β^{-1}, that $\lambda_\alpha\rho_\beta^{-1}s = \rho_\beta^{-1}\lambda_\alpha s$.

If $\sigma\beta = \alpha$, one can suppose $v = 0$, according to lemma 7.2, (b). Then, for $g = x_\alpha(u)w(\sigma)$ with $u \neq 0$, we have

$$\nu(gw_\beta(-1)) = \nu(g)w_\beta(-1) \quad \text{and} \quad \nu(w_\alpha(-1)gw_\beta(-1)) = h_\alpha(u^{-1})\nu(gw_\beta(-1));$$

whence

$$\begin{aligned}\tilde{\nu}(\lambda_\alpha\rho_\beta^{-1}[g, \tilde{w}(\sigma)]) &= \tilde{h}_\alpha(u^{-1})\tilde{w}(\sigma)\tilde{w}_\beta(-1) = \tilde{h}_\alpha(u^{-1})\tilde{w}(\sigma)\tilde{h}_\beta(-1)\tilde{w}_\beta(1)\\ &= \tilde{h}_\alpha(u^{-1})\tilde{h}_\alpha(-1)\tilde{w}(\sigma\sigma_\beta);\end{aligned}$$

on the other hand, we have in an analogous way,

$$\begin{aligned}\tilde{\nu}(\rho_\beta^{-1}\lambda_\alpha[g, \tilde{w}(\sigma)]) &= \tilde{w}_\alpha(-1)\tilde{w}(\sigma)\tilde{h}_\beta(u) = \tilde{h}_\alpha(-1)\tilde{w}_\alpha(1)\tilde{h}_\alpha(u)\tilde{w}(\sigma)\\ &= \tilde{h}_\alpha(-1)\tilde{h}_\alpha(u^{-1})\tilde{w}(\sigma_\alpha\sigma).\end{aligned}$$

Lemma 7.2, (b) thus shows that $\lambda_\alpha\rho_\beta^{-1}s = \rho_\beta^{-1}\lambda_\alpha s$.

Suppose finally that $\sigma\beta = -\alpha$. If $uv(1 - uv) = 0$, we reduce to the previous case by lemma 7.1, (d). If $uv(1 - uv) \neq 0$, then, for $g = x_\alpha(u)w(\sigma)x_\beta(v)$, we have

$$\begin{aligned}gw_\beta(-1) &= x_\alpha(u)w(\sigma\sigma_\beta)x_\beta(-v^{-1})w_\beta(1)h_\beta(v)x_\beta(-v^{-1})\\ &\equiv x_\alpha(u - v^{-1})w(\sigma)h_\beta(v)x_\beta(-v^{-1});\end{aligned}$$

whence

$$\begin{aligned}\tilde{\nu}(\lambda_\alpha\rho_\beta^{-1}[g, \tilde{w}(\sigma)]) &= \tilde{h}_\alpha((u - v^{-1})^{-1})\tilde{w}(\sigma)\tilde{h}_\beta(v)\\ &= \tilde{h}_\alpha((u - v^{-1})^{-1})\tilde{h}_\alpha(v^{-1})\tilde{w}(\sigma);\end{aligned}$$

on the other hand, we have in an analogous way,

$$\begin{aligned}\tilde{\nu}(\rho_\beta^{-1}\lambda_\alpha[g, \tilde{w}(\sigma)]) &= \tilde{h}_\alpha(u^{-1})\tilde{w}(\sigma)\tilde{h}_\beta(v - u^{-1})\\ &= \tilde{h}_\alpha(u^{-1})\tilde{h}_\alpha((v - u^{-1})^{-1})\tilde{w}(\sigma).\end{aligned}$$

Lemma 7.2 thus implies that $\lambda_\alpha\rho_\beta^{-1}s = \rho_\beta^{-1}\lambda_\alpha s$.

We therefore know that the elements λ_α of E commute with the elements ρ_β of E^*. It follows, taking lemma 7.1, (a) into account, that every element of E commutes with the elements of E^*. Since E and E^* are transitive on S, we conclude that E is simply transitive on S.

Then, as the elements $\tilde{w}_\alpha(-1)$ $(\alpha \in \Delta)$ satisfy the relations $(W1)$ and $(W2)$, we see immediately that the same holds for the elements λ_α of E.

In the case of rank 1, since E is simply transitive on S, the Bruhat decomposition of G_k indicates that E is the union of $\lambda(U^+)\lambda(\tilde{H})$ and of $\lambda(U^+)\lambda(\tilde{H})\lambda_\alpha\lambda(U^+)$. One immediately concludes that the kernel of the canonical morphism π from E onto G_k is reduced to $\lambda(A)$. The group E is thus a central extension of G_k by $\lambda(A)$. By setting

$$\xi_\alpha(u) = \lambda(x_\alpha(u)) \quad \text{and} \quad \xi_{-\alpha}(u) = \lambda_\alpha\xi_\alpha(-u)\lambda_\alpha^{-1},$$

one easily verifies that

$$\omega_\alpha(t) = \xi_\alpha(t)\xi_{-\alpha}(-t^{-1})\xi_\alpha(t) = \lambda(\tilde{h}_\alpha(t))\lambda_\alpha^{-1}$$

for all $t \in k^\times$. Now, according to lemmas 6.7, (a), 5.2, (c) and 5.1, ξ_α and $\xi_{-\alpha}$ give the isomorphisms from k into $\pi^{-1}(U^\alpha)$ and $\pi^{-1}(U^{-\alpha})$ which serve to define the Steinberg cocycle c_E at α of E relative to the isomorphisms x_α and $x_{-\alpha}$. One concludes that $c_E = \lambda \circ c_\alpha = \lambda \circ c$, where c is the Steinberg cocycle initially given.

We have thus completed the proof of Lemma 6.8.

Let us add that Lemma 7.2, (b) is in fact valid, with the notation of no. 5, regardless of the type of G and regardless of the central extension E of G_k.

8.

In this section, we will study the additional conditions to be imposed in the topological case on Steinberg cocycles.

Let k be an infinite topological field and A a topological abelian group, which are assumed, of course, to be Hausdorff. The case which interests us particularly is that where k and A are locally compact.

The group G_k is then endowed with a natural structure of a topological group. The subgroups H_k, N_k, U_k^+, U_k^-, U_k^α ($\alpha \in \Phi$) are all closed in G_k. The map $(u, h, u') \mapsto uhu'$ from $U_k^+ \times H_k \times U_k^-$ into G_k induces a homeomorphism from the product space onto an open subset Ω_k of G_k.

Let E be a central extension of G_k by A. By this we mean the data of an exact sequence: $1 \to A \xrightarrow{i} E \xrightarrow{\pi} G_k \to 1$, where i is an isomorphism from A onto a closed central subgroup of E and where π induces an isomorphism from $E/i(A)$ onto G_k.

We are going to consider, as in no. 5, the liftings $\tilde{U}^\alpha$ of the root groups U_k^α ($\alpha \in \Phi$), the maps $\tilde{x}_\alpha$, $\tilde{w}_\alpha$, $\tilde{h}_\alpha$ and the subgroups $\tilde{U}^+$, $\tilde{U}^-$, $\tilde{H}$ and $\tilde{N}$ of E.

We have the following lemma:

Lemma 0.40. *8.1. — (a) For every $\alpha \in \Phi$, $\tilde{x}_\alpha$ is an isomorphism from k onto the closed subgroup $\tilde{U}^\alpha$ of E. The group $\tilde{U}^+$ [resp. $\tilde{U}^-$] is a closed subgroup of E, from which π induces an isomorphism onto U_k^+ [resp. U_k^-].*

(b) For every $\alpha \in \Phi$, $\tilde{h}_\alpha$ is a homeomorphism from $k^\times$ onto a closed subset of E.

(c) For every $\alpha \in \Phi$, the Steinberg cocycle c_α at α is a continuous map from $k^\times \times k^\times$ into A.

(d) The map $(u, h, u') \mapsto uhu'$ defines a homeomorphism from $\tilde{U}^+ \times \tilde{H} \times \tilde{U}^-$ onto an open set Ω_E of E.

(e) When we agree that $c_\alpha(0, 1) = c_\alpha(1, 0) = 1$, the function $c_\alpha(u, 1 - uv)$ of (u, v), defined on a neighborhood of $(0, 0)$ in $k \times k$, is continuous at $(0, 0)$.

Proof. Let us first verify statement (a). According to Lemma 5.2, (c), we have

$$\tilde{x}_\alpha(u(t^2 - 1)) = \tilde{h}_\alpha(t)s(x_\alpha(u))\tilde{h}_\alpha(t)^{-1}s(x_\alpha(u))^{-1},$$

where t is a fixed element of $k^\times$ such that $t^2 \neq 1$ and where s is any lift of U_α^k into E. Let $\mathfrak{V}$ be a neighborhood of e in E. There then exists an open neighborhood $\mathfrak{V}'$ of e in E such that

$$\tilde{h}_\alpha(t)\tilde{z}\tilde{h}_\alpha(t)^{-1}\tilde{z}^{-1} \in \mathfrak{V} \text{ for every } \tilde{z} \in \mathfrak{V}'.$$

The image $\pi(\mathfrak{V}')$, being an open set of G_k, contains $x_\alpha(V^0)$ for a neighborhood V^0 of 0 in k. Consequently, if u is in V^0, the element $\tilde{x}_\alpha(u(t^2 - 1))$ belongs to $\mathfrak{V}$. This shows that the isomorphism $\tilde{x}_\alpha$ is continuous at 0, and therefore on the whole of k. Given the epimorphism π, then $\tilde{x}_\alpha$ induces an isomorphism from k onto a closed subgroup of E. From this, the assertion on $\tilde{U}^+$ and $\tilde{U}^-$ and statements (b), (c), and (d) follow easily. We have a continuous map $(u, v) \mapsto \tilde{x}_{-\alpha}(u)\tilde{x}_\alpha(v)$ from $k \times k$ into E. The element $\tilde{x}_{-\alpha}(u)\tilde{x}_\alpha(v)$ is in $\tilde{U}^+\tilde{H}\tilde{U}^-$ if $1 + uv \neq 0$, and according to Lemma 5.9, we have

$$\begin{aligned}\tilde{x}_{-\alpha}(u)\tilde{x}_\alpha(v) = {} & \tilde{x}_\alpha(v(1 + uv)^{-1})\tilde{h}_\alpha((1 + uv)^{-1}) \\ & \times [c_\alpha(-u, 1 + uv)c_\alpha(1 + uv, -1)^{-1}]\tilde{x}_{-\alpha}(u(1 + uv)^{-1});\end{aligned}$$

with our convention, this formula is valid as soon as $1+uv \neq 0$. Continuity then implies that $c_\alpha(-u, 1+uv)$ tends to 1 when (u, v) tends to $(0, 0)$. Hence assertion (e). □

We say in this section that a Steinberg cocycle c on $k^\times$ with values in A is **topological**, if it satisfies the following condition:

(S0) *The function $c(x,y)$ is continuous in $k^\times \times k^\times$, and, when we agree that $c(1,0) = c(0,1) = 1$, the function $c(x, 1-xy)$ of (x,y), defined on a neighborhood of $(0,0)$ in $k \times k$, is continuous at $(0,0)$.*

Let us denote, provisionally, $St(k^\times, A)$ the group of topological Steinberg cocycles on $k^\times$ with values in A. Since a Steinberg cocycle c satisfies the identity $c(1-uv, u) = c(v, 1-uv)$, if c belongs to $St(k^\times, A)$, the same holds for $\check{c}$ and $c^\natural$, with the notation of Proposition 5.7.

We are going to prove the topological version of Theorem 5.10.

Theorem 0.41. *8.2. — Let k be an infinite topological field, A a topological abelian group and G_k the topological group of rational points of a simple simply connected group G split over k. Then, the classes of central extensions of G_k by A are in one-to-one correspondence with the topological Steinberg cocycles (and bilinear if G is of non-symplectic type) on $k^\times$ with values in A. The multiplicative group of the latter is isomorphic to the group $H^2(G_k, A)$ formed by the former with Baer multiplication.*

Proof. Indeed, according to Theorem 5.10 and Lemma 8.1, one obtains a homomorphism from $H^2(G_k, A)$ into $St(k^\times, A)$, by associating to a central extension E of G_k by A the Steinberg cocycle c_E belonging to E. First, this homomorphism is injective. Indeed, according to Lemma 8.1, if c_E is trivial, the root groups $\tilde{U}^\alpha$ generate a closed subgroup of E which is isomorphic to G_k via π.

Given Theorem 5.10, it remains for us to show, given a central extension E of G_k by A in the abstract sense, that if c_E satisfies condition (S0), E can be equipped with a topology that makes it an extension of G_k by A in the topological sense.

First, we note that, according to Lemma 5.8 and the remark preceding our theorem, the Steinberg cocycles c_α in α ($\alpha \in \Phi$) all satisfy condition (S0). Let us equip $\tilde{H}$ with a topology by saying that the map

$$\left(\prod_{\alpha\in\Delta} h_\alpha(t_\alpha), a\right) \mapsto \left(\prod_{\alpha\in\Delta} \tilde{h}_\alpha(t_\alpha)\right) a$$

is a homeomorphism from $H_k \times A$ onto $\tilde{H}$. This topology makes $\tilde{H}$ a topological extension of H_k, since the cocycles c_α are continuous in $k^\times \times k^\times$. On the other hand, we transport onto $\tilde{U}^+$ [resp. $\tilde{U}^-$] the topology of U_k^+ [resp. of U_k^-] by means of the morphism π from E onto G_k.

Next, let us consider the subset $\Omega_E = \tilde{U}^+\tilde{H}\tilde{U}^-$ of E, which is equipped, by this decomposition, with the product topology. One then defines the neighborhoods of the identity element e in E as the subsets of E intersecting Ω_E along a neighborhood of e in Ω_E. We are going to show that these data at the origin make E a topological group, by verifying that the map $(x,y) \mapsto x^{-1}y$ is continuous at (e,e) and that $\mathrm{Ad}(z)$ and $\mathrm{Ad}(z^{-1})$ are continuous at (e,e) for every element z of a generating set of E. In the rank 1 case, this is verified immediately, taking into account Lemma 5.9 and taking as generators A, $\tilde{U}^\alpha$, and $\tilde{w}_\alpha(1)$. In the general case, one sees immediately that $(x,y) \mapsto x^{-1}y$ is continuous at (e,e) □

for $x \in \tilde{U}^+\tilde{H}$ and $y \in E$. On the other hand, by reducing to the rank 1 case, one verifies that $\mathrm{Ad}(z)$ is continuous at e for $z \in \tilde{U}^\alpha$ or $z \in \tilde{U}^{-\alpha}$, α being an element of Δ; consequently, $\mathrm{Ad}(z)$ is continuous at e for every element z of E. Let β be an element of Φ; there then exists an element n of $\tilde{N}$ such that $\tilde{U}^\beta = n\tilde{U}^\alpha n^{-1}$ with a positive root α; the identity $(nxn^{-1})^{-1}y = n(x^{-1}(n^{-1}yn))n^{-1}$ shows that $(x,y) \mapsto x^{-1}y$ is continuous at (e,e) for $x \in \tilde{U}^\beta$ and $y \in E$. Taking into account the decomposition of Ω_k, this immediately implies that $(x,y) \mapsto x^{-1}y$ is continuous at (e,e). Thus E is a topological group. It is now clear that A is a closed subgroup of E and that π induces an isomorphism from E/A onto G_k.

This completes the proof of Theorem 8.2.

9. Examples of Steinberg cocycles

In what follows, k is a topological field and A a topological abelian group, and by Steinberg cocycle one shall mean a topological Steinberg cocycle. We will denote by $S(k^\times, A)$ the group of Steinberg cocycles on $k^\times$ with values in A.

First, let us note some obvious functorial properties of the groups $S(k^\times, A)$.

a. If j is a continuous embedding of a topological field k_1 into k, we have a homomorphism j^* from $S(k^\times, A)$ to $S(k_1^\times, A)$, defined by

$$j^*(c)(x, y) = c(j(x), j(y)) \quad \text{for } c \in S(k^\times, A) \text{ and } x, y \in k_1^\times.$$

b. Let B be a topological abelian group and $\mathrm{Hom}(A, B)$ the group of continuous morphisms from A to B. For $c \in S(k^\times, A)$ and $\lambda \in \mathrm{Hom}(A, B)$, the composite $\lambda \circ c$ is a Steinberg cocycle on $k^\times$ with values in B. If we regard $S(k^\times, A)$, $S(k^\times, B)$ and $\mathrm{Hom}(A, B)$ as $\mathbf{Z}$-modules, we thus obtain a homomorphism of the tensor product $S(k^\times, A) \otimes_{\mathbf{Z}} \mathrm{Hom}(A, B)$ into $S(k^\times, B)$.

The theory of simple algebras, combined with the theory of Kummer fields, provides an example of a bilinear Steinberg cocycle. Suppose that k possesses n distinct n-th roots of unity, and let $B(k)$ denote the Brauer group of k and $B(k)_n$ its subgroup formed by elements of order dividing n. Then, choosing a primitive n-th root of 1 in k, one defines the Hilbert symbol $\{x, y\}_n$ (cf. A. Weil [33], p. 185); it is a bilinear map from $k^\times \times k^\times$ into $B(k)_n$, which satisfies $\{x, 1-x\}_n = 1$ for $x \in k^\times, x \neq 1$. The symbol $\{x, y\}_n$ therefore satisfies the equations of a bilinear Steinberg cocycle. For $\{x, y\}_n$ to satisfy the topological condition (S0), $B(k)_n$ being a discrete group, it suffices, for example, that the set $(k^\times)^n$ formed by the x^n $(x \in k^\times)$ be an open subset of $k^\times$.

Another example of a Steinberg cocycle, this one non-bilinear, comes from the theory of ordered fields (cf. N. Bourbaki [7]). Let k be an ordered field, archimedean or not. Let us define a map c from $k^\times \times k^\times$ into $\mathbf{Z}$ as follows: $c(x, y) = 1$ if $x < 0, y < 0$; $c(x, y) = 0$ otherwise. One sees that c is a Steinberg cocycle on $k^\times$ for the topology of k defined from the total order on k. From this, one deduces, by passing to the quotient, a bilinear Steinberg cocycle c^0 on $k^\times$ with values in $\mathbf{Z}/2\mathbf{Z}$. In fact, c^0 comes from a Hilbert symbol. Indeed, let $\bar{k}$ be an algebraic closure of k and k_1 a maximal ordered extension field of k contained in $\bar{k}$; we then have $\bar{k} = k_1(\sqrt{-1})$, and the Brauer group $B(k_1)$ of k_1 is of order 2. Moreover, k_1 possesses exactly two roots of 1, and one has a Hilbert symbol $\{x, y\}_{k_1}$ on $k_1^\times$ with values in $B(k_1)$. The cocycle c^0 is none other than the restriction of $\{x, y\}_{k_1}$ to $k^\times \times k^\times$, $B(k_1)$ being identified with $\mathbf{Z}/2\mathbf{Z}$.

We will explain what these examples give in the case of a non-discrete locally compact field.

If $k = \mathbf{R}$, the group G_k is a connected Lie group. Since k is an ordered field, G_k possesses a central extension E by $\mathbf{Z}$ [resp. $\mathbf{Z}/2\mathbf{Z}$] if G is of symplectic type [resp. non-symplectic]. The group E is locally isomorphic to G_k and is generated by one-parameter subgroups. Consequently, E is a connected Lie group. It is known that E is the universal covering of the connected Lie group G_k (É. Cartan [11]).

Let k be an ultrametric local field. The roots of 1 in k form a finite cyclic group μ_k, say, of order m; the set $(k^\times)^m$ is then an open subset of $k^\times$. According to local class field theory ([1], [27], [33]), the group $B(k)_m$ is cyclic of order m, and one can replace the Hilbert symbol $\{x, y\}_m$ by the norm residue symbol $(x, y)_{m,k}$, which is a bilinear map from $k^\times \times k^\times$ into μ_k canonically defined (see no. 11). It is also known that $(x, y)_{m,k}$ has the entire μ_k as its image. We thus have the canonical cocycle $(x, y)_{m,k}$ on $k^\times$ with values in μ_k, and, by Theorem 8.2, a central extension E_0 of G_k by μ_k. The group E_0 is locally isomorphic to G_k, and is equal to its own commutator subgroup; one could therefore say that E_0 is an m-sheeted covering of G_k. Let n be a divisor of m and $\mu_{k,n}$ the subgroup of μ_k formed by elements of order dividing n. Then, by passing to the power, we have the cocycle $(x, y)_{n,k} = (x, y)_{m,k}^{m/n}$ on $k^\times$ with values in $\mu_{k,n}$, and therefore a central extension of G_k by $\mu_{k,n}$.

A. Weil [32] noted the existence of such coverings for symplectic groups and $n = 2$. The existence of the covering E_0 was proven, for $G = SL_2$, independently by T. Kubota [19] and C. Moore [26], and the latter conjectured it in the general case. H. Bass, J. Milnor and J.-P. Serre [4] proved this conjecture for the groups SL_n and Sp_{2n} in characteristic 0.

CHAPTER III.
RECALL OF RESULTS OF C. MOORE AND CONCLUSION.

10. In this section, we present, according to C. Moore ([25], [26]), some generalities on central extensions of locally compact groups. All the statements in this section are essentially proven in these two memoirs.

In what follows, we will assume that the locally compact groups involved are all *separable*, that is, they have a countable basis of open sets.

Let G be a locally compact group and A a locally compact G-module. C. Moore [25] defined the cohomology groups $H^n(G, A)$ $(n \geq 0)$ in the topological sense, which have a usual interpretation in low dimensions. We are interested exclusively in the case where G acts on A trivially. In that case, $H^1(G, A)$ is the group of morphisms from G to A, and $H^2(G, A)$ is the group formed, with the Baer multiplication, by the classes of central extensions of G by A. By a central extension of G by A, we mean the datum of an exact sequence of locally compact groups

$$1 \to A \xrightarrow{i} E \xrightarrow{p} G \to 1,$$

where i is an isomorphism of A onto a central subgroup of E and where p induces an isomorphism of $E/i(A)$ onto G.

A locally compact group G is said to be *simply connected* if G satisfies the following condition: for every central extension E of G with a locally compact kernel, there exists one and only one morphism φ from G into E such that $p \circ \varphi = \mathrm{id}_G$, where p is the projection from E onto G and id_G is the identity map of G.

We then have the following criterion (cf. [26]):

Lemma 0.42. *10.1. — A locally compact group G is simply connected if and only if one has $H^1(G, \mathbf{T}) = H^2(G, \mathbf{T}) = \{0\}$, where $\mathbf{T}$ is the multiplicative group of complex numbers of modulus 1.*

Suppose that the group $[G, G]$ of commutators of G is dense in G, in other words that $H^1(G, \mathbf{T}) = \{0\}$. A central extension E of G will be called a *covering* of G if the group of commutators of E is dense in E. A covering E of G will be called *universal* if the group E is simply connected. The group G possesses, up to an isomorphism of extensions, at most one universal covering. The kernel of a universal covering of G will be called the *fundamental group* of G and will be denoted $\pi_1(G)$.

The following lemma gives a fundamental property of the universal covering [26]:

Lemma 0.43. *10.2. — Suppose that G possesses a universal covering E_0:*

$$1 \to \pi_1(G) \to E_0 \xrightarrow{\pi} G \to 1.$$

Then:

(a) For every central extension E of G, there exists a unique morphism φ from E_0 into E such that $\pi = p \circ \varphi$, where p is the projection of E onto G.

(b) If A is a locally compact abelian group, $H^2(G, A)$ is isomorphic to the group $\mathrm{Hom}(\pi_1(G), A)$ of morphisms from $\pi_1(G)$ into A.

In particular, if G possesses a universal covering, the group $H^2(G, \mathbf{T})$ can be equipped with a topology such that it is the dual of $\pi_1(G)$.

Let j be a morphism from a locally compact group H into another G. Then, to every central extension E of G by A, one associates, according to j, a central extension $j^*(E)$ of H by A; one thus obtains a morphism j^* from $H^2(G, A)$ into $H^2(H, A)$. The extension $j^*(E)$ of H is trivial if and only if there exists a morphism φ^* from H into E such that $j = p \circ \varphi^*$, where p is the projection of E onto G.

Let us recall some results on the existence of universal coverings for certain locally compact groups [26].

If G is a discrete group such that $G = [G, G]$, there exists a universal covering of G, which is itself a discrete group.

In order for a connected Lie group G to be simply connected in our sense, it is necessary and sufficient that G be simply connected as a differentiable manifold and that the cohomology groups $H^1(\mathfrak{g}, \mathbf{R})$ and $H^2(\mathfrak{g}, \mathbf{R})$ vanish, where $\mathfrak{g}$ is the Lie algebra of G and $\mathbf{R}$ the trivial $\mathfrak{g}$-module. A connected Lie group G possesses a universal covering E if and only if G is equal to its commutator group, that is to say, $\mathfrak{g}$ is equal to its derived algebra. If this is the case, E is itself a connected Lie group and $\pi_1(G)$ is the direct sum of the fundamental group of the manifold underlying G and the dual of $H^2(\mathfrak{g}, \mathbf{R})$.

Let K be a compact group. First, if K is a Lie group, $H^2(K, \mathbf{T})$ is a finite group. If K is a projective limit of compact Lie groups K_α, where α ranges over a countable filtering set, then $H^2(K, \mathbf{T})$

is isomorphic to the inductive limit of the groups $H^2(K_\alpha, \mathbf{T})$. Suppose that $[K, K]$ is dense in K. Then, K possesses a universal covering, which is itself compact; $\pi_1(K)$ is a profinite group, being the dual of $H^2(K, \mathbf{T})$.

Now let j be an injective morphism from a locally compact group H into another G, and suppose that $[G, G]$ is dense in G. A central extension E of G is said to be trivial above H, if $j^*(E)$ ...

is trivial. A covering E_0 of G will be called a *universal relative covering* of G with respect to H if E_0 satisfies the following condition: the extension E_0 is trivial over H and, for every central extension E trivial over H, there exists a morphism φ from E_0 into E such that $p \circ \varphi = \pi$, where π [resp. p] is the projection of E_0 [resp. of E] onto G. There exists, up to an isomorphism of extensions, at most one such covering of G. The kernel of the extension E_0 will be called the *relative fundamental group* of G with respect to H and will be denoted by $\pi_1(G, H)$. Then, if A is a locally compact abelian group, the kernel of the restriction morphism $j^\star$ from $H^2(G, A)$ into $H^2(H, A)$ is isomorphic to $\mathrm{Hom}(\pi_1(G, H), A)$.

Let j be an injective morphism from H into G, and suppose that G and H possess universal coverings E and F respectively. There exists then a morphism $j_\star$ from F into E such that $\pi \circ j_\star = j \circ \pi'$, where π [resp. π'] is the projection of E [resp. of F] onto G [resp. H]. The morphism $j_\star$ maps $\pi_1(H)$ into $\pi_1(G)$; it is the dual of the morphism $j^\star$ from $H^2(G, \mathbf{T})$ into $H^2(H, \mathbf{T})$. Let $\overline{j_\star(\pi_1(H))}$ be the closure of $j_\star(\pi_1(H))$ in $\pi_1(G)$; one obtains, by passing to the quotient, an extension $E/\overline{j_\star(\pi_1(H))}$ of G by $\pi_1(G)/\overline{j_\star(\pi_1(H))}$. This extension is a universal relative covering of G with respect to H [26]. We thus have the following lemma:

Lemma 0.44. *10.3. — Let j be an injective morphism from H into G, and suppose that G and H possess universal coverings. Then, there exists a universal relative covering of G with respect to H, and $\pi_1(G, H)$ is isomorphic to the quotient of $\pi_1(G)$ by $\overline{j_\star(\pi_1(H))}$.*

Let us add that $\pi_1(G, H)$ is a discrete group if H is moreover an open compact subgroup of G.

Finally, let us treat the transition to the restricted product. Let $\{G_\alpha\}$ be a family of locally compact groups, α ranging over a countable set of indices; let K_α be an open compact subgroup of G_α defined for almost all α. Let G denote the restricted product of the groups G_α with respect to the K_α.

Suppose that G_α possesses a universal covering E_α for all α and that $[K_\alpha, K_\alpha]$ is dense in K_α for almost all α. When this last hypothesis holds for α, let L_α be a universal covering of K_α, and $j_\alpha^\star$ be the morphism from L_α into E_α lifted from the embedding j^α of K_α into G_α. The image under $j_\alpha^\star$ of L_α [resp. of $\pi_1(K_\alpha)$] is open and compact in E_α [resp. $\pi_1(G_\alpha)$], and $j_\alpha^\star(L_\alpha)$ is equal to the closure of $[p_\alpha^{-1}(K_\alpha), p_\alpha^{-1}(K_\alpha)]$ in E_α, p_α being the projection of E_α onto G_α. We then have the following result [26]:

Lemma 0.45. *10.4. — The restricted product of the groups E_α with respect to $j_\alpha^\star(L_\alpha)$ is a universal covering of G. The fundamental group $\pi_1(G)$ of G is the restricted product of the groups $\pi_1(G_\alpha)$ with respect to $j_\alpha^\star(\pi_1(K_\alpha))$.*

11. In this section, we recall the arithmetic results of C. Moore [26] on the determination of Steinberg cocycles for a local field and on the uniqueness of reciprocity formulas in a global field.

Let k be a non-discrete locally compact field. We first recall the definition of the norm residue symbol (Artin-Tate [1], Serre [27], Weil [33]). Let $\bar{k}$ be an algebraic closure of k, k_{ab} the maximal abelian extension of k contained in $\bar{k}$ and $\mathfrak{A}$ the Galois group of the extension k_{ab} of k. Let α_k be the canonical morphism from $k^\times$ into $\mathfrak{A}$. Then, if k contains n distinct n-th roots of unity, the Hilbert symbol $(x, y)_{n,k}$ is defined as follows:

$$\text{for all } x, y \in k^\times, \quad (x, y)_{n,k} = \xi^{-1}\xi^{\alpha_k(y)}$$

where ξ is an n-th root of x. This is a bilinear map from $k^\times \times k^\times$ into the group μ_k of roots of unity contained in k, and we have $(x, 1-x)_{n,k} = 1$ for all $x \in k^\times, x \neq 1$.

Since $(k^\times)^n$ is open in $k^\times$, the Hilbert symbol $(x, y)_{n,k}$ is a locally constant map and is a Steinberg cocycle on $k^\times$ with values in μ_k. Let $S(k^\times, \mu_k)$ denote the group of Steinberg cocycles on $k^\times$ with values in μ_k, and $H(k^\times, \mu_k)$ its subgroup generated by the Hilbert symbols. If $k = \mathbf{C}$, $H(k^\times, \mu_k)$ is trivial. If k is the field $\mathbf{R}$ or an ultrametric field, μ_k is cyclic of finite order m; the group $H(k^\times, \mu_k)$ is also of order m and is generated by the symbol $(x, y)_{m,k}$.

In this last case, let $c^0(x, y) = (x, y)_{m,k}$. For $k = \mathbf{R}$, let us define a Steinberg cocycle c_0 on $k^\times$ with values in $\mathbf{Z}$ as follows: $c_0(x, y) = 1$ if $x < 0, y < 0$; $c_0(x, y) = 0$ otherwise.

This being said, the following theorem, due to C. Moore [26], allows for the determination of the Steinberg cocycles on $k^\times$ with values in a locally compact abelian group, whether separable or not.

Theorem 0.46. *11.1. — Let k be a local field and A a locally compact abelian group. Let $S(k^\times, A)$ [resp. $S^0(k^\times, A)$] denote the group of Steinberg cocycles [resp. bilinear Steinberg cocycles] on $k^\times$ with values in A.*

(a) If $k = \mathbf{C}, S(k^\times, A) = \{1\}$.

(b) If $k = \mathbf{R}$, the map $\lambda \mapsto \lambda \circ c_0$ [resp. $\lambda \mapsto \lambda \circ c^0$] is an isomorphism from $Hom(\mathbf{Z}, A)$ [resp. from $Hom(\mu_k, A)$] onto $S(k^\times, A)$ [resp. $S^0(k^\times, A)$].

(c) If k is ultrametric, the map $\lambda \mapsto \lambda \circ c^0$ is an isomorphism from $Hom(\mu_k, A)$ onto $S(k^\times, A)$. Every element of $S(k^\times, A)$ is bilinear.

In fact, Moore shows, without assuming the continuity of $c(x, 1-x)$ at $(x, y) = (0, 0)$, that one thus obtains all continuous maps from $k^\times \times k^\times$ into A satisfying equations (S1), (S2) and (S3) of no. 5.

As far as the proof is concerned, let us limit ourselves to the following indications. The general result is easily deduced from the particular case where $A = \mathbf{T}$. Let us then consider two maps $c \mapsto h(c)$ and $c \mapsto c^\natural$ from $S(k^\times, \mathbf{T})$ respectively into $H^2(k^\times, \mathbf{T})$ and $S^0(k^\times, \mathbf{T})$: $h(c)$ is the cohomology class of c and $c^\natural$ is defined by $c^\natural(x, y) = c(x, y^2)$. These two homomorphisms have as a common kernel the subgroup $S_0(k^\times, \mathbf{T})$ formed by elements c such that $c(x, y^2) = 1$ for all $x, y \in k^\times$. The group $H^2(k^\times, \mathbf{T})$ is trivial for $k = \mathbf{C}$ or $\mathbf{R}$, and is a torsion group for ultrametric k. This immediately entails the theorem for $\mathbf{C}$ and $\mathbf{R}$. In the ultrametric case, one sees from this that $S(k^\times, \mathbf{T})/S_0(k^\times, \mathbf{T})$ is a torsion group. One then must determine $S_0(k^\times, \mathbf{T})$, $S^0(k^\times, \mathbf{T})$ and the image of the morphism $c \mapsto c^\natural$. This is done with the help of essential results from local class field theory (see C. Moore [26]).

Recall also the case of finite fields, due to R. Steinberg [30]:

Lemma 0.47. *11.2. — Let $\mathbf{F}_q$ be a finite field with q elements. If q is odd, $\mathbf{F}_q$ possesses an element x such that neither x nor $1 - x$ is a square in $\mathbf{F}_q^\times$. Every Steinberg cocycle on $\mathbf{F}_q^\times$ is trivial.*

We are now going to consider the adelic case. Let k be a finite extension of $\mathbf{Q}$ or a field of algebraic functions of dimension 1 over a finite field. We denote by μ_k the group of roots of unity contained in k. For each place v of k, we denote by k_v the completion of k at v, and by μ_v the group of roots of unity contained in k_v. If v is non-archimedean, $\mathfrak{o}_v$ denotes the maximal compact subring of k_v and $\mathfrak{o}_v^\times$ the multiplicative group of invertible elements of $\mathfrak{o}_v$.

Let $\mathbf{A}_k^\times$ be the group of ideles of k, that is to say the restricted product of the groups $k_v^\times$ relative to $\mathfrak{o}_v^\times$, v running through the set of places of k. The group $k^\times$ is canonically a discrete subgroup of $\mathbf{A}_k^\times$. It is well known (and moreover results immediately from lemma 11.2) that a Hilbert symbol $(x, y)_{n,k_v}$ of an ultrametric field k_v is trivial on $\mathfrak{o}_v^\times \times \mathfrak{o}_v^\times$ if n is coprime to the residual characteristic p_v of k_v. It is also known that the order of μ_v is coprime to p_v for almost every place v. Consequently, if k contains n distinct n-th roots of unity, one can define on $\mathbf{A}_k^\times \times \mathbf{A}_k^\times$ the product $(x, y)_n$ of local symbols $(x_v, y_v)_{n,k_v}$:

$$(x, y)_n = \prod_v (x_v, y_v)_{n,k_v}$$

for $x = (x_v), y = (y_v)$ in $\mathbf{A}_k^\times$. This is a cocycle, locally constant, on $\mathbf{A}_k^\times$ with values in μ_k. The reciprocity law then asserts that $(x, y)_n = 1$ for all x, y in $k^\times$.

Let us denote by $S(\mathbf{A}_k^\times, \mathbf{T})$ [resp. $S^0(\mathbf{A}_k^\times, \mathbf{T})$] the product of the groups $S(k_v^\times, \mathbf{T})$ [resp. $S^0(k_v^\times, \mathbf{T})$]. Each element $c = (c_v)$ of $S(\mathbf{A}_k^\times, \mathbf{T})$ defines a cocycle, locally constant, on $\mathbf{A}_k^\times$ with values in $\mathbf{T}$. One thus identifies $S(\mathbf{A}_k^\times, \mathbf{T})$ with the group of cocycles on $\mathbf{A}_k^\times$. To the canonical embedding j of $k^\times$ into $\mathbf{A}_k^\times$,

we associate a restriction morphism $j^\star$ from $S(A_k^\times, \mathbf{T})$ into $S(k^\times, \mathbf{T})$, where k is regarded, of course, as a discrete field. By virtue of the reciprocity formula $(x, y)_m = 1$ $(x, y \in k^\times)$, where m is the order of μ_k, we see that the kernel of $j^\star$ contains a subgroup of order m, dual to μ_k.

Theorem 0.48. *11.3. — The kernel of the restriction morphism $j^\star$ from $S(A_k^\times, \mathbf{T})$ into $S(k^\times, \mathbf{T})$ is isomorphic to the dual of the group μ_k of roots of unity contained in k.*

This constitutes, in class field theory, a uniqueness theorem for reciprocity formulas in k.

Now let S be a finite set of places of k, and $A_k^\times(S)$ the group of S-ideles of k, that is to say the restricted product of the groups $k_v^\times$ for the places $v \notin S$. Let $S(A_k^\times(S), \mathbf{T})$ denote the product of the groups $S(k_v^\times, \mathbf{T})$ for the places $v \notin S$. The restriction morphism from $S(A_k^\times(S), \mathbf{T})$ into $S(k^\times, \mathbf{T})$, associated with the canonical embedding of $k^\times$ into $A_k^\times(S)$, is none other than the restriction to $S(A_k^\times(S), \mathbf{T})$ of the morphism $j^\star$ from $S(A_k^\times, \mathbf{T})$ into $S(k^\times, \mathbf{T})$. We thus have the following corollary:

Corollary 0.49. *11.4. — The kernel of the restriction morphism from $S(A_k^\times(S), \mathbf{T})$ into $S(k^\times, \mathbf{T})$ is isomorphic to the dual of μ_k if S consists entirely of imaginary places. Otherwise, it is reduced to the identity element.*

12. This section is devoted to our main theorems for split simply connected simple groups G. It is a matter of gathering the results of the preceding sections, in order to arrive at a solution to the congruence subgroup problem for G of rank ≥ 2.

We resume the notation of Chapters I and II, relating to the Chevalley-Demazure scheme $\mathfrak{G}$ associated with a simply connected simple group G; in particular, the torus H, the root system Φ, the root subgroups U^α, the system of simple roots Δ.

Theorems 8.2 and 11.1 lead to the following result on the universal coverings of the groups G_k over a local field k:

Theorem 0.50. *12.1. — Let k be a local field and G_k the group of rational points of a split simply connected simple group over k. The locally compact group G_k possesses a universal covering E_0.*

(a) If $k = \mathbf{C}$, the group G_k is itself simply connected.

(b) If $k = \mathbf{R}$, the fundamental group $\pi_1(G_k)$ of G_k is isomorphic to $\mathbf{Z}$ [resp. to $\{\pm 1\}$] when G is of symplectic type [resp. non-symplectic].

(c) If k is ultrametric, $\pi_1(G_k)$ is isomorphic to the group μ_k of roots of unity contained in k.

Proof. Indeed, the case of $\mathbf{C}$ is an immediate consequence of Theorems 8.2 and 11.1. Let us demonstrate the statement for the ultrametric case; the case of $\mathbf{R}$ is analogous and, moreover, well known. Let us take the norm residue symbol $(x, y)_{m,k}$, m being the order of μ_k; it is the canonical Steinberg cocycle c^v used in Theorem 11.1. Let E_0 be the central extension of G_k by μ_k corresponding to c^v by Theorem 8.2: $1 \to \mu_k \to E_0 \xrightarrow{\pi} G_k \to 1$. The group E_0 is equal to its commutator subgroup. Let E be a central extension of E_0 by a locally compact group $A : 1 \to A \to E \xrightarrow{p} E_0 \to 1$. It is then a matter of verifying that this extension is trivial. View E as an extension of G_k by $B = p^{-1}(\mu_k)$; one easily sees that B is central in E. Let c_E be the Steinberg cocycle belonging to the extension E of G_k. On one hand, we obviously have $c^v = p \circ c_E$; on the other hand, according to Theorem 11.1, there exists a morphism λ from μ_k into B such that $c_E = \lambda \circ c^v$. We thus have $p \circ \lambda = \mathrm{id}_{\mu_k}$, and B is the product of $\lambda(\mu_k)$ by A. Consequently, according to no. 8, the elements $\tilde{x}_\alpha(u)$ of E generate a subgroup E' such that $E' \cap B = \lambda(\mu_k)$, and p induces an isomorphism from E' onto E_0. The extension E of E_0 is therefore trivial. The theorem is thus demonstrated. □

Let us examine the ultrametric case again. Let $\mathfrak{o}$ be the maximal compact subring of the ultrametric field k. Then $G_\mathfrak{o}$ is an open compact subgroup of G_k, and is generated by the root subgroups $U_\alpha^\mathfrak{o}$: by lemma 5.2, (c), $G_\mathfrak{o}$ is therefore its own commutator subgroup if the residue field of k possesses at least four elements. Let us consider in the universal covering E_v of G_k the subgroup $F_\mathfrak{o}$ generated by the elements $\tilde{x}_\alpha(u)$ ($\alpha \in \Phi, u \in \mathfrak{o}$). By using the Bruhat decomposition of $G_\mathfrak{o}$ ([17], [10]) and lemma 5.9, one sees without difficulty that $F_\mathfrak{o}$ is an open compact subgroup of E_v, which is a central extension of $G_\mathfrak{o}$ by the p-Sylow subgroup of μ_k, where p is the residue characteristic of k. In particular, the extension E_v is trivial above $G_\mathfrak{o}$ if the order of μ_k is coprime to p.

Now let k be a global field and $\mathbf{A}_k$ the ring of adeles of k. The adelized group $G_{\mathbf{A}_k}$ is the restricted product of the groups G_{k_v} relative to the $G_{\mathfrak{o}_v}$, where $G_{\mathfrak{o}_v}$ is defined for every non-archimedean place. For each v, let E_v be the universal covering of G_{k_v} defined from the canonical cocycle c^v or c_v of k_v. If v is non-archimedean, $F_{\mathfrak{o}_v}$ denotes the subgroup of E_v generated by the elements $\tilde{x}_\alpha(u)$ ($\alpha \in \Phi, u \in \mathfrak{o}_v$). According to what precedes, for almost all v, $G_{\mathfrak{o}_v}$ is equal to its commutator subgroup and $F_{\mathfrak{o}_v}$ is isomorphic to $G_{\mathfrak{o}_v}$.

The following theorem is therefore a consequence of lemma 10.4:

Theorem 0.51. *12.2. — The adelized group $G_{\mathbf{A}_k}$ possesses a universal covering. The fundamental group $\pi_1(G_{\mathbf{A}_k})$ of $G_{\mathbf{A}_k}$ is isomorphic to the direct sum of the groups $\pi_1(G_{k_v})$.*

ARITHMETIC SUBGROUPS

We now see that the compact group $S(A_k^\times, \mathbf{T})$ or $S^0(A_k^\times, \mathbf{T})$, defined in the preceding section, is precisely the dual of $\pi_1(G_{A_k})$.

We will then consider the embedding j of G_k into G_{A_k}. Let E_k [resp. E_{A_k}] be the universal cover of G_k [resp. of G_{A_k}], and let j_* denote the morphism, a lift of j, from E_k into E_{A_k}. According to Lemma 10.3, there exists a relative universal cover of G_{A_k} with respect to G_k, and $\pi_1(G_{A_k}, G_k)$ is isomorphic to $\pi_1(G_{A_k})/j_*(\pi_1(G_k))$. Now, it is easily verified that the morphism j_* from $\pi_1(G_k)$ into $\pi_1(G_{A_k})$ has as its dual the morphism j^*, defined in no. 11, from $S(A_k^\times, \mathbf{T})$ [resp. from $S^0(A_k^\times, \mathbf{T})$] into $S(k^\times, \mathbf{T})$ [resp. $S^0(k^\times, \mathbf{T})$]. Theorem 11.3 then states that $\pi_1(G_{A_k}, G_k)$ is isomorphic to μ_k. We thus have the following theorem:

Theorem 0.52. *12.3. — There exists a relative universal cover $\tilde{G}_{A_k}$ of G_{A_k} with respect to G_k. The relative fundamental group $\pi_1(G_{A_k}, G_k)$ is isomorphic to the group μ_k of roots of unity contained in k.*

Furthermore, one can construct $\tilde{G}_{A_k}$ directly from the canonical cocycle $c = (c_v)$ on $A_k^\times$ with values in μ_k, given by $c_v(x, y) = (x, y)_{m,k_v}$ for all v, where m is the order of μ_k.

Let S now be a finite set of places of k, and $A_k(S)$ the ring of S-adeles of k. For the adelized group $G_{A_k(S)}$ and the canonical embedding of G_k into $G_{A_k(S)}$, there is a result analogous to the two preceding theorems. Notably, Corollary 11.4 implies the following theorem:

Theorem 0.53. *12.4. — There exists a relative universal cover $\tilde{G}_{A_k(S)}$ of $G_{A_k(S)}$ with respect to G_k. The relative fundamental group $\pi_1(G_{A_k(S)}, G_k)$ is isomorphic to μ_k if S consists entirely of imaginary places; otherwise, it is reduced to the identity element.*

We are now in a position to answer the congruence subgroup problem for G of rank ≥ 2. Let S be a non-empty finite set of places of k containing all archimedean places, and $\mathfrak{o}$ the Dedekind ring of arithmetic type defined from S. Let us recall the exact sequence studied in no. 3:

$$1 \to C^S(G_k) \to \widehat{G_k(S)} \xrightarrow{\pi} G_{A_k(S)} \to 1.$$

The preceding theorem and Theorem 3.3 allow for the determination of the kernel $C^S(G_k)$.

Theorem 0.54. *12.5. — Suppose that G is of rank ≥ 2. Then, the group $\widehat{G_k(S)}$ constitutes a relative universal cover of $G_{A_k(S)}$ with respect to G_k. The kernel $C^S(G_k)$ is isomorphic to the group μ_k of roots of unity contained in k if S consists entirely of imaginary places. Otherwise, $C^S(G_k)$ is reduced to the identity element.*

Indeed, according to the two cited theorems, there exists a morphism ϕ from $\widehat{G_k(S)}$ into $\widetilde{G}_{A_k(S)}$ and another ψ from $\widetilde{G}_{A_k(S)}$ into $\widehat{G_k(S)}$ such that $\pi = \pi' \circ \phi$ and $\pi' = \pi \circ \psi$, where π' is the projection of $\widetilde{G}_{A_k(S)}$ onto $G_{A_k(S)}$. One deduces from this that ψ is the inverse of ϕ and thus that the two coverings of $G_{A_k(S)}$ are isomorphic.

Corollary 0.55. *12.6. — Suppose that G is of rank ≥ 2. If S contains at least one non-imaginary place of k, every S-arithmetic subgroup of G_k is an S-congruence subgroup.*

In section 3, we used the subgroups $\Gamma_\mathfrak{q}$ and $E_\mathfrak{q}$ of $G_\mathfrak{o}$ defined for every ideal $\mathfrak{q}$ of $\mathfrak{o}$. We recall that $C^S(G_k)$ is the projective limit, over non-zero ideals $\mathfrak{q}$, of the separated completions $\widehat{C}_\mathfrak{q}$ of the groups $\Gamma_\mathfrak{q}/E_\mathfrak{q}$. Theorem 12.5 thus implies that the $\widehat{C}_\mathfrak{q}$ are finite cyclic groups. In fact, for $G = \mathrm{SL}_3$ or Sp_4, Bass, Milnor and Serre [4] have shown that the $\Gamma_\mathfrak{q}/E_\mathfrak{q}$ are themselves finite groups and notably that $G_\mathfrak{o}$ is equal to $E_\mathfrak{o}$. By corollary 4.6, this is then generally the case for G of rank ≥ 2. We thus have the following theorem:

Theorem 0.56. *12.7. — Suppose that G is of rank ≥ 2. Then, the group $G_\mathfrak{o}$ is generated by the root subgroups $U_\alpha^\mathfrak{o}$, and every S-arithmetic subgroup of G_k has a subgroup of finite index generated by unipotent elements.*

For $G = \mathrm{SL}_2$ and a Dedekind ring $\mathfrak{o}$ of arithmetic type, J.-P. Serre [29] has shown that the conclusion of theorem 12.5 remains valid if S consists of at least two places of k (see also J. Mennicke [24]). On the other hand, $C^S(\mathrm{SL}_2(k))$ is an infinite group if $\mathfrak{o}$ is the ring of integers of $\mathbf{Q}$ or of an imaginary quadratic field; the first case is due to F. Klein, and the second to Serre [29].

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It would be of interest to generalize the results of this memoir to non-split simply connected semi-simple groups G. This is essentially a local problem relative to $\pi_1(G_{k_v})$, an adelic problem relative to $\pi_1(G_{A_k}, G_k)$ and the congruence subgroup problem for G_k, where k_v is a local field and k a global field.

The only systematic result that we know of in this direction is that of É. Cartan on the universal coverings of real semi-simple groups.

Soit G be a simply connected semi-simple group defined over $\mathbf{R}$ and $\mathbf{R}$-simple. The following theorems are due to É. Cartan [11]:

a. The group $G_{\mathbf{R}}$ is topologically connected;

b. The fundamental group $\pi_1(G_{\mathbf{R}})$ of $G_{\mathbf{R}}$ is isomorphic to $\{0\}$, $\mathbf{Z}/2\mathbf{Z}$ or $\mathbf{Z}$.

Several proofs of these theorems are known from a "compact" point of view. In fact, the method of this memoir, based on the theory

relative of algebraic groups [6], allows one to retrieve these results from an "anticompact" point of view and without performing case-by-case verification. One also finds there the following criterion for the order of $\pi_1(G_{\mathbf{R}})$:

Proposition 0.57. *c. Let H^d be a maximal split torus of G over $\mathbf{R}$ and H^c be a compact torus of G over $\mathbf{R}$ such that $H = H^d H^c$ is a maximal torus of G. Then, $\pi_1(G_{\mathbf{R}})$ is trivial if and only if no long root of G relative to H vanishes on H^c. Suppose that certain long roots of G vanish on H^c. Then, for $\pi_1(G_{\mathbf{R}})$ to be infinite, it is necessary and sufficient that two such roots, not opposite, are necessarily orthogonal, in other words that G over $\mathbf{R}$ is of type C_n or BC_n $(n \geq 1)$.*

The local problem is thus completely resolved in the real case. In the case of an ultrametric local field k, it is natural to conjecture that, if G is a simply connected simple group of relative rank ≥ 1 over k, the group G_k possesses a universal covering with a finite fundamental group $\pi_1(G_k)$.

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